

Exact T -counts of Toffoli layers from an isotropy bound

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Abstract

The T -count is the dominant cost of fault-tolerant Clifford+ T computation. We prove that a layer of m disjoint CCZ gates, the diagonal core of a parallel Toffoli layer, needs exactly $6m+1$ T gates in every Hadamard-free Clifford+ T circuit with clean ancillas. Campbell and Howard gave the matching construction. To our knowledge this is the first proof that it is optimal for general m (for $m = 1$ the value 7 is classical, and Campbell and Howard state the value 13 at $m = 2$ without proof). On every controlled diagonal Clifford gate, the class of their controlled-unitary theorem, the same floor comes within one of their exact count, and it recovers their $4m + 3$ for a fan-out of m Toffolis from one control. The proof rests on an isotropy constraint: for a pure-cubic phase, the vectors recording which T gates touch each qubit span a totally isotropic subspace. In general the constraint gives the isotropy floor $\delta \geq 2(n - d^*) - r$ for every diagonal level-three gate, where d^* is the dimension of the radical of the phase's moment tensor and r the rank of its quadratic moments. Both are computed from the phase polynomial in polynomial time. The floor is never below stabilizer nullity ν , which equals $n - d^*$ on this class, and a separate parity argument raises it to $2\nu + 1$ on non-Clifford pure-cubic gates. On the output of PyZX's TODD-based optimizer for the 24 benchmark circuits with input T -count at most 400 that it completes, the floor certifies 193 of its 311 merged phase-polynomial blocks optimal for the optimizer's Hadamard positions and block partition (186 to 193 across five optimizer seeds), against 113 for nullity. The floor also holds, under stated conditions, for circuits whose only internal Hadamards form unitarily uncomputed temporary AND blocks, while under general adaptive feedforward only $t \geq \nu$ is proved.

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1 Introduction

The number of T gates is the standard cost measure for fault-tolerant Clifford+ T computation. Each T consumes a distilled magic state, and in a surface-code architecture the factories that produce those states are a large share of the physical-qubit and runtime footprint [1]. Whether a planned computation fits on a given machine is therefore largely a question about its T -count. Any compiled circuit gives an upper bound on that count, but only for the compiler that produced it. A lower bound constrains every circuit, and it is useful for resource estimation when it is efficiently computable and close to the true count.

In many algorithms much of that T -count comes from Toffoli gates, which usually occur in layers: reversible arithmetic, phase oracles and the control trees of select operations all apply many Toffolis in parallel. A single Toffoli has T -count seven among ancilla-free Clifford+ T circuits [2]. We show that the diagonal core of a parallel Toffoli layer, a layer of m disjoint CCZ gates, has T -count $6m + 1$ in the model of Hadamard-free Clifford+ T circuits, built from CNOT and T with diagonal Clifford gates free, with clean ancillas. In this model the value seven at $m = 1$ follows from the characterization of Amy and Mosca [3] together with Corollary 84.

The lower bounds known for this class are either expensive or loose. Amy and Mosca [3] showed that the exact $\{\text{CNOT}, T\}$ T -count of a diagonal level-three gate is a minimum-distance decoding problem for a Reed–Muller code, for which no polynomial-time algorithm is known, and optimizing the T -count of a general Clifford+ T circuit is NP-hard [4]. Meet-in-the-middle synthesis [5] and exact counting [6] are also exact but take exponential time. Robustness of magic [7], channel robustness [8], stabilizer extent [9] and the stabilizer Rényi entropy [10, 11] all give T -count lower bounds. On a general gate, computing any of them exactly means either optimizing over stabilizer decompositions or summing over all Pauli operators, and both cost exponential time. Stabilizer nullity is efficiently computable on this class through the closed form below, but for CCZ, the diagonal core of the Toffoli, it gives three against an ancilla-free count of seven [2]. No magic monotone under stabilizer operations with feedforward can close that gap, since such a monotone also bounds circuits that use measurement and feedforward, and with those CCZ costs four T gates [12] (Remark 29).

We show that a bound which is not a monotone can be exact on families where stabilizer nullity, the standard efficiently computable monotone, is off by a factor of two. Each T gate of a $\{\text{CNOT}, T\}$ circuit acts on a parity of the qubits. When the target’s phase is pure-cubic, the vectors recording which T gates touch each qubit span a totally isotropic subspace of $\mathbb{F}_2^t$: each qubit is touched by an even number of them, and so is each pair of qubits. Such a subspace has dimension at most

$t/2$. With vanishing quadratic part this is the self-orthogonality of a gate-synthesis matrix given by Campbell and Howard’s Lemma 3 [13]. We show that this dimension is at least $n - d$, where n is the number of qubits and d the dimension of the cubic radical, the space of directions along which the cubic moments vanish. For a non-Clifford target a parity step adds one, so $t \geq 2(n - d) + 1$. We call the invariant behind this the cubic moment form and the bound the isotropy floor. It is computed by Gaussian elimination, with no search.

1.1 Contributions

The gates in question are the diagonal ones of the third Clifford level, whose phase polynomial over $\mathbb{Z}_8$ has degree at most three (T contributes a linear term, CS a quadratic one and CCZ a cubic one). They apply a classical function as a phase at $\pi/4$ angles and occur throughout reversible arithmetic and phase oracles.

Given the phase polynomial of a diagonal level-three gate U on n qubits, we compute in $\text{poly}(n)$ time a lower bound on the T -count $\delta(U)$ of every $\{\text{CNOT}, T\}$ circuit for U , and we name the invariant of the phase polynomial that forces it. Any circuit that meets the bound is then certified optimal.

The floor depends only on the order- ≤ 3 moments of the phase polynomial, and every rank it needs follows from them by Gaussian elimination in $O(n^3)$ time, counted as word operations on n -bit rows. The moments are easy to obtain in the two cases that arise in resource estimation:

- From a $\{\text{CNOT}, T\}$ circuit they come from propagating parities through the CNOTs, in time polynomial in the size of the circuit (Appendix A).
- From a Boolean function or phase polynomial (a Toffoli network, a phase oracle, a QAOA cost layer or a field-arithmetic block) they take a short calculation for each of the $O(n^3)$ sets of at most three qubits. There the phase polynomial is the specification, so the bound is available before any circuit exists.

We do not treat an opaque Clifford+ T circuit with internal Hadamards, whose phase polynomial is as hard to recover as its amplitudes. Stabilizer nullity is defined on the unitary (or its Choi state), and on this class it too is computed from the phase polynomial (Proposition 34), so both bounds take the same input.

The base model is $\{\text{CNOT}, T\}$ circuits with arbitrarily many clean ancillas, in which every lower bound below holds without further hypotheses (Corollary 84). The fourth item below relaxes it. The model is incomparable to the ancilla-free Clifford+ T setting in which exhaustive search found the T -count of the single Toffoli [2], since that setting allows internal Hadamards and forbids ancillas.

1. The floor and how tight it is. Every diagonal level-three gate obeys $\delta(U) \geq 2(n - d^*) - r$, where d^* is a kernel dimension and r a rank of the phase polynomial’s moments, both computed in $O(n^3)$ time, and a non-degenerate pure-cubic gate obeys $\delta(U) \geq 2n + 1$ (Theorems 27 and 26). Stabilizer nullity has the closed form $\nu = n - d^*$ on this class (Proposition 34), so the shift group of [14] is computed by Gaussian elimination. The floor is never below it and equals $2\nu + 1$ on non-Clifford pure-cubic gates (Corollary 35 and Proposition 32). Since the code of relations, the parity sets whose T gates multiply to a diagonal Clifford, has minimum distance 15, a residue of weight at most seven is already optimal on any number of qubits, and above that weight the T -count can fail to be additive (§2.1.3). An $O(n^3)$ ceiling brackets every gate. On non-degenerate gates ($d^* = 0$) the bracket stays within a constant factor under bounded locality, and it opens as $\Omega(n)$ in the ancilla-free worst case, where no bound linear in n can do better (§2.2.4 and

Remark 36). Exhaustive classification shows the floor exact on every pure-cubic class for $n \leq 5$ and on 39.4% of pure-cubic gates at $n = 6$ (§2.4.1). The floor extends to pure level- k gates at every level of the Clifford hierarchy and bounds CS and CCZ counts, although its exactness is specific to level three (§2.4.2). From level four on it bounds the number of $R_Z(2\pi/2^k)$ rotations, with lower-level phases free.

2. Exact T -counts. A layer of m disjoint CCZ gates costs $6m + 1$ for every m (Theorem 40). This matches a construction of Campbell and Howard [13, Thm. 6, Cor. 1], and to our knowledge the lower bound is new for $m \geq 2$. Controlled-parity layers cost the same (Corollary 41). Layers whose blocks share single qubits cost $6m + 1 - 2s$, with $s = 3m - n$ the incidences in excess of one per qubit, whenever the merges can be routed along a forest of blocks in which each block has at most three neighbors (Proposition 47). For the fan-out this recovers the count $4m + 3$ (Theorem 43), which follows from the controlled-unitary theorem of Campbell and Howard [13, Thm. 5] for ancilla-free circuits and is written out there at $m = 2$. The floor extends its optimality to circuits with clean ancillas. On every controlled unitary of that theorem, a control on a diagonal Clifford, the floor is within one of their exact count (Proposition 44). Every cubic $\ell \cdot q$ that factors into a linear form and a quadratic one is a controlled unitary after a relabelling, so the floor is exact on it (Corollary 53, a case of the controlled-unitary theorem). This includes the star and the complete cubic (§2.2.3).
3. Applications and certification. A Toffoli network in which no target is ever used as a control is Clifford-equivalent to the CCZ layer on its triples, so the floors bound every synthesis whose Hadamards lie in the target layers (Proposition 62, in the setting of [15]). They certify such a synthesis optimal when a construction meets them, as for linear overlap with forest routing (Corollary 63). Otherwise they still bound that layer, the diagonal shadow, whose T -count for the Gottesman–Mochon gate lies in [15, 28] (Proposition 65). Non-degenerate cubic phase oracles cost at least $2n + 1$ per call, an Ising cost layer is bracketed by its Hamiltonian alone, and the $\text{GF}(2^k)$ multiplication oracle costs $\Theta(k)$ asymptotically, via bilinear complexity, at least $6k + 1$ and exactly 17, the known value [16], at $k = 2$ (§2.3.3). On the output of TODD [17, 18] the floor certifies 193 of 311 merged blocks (186 to 193 across optimizer seeds) against 113 for nullity on the 24 circuits it completes, on the output of phase teleportation [19] 252 of 337 against 141 on 27 circuits, and on all 515 decodable blocks it is within 2 of the true T -count (§2.3.4).
4. Beyond the exact model. With internal Hadamards the floor gains one correction term (Proposition 4), which we bound for circuits whose Hadamards occur only inside temporary ANDs [20], ancillas that hold the AND of two bits. The ANDs must be uncomputed unitarily, with controls affine in the data and restored by each window. No odd phase may act on an AND value, no phase may involve two AND values, and each block must meet locality conditions on its windows (Corollary 86). Under relaxed block conditions that admit AND chains (Theorem 92), a counting argument valid at every level brackets the AND-compiled cost of $C^{k-1}Z$ on $k \geq 4$ qubits to within three T gates when the AND controls are single wires (Corollary 93). Block-local optimizers are safe exactly where the floor holds on every layered circuit, which remains open (Proposition 94). A circuit close enough to the target implements it exactly, so every floor survives approximation, in the regimes where it is proved, below a computable though very small accuracy (Theorem 97). Within the same regimes, non-adaptive measurement, Pauli-frame tracking with a stabilizer record and, for $\{\text{CNOT}, T\}$ circuits, catalysts without translation symmetry keep the full floor, while under adaptive feedforward only $t \geq \nu$ is proved and no bound $t \geq c\nu$ with $c > 4/3$ holds (Proposition 5).

Section 2 explains the method and states informal versions of the main results, on the layer, the

floor, internal Hadamards and measurement (Theorems 2 and 3 and Propositions 4 and 5). Each points to its formal version in the appendices, which contain the proofs. Section 4 lists the open problems.

2 Technical overview

2.1 Warm-up: the T -count of CCZ

We derive $\delta(\text{CCZ}) = 7$ from first principles. §2.1.3 gives a second proof, shows where additivity fails and extends the argument to a layer of m gates.

2.1.1 The mechanism, on one gate

1. The gate CCZ acts on basis states by $|x\rangle \mapsto (-1)^{x_1 x_2 x_3} |x\rangle$. Since $T = \text{diag}(1, e^{2\pi i/8})$, every phase a $\{\text{CNOT}, T\}$ circuit produces is an eighth root of unity, and the sign is one of them: $(-1)^b = e^{2\pi i(4b)/8}$. A diagonal gate assigns one phase to each basis state, so it is a function $\phi: \mathbb{F}_2^n \rightarrow \mathbb{Z}_8$, with $|x\rangle$ picking up $e^{2\pi i\phi(x)/8}$. For CCZ,

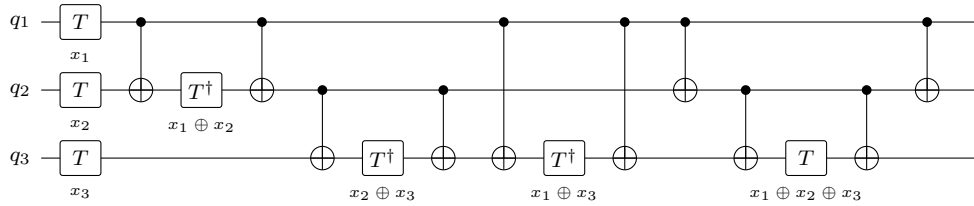
$$\phi(x) = 4x_1 x_2 x_3, \quad (1)$$

and this polynomial is the input.

2. A T applied to the parity y multiplies the state by $e^{2\pi i(y \cdot x)/8}$, where the bit $y \cdot x$ is read as the integer 0 or 1. Building a $\{\text{CNOT}, T\}$ circuit for (1) means choosing a coefficient $c_y \in \mathbb{Z}_8$ for each parity with $\sum_y c_y (y \cdot x) \equiv \phi(x)$ for every x . One choice works:

$$4x_1 x_2 x_3 \equiv x_1 + x_2 + x_3 - (x_1 \oplus x_2) - (x_1 \oplus x_3) - (x_2 \oplus x_3) + (x_1 \oplus x_2 \oplus x_3) \pmod{8}. \quad (2)$$

A T adds 1 to a coefficient, an S adds 2 and a Z adds 4, so a parity costs a T exactly when its coefficient is odd. The three minus signs above are the coefficient 7, that is $T^\dagger$. Each term is therefore one T : a T on each single qubit, a $T^\dagger$ on each pair, a T on the triple. CNOTs route a parity onto a wire and route it back, so the circuit is



The circuit uses ten CNOTs and seven T gates, and the parity each T acts on is written beneath it.

3. Some sets of parities contribute nothing that costs a T : their T gates multiply to a diagonal Clifford, which is free. Adding such a set changes the circuit and leaves the gate alone up to free gates, so one specification has many circuits, and the T -count is the fewest odd coefficients any of them uses. The condition for this is linear, so those sets are closed under combination and form a code. Amy and Mosca [3] identified it as a punctured Reed–Muller code. The circuits for one specification are therefore a single coset of it, and the T -count is the lightest member of that coset, a minimum-distance decoding problem with no known polynomial-time algorithm. The remaining steps bound the decoding optimum from below without solving it.

4. The circuit now becomes a matrix. Take the parities with odd coefficients, since only those cost a T , and make them the rows of a matrix over $\mathbb{F}_2$: one row per T gate, one column per qubit. Reading (2) term by term traces out the seven,

parity y	as a row	c_y	gate
x_1	100	1	T
x_2	010	1	T
x_3	001	1	T
$x_1 \oplus x_2$	110	7	$T^\dagger$
$x_1 \oplus x_3$	101	7	$T^\dagger$
$x_2 \oplus x_3$	011	7	$T^\dagger$
$x_1 \oplus x_2 \oplus x_3$	111	1	T

and stacking the entries of its second column, in that order, gives the matrix

$$Y = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad (3)$$

so t , the number of T gates, is 7 here.

5. Write $c_a \in \mathbb{F}_2^t$ for column a of (3): it marks the T gates that touch qubit a . The weight of c_a is how many of the chosen parities contain a , and the overlap $\langle c_a, c_b \rangle$ is how many contain both a and b . Taken mod 2, those two counts are the gate's moments of order one and order two. They depend only on the gate and not on which circuit produced it (Fact 21), so any one circuit may be used to compute them. Call the phase *pure-cubic* when every one of these moments is even. The phase of CCZ is, since it has no linear or quadratic term, so

$$\langle c_a, c_b \rangle = 0 \quad \text{for all } a, b, \text{ the case } a = b \text{ included.} \quad (4)$$

Take CCZ's first column, $c_1 = (1, 0, 0, 1, 1, 0, 1)^\top$. Qubit 1 appears in four of the seven parities, namely x_1 , $x_1 \oplus x_2$, $x_1 \oplus x_3$ and $x_1 \oplus x_2 \oplus x_3$. It shares two of those with qubit 2, namely $x_1 \oplus x_2$ and $x_1 \oplus x_2 \oplus x_3$. Every weight here is four and every overlap two, so (4) holds.

6. Self-orthogonality gives $t \geq 2 \text{rank } Y$. Put $\mathcal{C} = \text{span}\{c_1, \dots, c_n\}$ inside $\mathbb{F}_2^t$. Then (4) says $\mathcal{C} \subseteq \mathcal{C}^\perp$, and $\dim \mathcal{C}^\perp = t - \dim \mathcal{C}$, so

$$\dim \mathcal{C} \leq t - \dim \mathcal{C}, \quad \text{hence} \quad t \geq 2 \dim \mathcal{C} = 2 \text{rank } Y. \quad (5)$$

7. The cubic part forces the rank up. Count once more, now three indices at a time: T_{abc} records whether an odd number of the chosen parities contain all of a , b and c . For distinct a , b , c , in the columns of (3),

$$T_{abc} = \sum_{j=1}^t (c_a)_j (c_b)_j (c_c)_j \pmod{2}. \quad (6)$$

Equation (4) said the counts for one and for two indices are even, so the order-three counts are the only ones left. The counts of orders one to three together form the full moment tensor, and

its order-three part is the cubic moment form. On three qubits there is a single triple to count, and one of the seven parities contains it, namely $x_1 \oplus x_2 \oplus x_3$. One is odd, so $T_{123} = 1$ is the only nonzero entry of the form. Each $w \in \mathbb{F}_2^n$ gives a matrix $T(w)$ with entries $T(w)_{bc} = \sum_a w_a T_{abc}$, and $w \mapsto T(w)$ is linear. The radical R is its kernel, and $d = \dim R$. For CCZ,

$$T(w) = \begin{pmatrix} 0 & w_3 & w_2 \\ w_3 & 0 & w_1 \\ w_2 & w_1 & 0 \end{pmatrix},$$

which is the zero matrix only when $w = 0$, so $R = \{0\}$ and $d = 0$.

Write K for the kernel of $w \mapsto \sum_a w_a c_a$. If $w \in K$ then, by (6),

$$T(w)_{bc} = \sum_j (c_b)_j (c_c)_j \left(\sum_a w_a c_a \right)_j = 0,$$

so $w \in R$. Thus $K \subseteq R$ and $\dim K \leq d$, giving

$$\text{rank } Y = n - \dim K \geq n - d. \quad (7)$$

For CCZ this gives $\text{rank } Y = 3$.

8. Chaining (5) with (7) gives $t \geq 2(n - d)$ for every circuit, so $\delta \geq 2(n - d)$. For CCZ that says $t \geq 6$, and a trivial radical adds one (Theorem 26). Suppose some circuit had exactly six T gates. Its three columns are independent by (7), so $\mathcal{C}$ has dimension three, half of six, and (5) holds with equality: $\mathcal{C}$ is all of $\mathcal{C}^\perp$. The all-ones vector $\mathbf{1} \in \mathbb{F}_2^6$ is orthogonal to every column because every column has even weight. It therefore lies in $\mathcal{C}^\perp = \mathcal{C}$, and so it is the sum of the columns over some nonempty set S of qubits. Let w be the indicator of S . Then (6) gives $T(w)_{bc} = \langle c_b, c_c \rangle = 0$ for all b, c , so this nonzero w lies in the radical. The radical of CCZ is $\{0\}$, so no six-gate circuit exists (Lemma 25), and $t \geq 7$. The circuit above uses seven, so the floor is exact and $\delta(\text{CCZ}) = 7$.

The factor of two uses purity, through (4) of step 5, in steps 6 and 7. In step 7 the terms with coincident indices are overlaps $\langle c_b, c_c \rangle$, which vanish only on a pure-cubic phase, so off that locus the argument needs the radical R^* of the full moment tensor (Definition 1). The extra one of step 8 uses (4) again together with the trivial radical. When $d > 0$ the vector w may lie in the radical, and the argument stops at $2(n - d)$. Stripping the radical off first recovers the extra one for every non-Clifford pure-cubic gate (Proposition 32). Figure 1 summarizes the argument. §2.2.1 states the floor for a general phase, where the order-two moments need not vanish.

2.1.2 Definitions

The function ϕ of step 1 is the *phase polynomial* of the gate, and the vector $(c_y \bmod 2)_y$ of its coefficients in (2), the *residue*, records which parities carry a T gate.

Definition 1 (cubic moment form and radical). For distinct a, b, c the order-three moment T_{abc} is the parity of the number of odd parities containing all three, and $T_{abc} = 0$ when two indices coincide. The resulting symmetric trilinear form on $\mathbb{F}_2^n$ is the *cubic moment form*. Filling one slot with $w \in \mathbb{F}_2^n$ leaves the matrix $T(w)_{bc} = \sum_a w_a T_{abc}$, the *slice* of the form in direction w . The map $w \mapsto T(w)$ is linear, and its kernel is the *radical* R , with $d = \dim R$. The moments of orders one and two form the symmetric *quadratic moment matrix* Q , of rank r . Placing them at coincident indices gives the full moment tensor, whose radical R^* has dimension d^* .

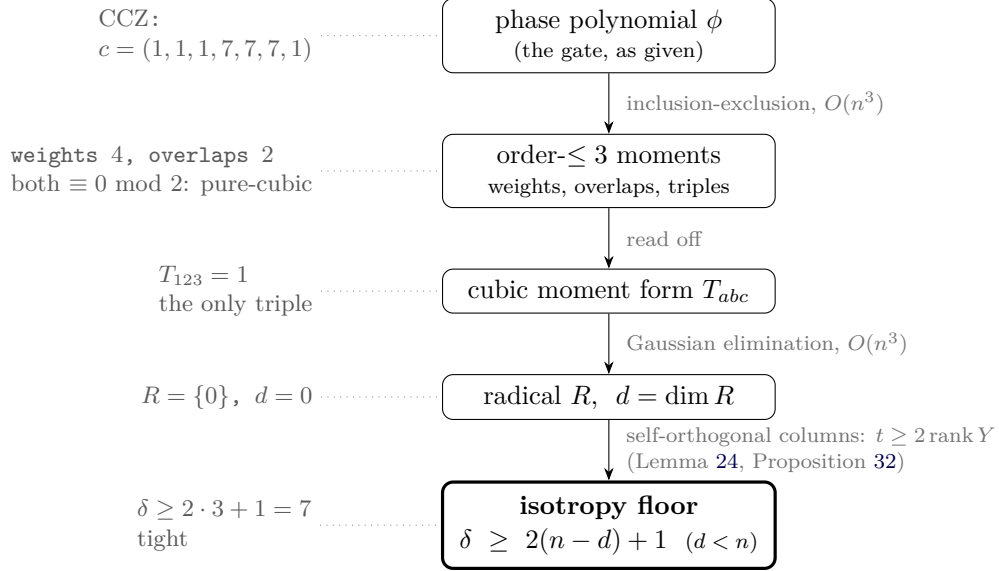


Figure 1: The isotropy floor, end to end, with the §2.1 example beside it. The figure shows the pure-cubic case, and §2.2.1 gives the general floor. At CCZ the floor returns 7 and the seven- T construction meets it, so $\delta(\text{CCZ}) = 7$.

These do not depend on which representation is used (Fact 21), and on a pure-cubic gate $Q = 0$, $r = 0$ and $d^* = d$. The full moment tensor is the signature tensor of Heyfron and Campbell, so δ is its symmetric rank over $\mathbb{F}_2$ (§3). The *isotropy floor* is $2(n - d) + 1$ for a non-Clifford pure-cubic gate ($d < n$) and $2(n - d^*) - r$ otherwise. Both are lower bounds on δ (Proposition 32 and Theorem 27).

2.1.3 Rigidity, and from one gate to a layer

A second argument gives $\delta(\text{CCZ}) = 7$ directly. Every nonzero codeword of step 3 has weight at least 15, so a residue of weight at most 7 is already the lightest in its coset, and its weight is the T -count on any number of qubits (Theorem 38). Above weight seven the T -count is not additive, as Campbell and Howard observed [13, Thm. 6]. A T beside a CCZ adds nothing, $\delta(\text{CCZ} \oplus T) = 7$, and two disjoint CCZs cost 13, because gates placed side by side share a larger code whose codewords reach into both blocks (Theorem 39). Counting block by block therefore never under-reports a Hadamard-free cost but can over-report it.

The argument is not specific to $n = 3$. Put m disjoint CCZs side by side on $n = 3m$ qubits: the moment form is block-diagonal, step 7's check repeats on each block, so again $d = 0$, and step 8 returns $2n + 1 = 6m + 1$. A construction uses that many T gates (Proposition 47, drawn at $m = 2$ in Figure 2 of Appendix C), matching a count of Campbell and Howard [13, Cor. 1].

Theorem 2 (the parallel Toffoli layer (informal version of Theorem 40)). *A layer of m disjoint CCZ gates has T -count exactly $6m + 1$, against $\{\text{CNOT}, T\}$ circuits with arbitrarily many clean ancillas.*

Stabilizer nullity is tensor-additive and gives only $3m$ here, one T gate per dimension (Table 1).

Table 1: Three lower bounds on the parallel Toffoli layer. The isotropy floor meets the construction at every m . The amortized SRE recovers about a quarter of the factor of two. The SRE rate per gate is $\sigma = (5 - \log_2 11)/(2 - \log_2 3) = 3.7118\dots$: the stabilizer Rényi entropy of the CCZ Choi state over the T gate’s amortized value, from the bound of Zhu et al. [11].

target	n	ν [14]	amortized SRE [11]	isotropy floor	δ (true)
CCZ	3	3	4	7	7
two disjoint CCZ	6	6	8	13	13
three disjoint CCZ	9	9	12	19	19
m disjoint CCZ	$3m$	$3m$	$\lceil \sigma m \rceil$	$6m + 1$	$6m + 1$
per-CCZ rate		3.00	3.712	6.00	6.00

2.2 The floor and its tightness

2.2.1 Nullity, entropy and the floor

Theorem 3 (the isotropy floor (informal version of Theorem 27)). *Every diagonal level-three gate U on n qubits satisfies $\delta(U) \geq 2(n - d^*) - r \geq \nu$, where d^* and r are computed from its phase polynomial in $O(n^3)$ time and $\nu = n - d^*$ is its stabilizer nullity.*

On non-Clifford pure-cubic gates $r = 0$, the expression is 2ν , and Proposition 32 adds one, so the isotropy floor is $2\nu + 1$. In general the expression exceeds nullity by $\nu - r$ (Corollary 35), so it meets nullity only when $r = \nu$. On pure-cubic gates nullity and the floor both depend only on the dimension $n - d$ spanned by the slices, one T gate per dimension against two. The amortized stabilizer Rényi entropy of Zhu et al. [11] depends instead on the ranks of the slices, so it is incomparable with nullity in general: it exceeds nullity on every non-Clifford pure-cubic gate with $n \leq 6$ and falls below it on the fan-out layer from $m = 4$. It lies below the floor on every non-Clifford pure-cubic gate at $n \leq 6$ and is not polynomial-time (Appendix K, Open Problem 9).

2.2.2 Sharing

Suppose any two blocks share at most one qubit, which we call *linear overlap*, and let $s = 3m - n$ count the merged incidences. The layer stays non-degenerate, so its floor is $2n + 1 = 6m + 1 - 2s$ (Lemma 42 and Corollary 45). Where two blocks share a qubit, two T gates act on one wire and combine into an S (Figure 3), and the splicing construction attains the floor whenever the merges can be routed along a forest of blocks in which each block has at most three neighbors (Proposition 47). Fanning m blocks out of one control puts all the sharing on one qubit, and this *fan-out* costs exactly $4m + 3$, the count Campbell and Howard proved optimal among ancilla-free circuits [13] (Theorem 43). Once the merges must close a cycle the floor can be loose. Three blocks in which each pair shares a different qubit, such as $\{0, 1, 2\}$, $\{0, 3, 4\}$ and $\{1, 3, 5\}$, have floor 13 and T -count 15 (Remark 48). Blocks that all share one pair of qubits are Clifford-equivalent to a single CCZ and cost 7 for every m (Remark 49).

2.2.3 Cubics that factor

The floor is also exact beyond layers. If a cubic factors as $f = \ell \cdot q$, a linear form times a quadratic one, and q restricted to $\ell = 0$ has an invertible adjacency matrix over $\mathbb{F}_2$, the gate is non-degenerate and $\delta = 2n + 1$ (Proposition 54). After a CNOT relabelling such a gate is a controlled unitary, so

this value also follows from the controlled-unitary theorem of Campbell and Howard [13, Thm. 5]. Invertibility forces n odd. The class contains the star $S_n = \sum_{0 < j < k < n} x_0 x_j x_k$ and the complete cubic $e_3 = e_1 \cdot e_2$, so the floor gives the T -count of both at every odd n (Corollary 55). At $n = 9$ the complete cubic has 84 blocks, 588 T gates counted block by block, against $\delta = 19$. Without the invertibility hypothesis every cubic that factors still attains its floor $2\nu + 1$ (Corollary 53).

2.2.4 An upper bound, and three regimes

Solving the moment equations downwards, triples first, gives a canonical residue whose weight $W(s)$ is a valid T -count, also in $O(n^3)$ time (Proposition 30, worked on two overlapping blocks in Example 31). Every instance therefore gets a certified bracket floor $\leq \delta \leq W(s)$, and for a 3-local Ising cost layer the ceiling is the number of Hamiltonian terms whose angle is an odd multiple of $\pi/8$ (Proposition 52). On the families above, explicit constructions meet the floor itself. Under bounded locality, for a non-degenerate gate ($d^* = 0$) with every qubit in at most D cubic or quadratic terms, the ratio of ceiling to floor is $O(D)$ and independent of n (Corollary 37). In the worst case it is $\Omega(n)$, because every floor here is at most $2n + 1$ while the worst-case ancilla-free T -count is $\Theta(n^2)$ (Remark 36).

2.3 Applications

The floors carry over to Toffoli networks, oracles and optimizer output, and Appendix J turns them into resource estimates.

2.3.1 Reversible circuits

Every Toffoli is a CCZ conjugated by Hadamards on its target. If no target of a Toffoli network is ever used as a control, the network is *mismatch-free*, every Hadamard slides to the two ends, and the network π is Clifford-equivalent to the CCZ layer D_π on the same triples, its *diagonal shadow*. Write δ_T for the Clifford+ T T -count of any gate, keeping δ for the Hadamard-free count of a diagonal gate. Then $\delta_T(\pi) = \delta_T(D_\pi) \leq \delta(D_\pi)$, and every floor above bounds every circuit for the network whose Hadamards lie in its target layers (Proposition 62, after [15]). On linear-overlap networks routed as in §2.2.2 this gives $\delta(D_\pi) = 6m + 1 - 2s$ (Corollary 63), and the m Fredkins of a swap test form the fan-out. Without mismatch-freedom the floor still bounds the shadow (Corollary 64). For the Gottesman–Mochon gate, the first gate of C_3 known not to be semi-Clifford (reported in [21]) and Clifford-conjugate to a permutation [15], the shadow’s T -count lies in [15, 28] against a nullity of 7 (Proposition 65).

2.3.2 Arithmetic

The Toffoli skeleton of an m -Toffoli ripple carry [22] is a chain in which consecutive blocks share one carry wire, so it has linear overlap with $s = m - 1$ and floor $4m + 3$, against $7m$ for the Toffolis counted one at a time (Corollary 45). The AND chains inside unary iteration [23] have the same structure (§J.1). Each carry is written and then read, so the chain is not mismatch-free, and the bound applies to its CCZ content, the quantity a Toffoli-count estimate counts, not to the adder itself. Work that lowers the Toffoli count, such as that of Motlagh and Pocrnic for QROM [24], is complementary.

2.3.3 Oracles and cost layers

A 3-local Ising cost layer $\exp(-i\gamma \sum_S w_S Z_S)$ with every γw_S a multiple of $\pi/8$ is a diagonal level-three gate whose parities are the terms of the Hamiltonian (a Max-3-XORSAT objective gives one with only 3-body terms), so both ends of the bracket are read off the interaction hypergraph, couplings and fields (Corollary 51 and §2.2.4). A non-degenerate Boolean oracle $(-1)^f$, with f a homogeneous cubic such as the cubic form $\sum_{\{a,b,c\} \in E} x_a x_b x_c$ of a 3-uniform hypergraph, costs at least $2n + 1$ per call (Corollary 50), and the same floor bounds the diagonal core of an IQP circuit (Remark 68). The multiplication oracle $U_k|x, y, z\rangle = (-1)^{z \cdot (x*y)}|x, y, z\rangle$ of $\text{GF}(2^k)$ on $3k$ qubits is non-degenerate for every k , so $\delta(U_k) \geq 6k + 1$ (Proposition 56), and the ranks of its slices raise this to about $7k$ (Proposition 60). At $k = 2$ exact decoding gives $\delta(U_2) = 17$, matching the best published circuits [25] and AlphaTensor-Quantum’s optimality proof [16] (Remark 57). Table 9 collects the application floors.

2.3.4 Certifying the optimizers

A heuristic T -count optimizer reports an upper bound with no bound on its gap to the optimum, and exact methods such as Reed–Muller decoding are exponential. The floor is as cheap to compute as nullity. We ran PyZX’s phase-block optimizer, based on TODD [17, 18], on the 24 circuits of the standard reversible benchmark suite [5, 26] with input T -count at most 400 that it finishes within four minutes, and computed the floor of every output block. A block is *certified* when its floor equals its T -count, since no $\{\text{CNOT}, T\}$ resynthesis of that block can then use fewer T gates. Appendix F.2 gives the method, the seed and every circuit.

Table 2: Certifying the output of the phase-block optimizer, for eight of the 24 circuits and the totals over all of them. Blocks certified is the number of output blocks whose floor equals their T -count, out of all blocks. By ν alone counts the blocks that stabilizer nullity certifies. $\sum$ block floors adds the floors of all blocks. It bounds from below every block-by-block resynthesis that keeps the optimizer’s Hadamard layering. T after merging is the output’s T -count once parities are merged inside each block, and T emitted is the count the optimizer itself produces.

circuit	n	blocks certified	by ν alone	$\sum$ block floors	T after merging	T emitted
csum_mux_9	30	0/3	0	73	76	76
vbe_adder_3	10	5/5	0	28	28	32
tof_10	19	15/18	8	84	87	87
mod_red_21	11	12/16	3	76	81	83
ham15-low	17	13/21	3	111	119	123
barenco_tof_10	19	16/33	8	129	146	174
grover_5	9	29/49	27	185	205	223
adder_8	24	8/20	5	214	246	250
all 24 circuits		193/311	113	1563	1800	1878

On the 24 circuits that TODD completes (Table 2), the floor certifies 193 of the 311 blocks against 113 for nullity, at the seed with the lowest total T -count and between 186 and 193 across five seeds. No certified block is pure-cubic, so every certificate comes from the general floor $2(n - d^*) - r$ of Theorem 27. With phase teleportation [19] as the optimizer it certifies 252 of 337 blocks against 141 on 27 circuits, and 248 of 318 on the same 24. On the 515 blocks of the two runs that are small enough to decode exactly, 239 and 276, the floor is within 2 of the true T -count. The certificate is relative to the optimizer’s Hadamard layering. It bounds every block-by-block resynthesis that keeps those Hadamards in place, and not one that moves them: on `csum_mux_9` the block floors

sum to 73, and AlphaTensor-Quantum reaches 71 on 42 qubits instead of 30 [16]. For the $\text{GF}(2^k)$ multipliers, which are mismatch-free as whole circuits, the slice-rank floor $G(k) \approx 7k$ lies a factor 1.2 to 19 below the best published counts at $k = 2$ to 256 (Proposition 60 and the table after it). At $k = 2$ the published count 17 is optimal, so there the factor is slack in the floor.

2.4 Limits and extensions

2.4.1 Where the floor is exact, and what separates the rest

Two gates related by a CNOT relabelling of the qubits have the same T -count, and below seven qubits there are few enough such classes to decode them all. The floor is exact on every pure-cubic class for $n \leq 5$ and on three of the five at $n = 6$, which is 39.4% of pure-cubic gates at $n = 6$ (Table 3). The three $n = 6$ classes with $d = 0$ cost 13, 15 and 17, while nullity and the floor depend only on the radical and cannot separate them. The ranks of the slices do (Appendix B.4 and Corollary 59), and separating all three exactly is open (Appendix K).

2.4.2 Other levels and other gate sets

For pure level- k gates the isotropy argument gives $\delta_k \geq 2(n - d_k)$ at every level k , where δ_k is Amy and Mosca's $R_Z(2\pi/2^k)$ -count [3], a T -count only at $k = 3$ (Proposition 70). The exactness results are specific to level three, since the slack on $C^{k-1}Z$ grows without bound past $k = 3$ (Proposition 72). Dividing a floor by 3 or 7 bounds the CS or CCZ count of a Hadamard-free circuit (Propositions 75 and 77). On the parallel Toffoli layer this gives m CCZ gates for every $m \leq 7$, so a CCZ count gains nothing from merging below $m = 8$ (Corollary 78).

2.4.3 Internal Hadamards

After a Hadamard the state is a sum over branches, and a later T gate acts on every branch at once, so the T gates could number fewer than the odd coefficients of the phase polynomial, and $\delta(U)$ is no longer automatically a lower bound. Expanding the h internal Hadamards, the branches on which the ancillas return clean form an affine space, the *fiber*, and the output rank ρ_G counts the independent branch combinations that reach the output. The odd parities of the expanded circuit that are constant on the fiber implement an ordinary phase ψ . The N_{ker} parities that vary must implement the *correction* $\kappa = \phi - \psi$. Remark 89 traces every quantity on the standard ancilla-based CCZ circuit.

Proposition 4 (internal Hadamards (informal version of Proposition 82)). *A measurement-free Clifford+ T circuit with clean ancillas that implements a diagonal level-three gate U uses $t \geq \delta(U) + (N_{\text{ker}} - \delta(\kappa))$ T gates.*

Its content is the inequality $N_{\text{ker}} \geq \delta(\kappa)$, under which $t \geq \delta(U)$. That holds with at most two internal Hadamards, at maximal output rank and for Clifford corrections (Corollary 83). The conclusion $t \geq \delta(U)$ also holds, by a direct proof, for unitarily uncomputed AND blocks (Corollary 86), which covers the temporary-AND compilations of the layers and fan-outs above but not AND chains. Every quantity in the inequality except the two T -counts is polynomial in the circuit, so $N_{\text{ker}} \geq \bar{\delta}(\kappa)$, with $\bar{\delta}$ the ceiling of §2.2.4, certifies the floor in polynomial time, and $t \geq \delta(U) - \bar{\delta}(\kappa)$ holds with no hypothesis, although it is vacuous when the right side is not positive (Proposition 91). The strata with $h \geq 3$ and $\rho_G < h/2$, among them $\rho_G = 1$, are open in general (Table 8 and Open Problem 6). Corollary 86 covers the standard ancilla-based CCZ circuit in it, and nested AND chains and odd phases on AND values remain, because the obstruction comes from mod-4 and

mod-8 carries that are invisible over $\mathbb{F}_2$ (Remark 88). Non-diagonal targets beyond permutations are open as well (Open Problem 7). Block-local optimizers are safe only as far as that hypothesis reaches. Summing optimal block counts never falls below the Hadamard-free optimum $\delta(U)$ exactly when $t \geq \delta(U)$ holds on every layered circuit (Proposition 94 and §H.3).

2.4.4 Approximation

Let V be a measurement-free circuit on N wires with t T gates that agrees with U to within ε on its clean block. If $\varepsilon < 2^{-(2N(t+1)+2)}$ then V implements U exactly, because the entries of Clifford+ T circuits are algebraic numbers of bounded height, so every floor applies to V in the regimes where it is proved (Theorem 97). With no hypothesis on ε ,

$$t \geq \min\left(\delta(U) - \bar{\delta}(\kappa), \frac{\log_2(1/\varepsilon) - 2}{2N} - 1\right),$$

and for the parallel Toffoli layer on its $N = 3m$ wires, without ancillas, the exact count $6m+1$ applies in the same sense for every $\varepsilon < \varepsilon_m^* = 2^{-(36m^2+6m+2)}$ (Remark 99). The randomized constructions of Gosset, Kothari and Zhang [27], which reach $O(\log(1/\varepsilon))$ T gates for the multiply-controlled Toffoli, are mixtures of circuits and lie outside this model. The same authors show that a purely unitary implementation needs $\Omega(n)$ T gates even at constant error [27] (Remark 99).

2.4.5 Measurement and feedforward

Measurement without feedforward leaves every floor intact in the regimes where it is proved, for circuits that leave their ancillas in a computational-basis state, since an outcome that conditions nothing can be postponed to the final readout. Adaptive feedforward lowers the provable floor to the nullity bound $t \geq \nu$ of Beverland, Campbell, Howard and Kliuchnikov [28] (Propositions 102 and 100), which the closed form of Proposition 34 makes efficient.

Proposition 5 (the measurement boundary (informal version of Proposition 101)). *Clean ancillas and non-adaptive measurement with computational-basis ancilla output leave every floor of this paper intact in the regimes where it is proved. Adaptive feedforward lowers the provable floor to the nullity $\nu = n - d^*$, and no bound $t_{\text{ad}} \geq c\nu$ with $c > 4/3$ holds on diagonal level-three gates.*

The Jones gadget [12] builds CCZ adaptively with four T gates against seven, so $t/\nu = 4/3$ is attained, and whether some c between 1 and $4/3$ holds is open (Open Problem 8). Feedforward helps because a controlled correction lies one level higher in the hierarchy than the correction itself. Deferring the measurement turns the Jones gadget’s conditioned CZ into a CCZ, and its four T gates into $4 + 7 = 11$ (Figure 6). Pauli-frame corrections, the model used in fault-tolerant software, keep the full floor when their record is a stabilizer state (Proposition 104), and arbitrary feedforward keeps the ε -robust nullity bound $t \geq \min(\nu, \log_2(1/\varepsilon)/(4N) - 2M - 2)$, with M the number of mid-circuit measurements (Theorem 106).

3 Related work

Beverland, Campbell, Howard and Kliuchnikov [28] introduced stabilizer nullity, the bound this paper is measured against, as a state monotone that also bounds circuits with adaptive feedforward. Jiang and Wang [14] defined its unitary form. To our knowledge there is no other efficiently computable lower bound on the whole class beyond the related monotones of [28]. On controlled

diagonal Cliffords, which include layers whose blocks share one control, the controlled-unitary theorem of Campbell and Howard [13, Thm. 5] gives the exact T -count through Lempel’s factorization, and the floor comes within one of it there (Proposition 44). Jiang and Wang also exhibit a factor of two on one family, between the nullity of the unitary and the nullity of a state it prepares, which separates two monotones. The Choi state is not such a state, since its nullity equals that of the unitary (Proposition 100). Our factor of two is between the T -count and the unitary nullity, on every non-Clifford pure-cubic gate and with equality on named families (Proposition 32).

The strongest exponential-time measure we compared against is the amortized stabilizer Rényi entropy of Zhu et al. [11], which lies strictly between nullity and the floor on every non-Clifford pure-cubic gate up to six qubits, although nullity overtakes it on the fan-out layer from $m = 4$ (Appendix K). Seddon and Campbell [8] define channel robustness and magic capacity, which bound the T -count of multi-qubit operations. Both are computed by optimization over stabilizer decompositions whose size grows superexponentially with n .

Campbell and Howard [13, Thm. 6, Cor. 1] gave a construction achieving $6m + 1$ for a layer of m disjoint CCZ gates. The best earlier lower bounds we know of were the nullity $3m$ [14] and about $3.71m$, which we obtain by evaluating the amortized stabilizer Rényi entropy bound of [11] on CCZ and using additivity over the blocks (Table 1). Theorem 40 supplies the matching one. Ruiz et al. [16] found the count 17 for the GF(4) multiplier and proved it optimal among symmetric decompositions of its signature tensor by search. Apart from such searches and the controlled-unitary theorem, reversible level-three primitives and field-multiplication oracles had constructions but, to our knowledge, no efficiently computable lower bound above nullity. On the fan-out layer the floor recovers the exact count $4m + 3$ of Campbell and Howard (Theorem 43).

In the exact characterization of Amy and Mosca [3], the floor is a rank argument that bounds the optimum of their decoding problem from below without solving it. It is not always tight, and §2.2 identifies gates on which the two agree. Kam et al. [29] bound the T -count left after exact post-processing by the denominator exponent of the synthesized matrix entries, a number-theoretic certificate. It gives nothing on this class, since the entries of a diagonal level-three gate are eighth roots of unity, which have no denominator.

Heyfron and Campbell [17] phrase T -count optimization as decomposing the signature tensor of a phase polynomial into few symmetric rank-one terms, and their TODD compiler and AlphaTensor-Quantum [16] search for such decompositions. Khoruzhii, Gelß and Pokutta [30] match or improve the counts of these searches by tensor decomposition over $\mathbb{F}_2$ at far lower computational cost. The signature tensor is the moment tensor used here, so δ is its symmetric rank over $\mathbb{F}_2$ and the floor bounds that rank from below. For symmetric matrices the minimal decomposition over $\mathbb{F}_2$ is known exactly from Lempel [31], and the $-r$ term of Theorem 27 rests on the same alternating-form count.

Using communication complexity, Girish, May, Parham and Yuen [32] prove an $\Omega(n)$ magic count for the n -controlled Toffoli, and Gosset, Kothari and Wu [33] bound state preparation through Pauli postselection. Both bounds are asymptotic, and the first covers adaptive circuits when measurements are counted along with magic. Ours are per-gate and non-asymptotic and are exact on named families, but the factor of two over nullity holds only without feedforward (§2.4.5).

In distillation and code design, Bravyi and Haah [34], Nezami and Haah [35] and Rengaswamy et al. [36] use related self-orthogonality conditions, triorthogonality and the CSS-T condition, to control distillation rates and transversal- T codes. These works bound code parameters. Here the pair-level version, even column weights and pairwise even overlaps of binary columns, yields a floor on the T -count of a given gate. The alternating rank of the cubic part, used by Khoruzhii, Gelß and Pokutta [37] to classify cubic forms, is a different invariant from the radical used here.

4 Discussion and open problems

4.1 Scope of the floor

The isotropy floor is not a magic monotone, so it need not behave well under composition or under stabilizer operations on arbitrary states, which allows it to be exact where nullity is not (§1). The floor extends to pure level- k gates at every level of the Clifford hierarchy, but its exactness results are specific to diagonal level-three gates (Propositions 70 and 72).

The floor does not determine the T -count of the multi-controlled gate $C^{n-1}Z$, for the reasons collected in Appendix K.3, although Corollary 93 brackets it to within three T gates among AND-compiled circuits whose AND controls are data inputs or earlier AND values, for $n \geq 4$.

4.2 Open problems

The open problems below concern internal Hadamards (the first two), measurement (the third) and the tightness of the floor (the last two).

Open Problem 6 (the last stratum). Does the floor hold on the strata (h, ρ_G) with $\rho_G = 1$ and $h \geq 3$, and on the other open strata of Table 8 (§2.4.3, Conjecture 87)? The standard ancilla-based CCZ circuit lies in the first case, and Corollary 86 settles it together with every AND-compiled circuit that meets the block hypotheses of that corollary (Remark 90). What remains are nested chains of temporary ANDs, in which one AND value controls another, and odd phases on AND values.

Open Problem 7 (one Hadamard layer). For $U = D_2 H^{\otimes n} D_1$ the nullity appears to add, $\nu(U) = \nu(D_1) + \nu(D_2)$, and at the minimal Hadamard count the T -count is bounded below by the coset distance of the lifted phase polynomial, which equals $\delta(D_1 \otimes D_2)$. Both rest on sketches and numerics (Appendix K.4). Do they hold, and does $\text{CCZ } H^{\otimes 3} \text{CCZ}$ cost 14 at three Hadamards, one above that floor, as the realizable parity sets suggest? With L layers the block-sum estimate grows about as $L(2n + 1)$ while nullity stays at most $2n$, so any floor there must use the shape of the circuit (Remark 114).

Open Problem 8 (the adaptive constant). Adaptive feedforward keeps only $t \geq \nu$, and the Jones gadget [12] rules out $t \geq c\nu$ for every $c > 4/3$ (Proposition 101). Does the bound hold for some c with $1 < c \leq 4/3$?

Open Problem 9 (entropy versus isotropy). The amortized SRE floor is at most $\approx 2.4094(n - 1)$ and the isotropy floor at most $2n + 1$, so from $n = 9$ the entropy could exceed the floor. A hill-climb over the rank profiles of the slices has not found such a gate. It reaches an entropy per qubit of 0.758 at $n = 11$, against the 0.868 a crossing would need there (Appendix K). Does the isotropy floor stay above the amortized SRE floor for every n ?

Open Problem 10 (the next field multiplier). The T -count of the GF(8) oracle lies between 20, its slice-rank floor (Proposition 60), and 23, the count of the best published circuit [25]. It is the smallest field multiplier whose T -count is open, since the count 17 at $k = 2$ is known (Remark 57). Is the GF(8) oracle's T -count 20, 23, or strictly between?

4.3 How the appendices are organized

The sections above can be read without the appendices, which contain the proofs in dependency order. Appendix A fixes notation, and Appendix B builds the floor and its ceiling. Appendices C–E treat the Toffoli layers, the oracles and the reversible circuits, and Appendix F catalogues the

remaining families together with the benchmark certification. Appendix G extends the floor to other levels and, in §G.2, to other gate sets. Appendices H and I treat internal Hadamards and the approximation and measurement models that §2.4 previews. Appendix J turns the floors into resource estimates, Appendix K collects further open questions and the one-Hadamard-layer case, and Appendix L lists the computations behind the tables.

A Preliminaries

We work over the field $\mathbb{F}_2 = \{0, 1\}$ with addition equal to XOR ($1 + 1 = 0$) and multiplication equal to AND. Vectors are elements of $\mathbb{F}_2^t$ and are added coordinatewise. A subspace $W \subseteq \mathbb{F}_2^t$ has $|W| = 2^{\dim W}$.

A.1 Bilinear forms and isotropic subspaces

Write $\langle u, v \rangle = \sum_{i=1}^t u_i v_i \bmod 2$ for the dot product on $\mathbb{F}_2^t$. It is symmetric, bilinear and non-degenerate, so that $\dim W + \dim W^\perp = t$ for $W^\perp = \{v : \langle v, w \rangle = 0 \ \forall w \in W\}$. Call W *totally isotropic* if $\langle u, v \rangle = 0$ for all $u, v \in W$, equivalently $W \subseteq W^\perp$. The property of characteristic 2 used throughout is that a vector can be orthogonal to itself: $\langle u, u \rangle = \sum_i u_i^2 = \sum_i u_i = \text{wt}(u) \bmod 2$, the parity of the Hamming weight.

Lemma 11. *If $W \subseteq \mathbb{F}_2^t$ is totally isotropic then $\dim W \leq t/2$.*

Proof. $W \subseteq W^\perp$ gives $\dim W \leq \dim W^\perp = t - \dim W$. $\square$

We also use the standard fact that a form's rank on a subspace equals the rank of the Gram matrix of any generating set [38].

Lemma 12. *Let $\mathcal{C} = \text{span}\{c_1, \dots, c_n\} \subseteq \mathbb{F}_2^t$ and let G be the $n \times n$ Gram matrix $G_{ab} = \langle c_a, c_b \rangle$. Then the rank of the dot product restricted to $\mathcal{C}$ equals $\text{rank}_{\mathbb{F}_2} G$, and the radical $\mathcal{C} \cap \mathcal{C}^\perp$ of the restricted form has dimension $\dim \mathcal{C} - \text{rank } G$.*

Proof. Pick a basis of $\mathcal{C}$ from among the c_a , and let B be the Gram matrix of the form in that basis. Writing the generating set in terms of the basis gives $G = P^\top B P$ with P of full row rank, so $\text{rank } G = \text{rank } B$, and the radical of a symmetric form has dimension $\dim \mathcal{C} - \text{rank } B$. $\square$

A.2 Boolean functions, parities, and cubic forms

Definition 13 (parity). Identify $y \in \mathbb{F}_2^n$ with the set $\{i : y_i = 1\} \subseteq [n]$. The parity indexed by y is the function $x \mapsto y \cdot x = \bigoplus_{i \in y} x_i$.

Definition 14 (algebraic normal form, degree, cubic form). Every $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ is uniquely a XOR of monomials, $f(x) = \bigoplus_{A \subseteq [n]} \hat{f}_A \prod_{i \in A} x_i$ with $\hat{f}_A \in \mathbb{F}_2$. Its degree is $\max\{|A| : \hat{f}_A = 1\}$. A *cubic form* is a homogeneous degree-3 function $\bigoplus_{a < b < c} T_{abc} x_a x_b x_c$, encoded by the symmetric tensor $T_{abc} \in \mathbb{F}_2$ (with $T_{abc} = 0$ whenever two indices coincide).

A.3 Phase polynomials and the T -count

We use Clifford+ T circuits. The Clifford group is generated by H , $S = \text{diag}(1, i)$, and CNOT. The non-Clifford generator is $T = \text{diag}(1, \omega)$ with $\omega = e^{i\pi/4}$. A Hadamard-free $\{\text{CNOT}, T\}$ circuit is diagonal and applies a phase that is a sum of T -rotations on parities of the input:

$$U|x\rangle = \omega^{P(x)}|x\rangle, \quad P(x) = \sum_{y \in \mathbb{F}_2^n} c_y (y \cdot x) \bmod 8, \quad c_y \in \mathbb{Z}_8. \quad (8)$$

A *diagonal level-three gate* is a unitary of this form, equivalently, up to a global phase, a diagonal gate in the third level of the Clifford hierarchy [39]. A T adds 1 to one coefficient c_y , an S adds 2, a Z adds 4, and two gates on the same parity merge by adding their coefficients. The T -count of the representation is the number of odd c_y , because only odd coefficients require a T and merging cannot increase the count of odd coefficients. Write $\delta(U)$ for the minimum T -count over all representations of U , counting T and $T^\dagger$ alike, with diagonal Clifford gates free.

For example, over $\mathbb{Z}$ and hence modulo 8, $4x_i x_j x_k = \sum_{\emptyset \neq S \subseteq \{i,j,k\}} (-1)^{|S|-1} (S \cdot x)$, so CCZ uses the seven parities indexed by the nonempty subsets of $\{i, j, k\}$, each with an odd coefficient.

Lemma 15 (Clifford bi-invariance). *Let C, C' lie in the group generated by CNOT, X and the diagonal Cliffords, and let CUC' be diagonal. Then $\delta(CUC') = \delta(U)$.*

Proof. Every element of the group is DP with D a diagonal Clifford and P an affine permutation $x \mapsto Ax \oplus b$ built from CNOT and X . Writing $C = D_1 P_1$ and $C' = P_2 D_2$, the product $CUC' = D_1 (P_1 U P_1^{-1}) (P_1 P_2) D_2$ is diagonal only if $P_1 P_2 = I$, so $CUC' = D_1 (P_1 U P_1^{-1}) D_2$. Conjugation by P_1 replaces the parity $y \cdot x$ by $(A^\top y) \cdot x \oplus y \cdot b$, a bijection on nonzero parities that changes a coefficient c_y at most to $-c_y$ and adds a global phase, so odd coefficients stay odd. A diagonal Clifford adds only even coefficients (S adds 2, Z adds 4, and CZ adds 2, 2, 6 on $x_i, x_j, x_i \oplus x_j$). The number of odd coefficients is therefore unchanged in every representation, and the minimum is preserved. $\square$

Lemma 16 (restriction monotonicity). *Let U be a diagonal level-three gate on n qubits and $S \subseteq [n]$, and write $U|_S$ for the gate induced by fixing the inputs outside S to 0. Then $\delta(U) \geq \delta(U|_S)$.*

Proof. Take any $\{\text{CNOT}, T\}$ representation of U , a multiset of parities y with coefficients c_y . Setting $x_j = 0$ for $j \notin S$ sends the parity $y \cdot x$ to $(y \cap S) \cdot x$, so the restricted multiset represents $U|_S$. Two parities may become equal, in which case their coefficients add and an odd pair can become even, or a parity may vanish outright, contributing nothing. Neither can raise the odd count, so the restricted representation has at most as many odd parities. Minimizing over representations gives $\delta(U|_S) \leq \delta(U)$. $\square$

A.4 Linear codes and Reed–Muller codes

Definition 17 (linear code, weight, distance, dual). A binary linear code of length L is a subspace $C \subseteq \mathbb{F}_2^L$. The Hamming weight $\text{wt}(v)$ is the number of 1s. The minimum distance is $d_{\min}(C) = \min_{0 \neq c \in C} \text{wt}(c)$. The dual is $C^\perp = \{v : \langle v, c \rangle = 0 \ \forall c \in C\}$, of dimension $L - \dim C$. The syndrome of a vector is its inner products against a basis of $C^\perp$. Two vectors have the same syndrome iff they differ by a codeword, that is, lie in the same coset $v + C$. A coset leader is a minimum-weight vector in a coset, and *minimum-distance decoding* asks for the nearest codeword to a given vector.

Definition 18 (puncturing and shortening). Puncturing C at a coordinate deletes that coordinate from every codeword. Shortening keeps only the codewords that are 0 at the coordinate, then deletes it. The two are exchanged by duality: the dual of C punctured at i is $C^\perp$ shortened at i . We write $\text{RM}(r, m)^*$ for $\text{RM}(r, m)$ punctured at the all-zeros coordinate.

Definition 19 (Reed–Muller code). $\text{RM}(r, m)$ is the evaluation code of all Boolean polynomials of degree at most r in m variables: list the value of each such polynomial at all 2^m points. It has dimension $\sum_{i \leq r} \binom{m}{i}$ and minimum distance 2^{m-r} , with minimum-weight codewords the indicators of $(m-r)$ -flats (cosets of $(m-r)$ -dimensional subspaces), and its dual is $\text{RM}(m-r-1, m)$.

A.5 The Amy–Mosca correspondence and the moment reformulation

Reduce the coefficients in (8) modulo 2: the *residue* $\text{Res}_2(U) = (c_y \bmod 2)_y \in \mathbb{F}_2^{2^n-1}$ (indexed by nonzero parities) records which parities carry an odd phase. Amy and Mosca [3] proved that two representations implement the same gate up to a diagonal Clifford iff their residues differ by a codeword of $\text{RM}(n-4, n)^*$, so

$$\delta(U) = \text{dist}(\text{Res}_2(U), \text{RM}(n-4, n)^*) = \min_{c \in \text{RM}(n-4, n)^*} \text{wt}(\text{Res}_2(U) \oplus c). \quad (9)$$

The degree $n-4$ has a short explanation. Expanding each $y \cdot x$ over $\mathbb{Z}$ modulo 8 leaves three carry orders, so a parity set implements a diagonal Clifford exactly when its indicator is orthogonal to every monomial $\prod_{i \in A} y_i$ with $1 \leq |A| \leq 3$. These span $\text{RM}(3, n)$ shortened at $y = 0$, whose dual is $\text{RM}(n-4, n)^*$.

Definition 20 (moments). For a set $\mathcal{S} \subseteq \mathbb{F}_2^n \setminus \{0\}$ of parities and $A \subseteq [n]$, the moment is

$$M_A(\mathcal{S}) = |\{y \in \mathcal{S} : A \subseteq y\}| \bmod 2.$$

The moment M_A is the syndrome coordinate of $\mathcal{S}$ against the monomial $\prod_{i \in A} y_i$, so the derivation above gives the following operational form of (9).

Fact 21 (moment form of Amy–Mosca). *The coset of a residue, and hence the gate it represents, is determined by the moments M_A with $1 \leq |A| \leq 3$, and*

$$\delta(U) = \min\{|\mathcal{S}| : M_A(\mathcal{S}) = s_A(U) \text{ for all } 1 \leq |A| \leq 3\},$$

where the syndrome $s_A(U)$ is M_A of any residue of U , well defined by the first sentence. The order-0 moment and all moments with $|A| \geq 4$ are unconstrained.

In the language of Heyfron and Campbell [17] the moments with $|A| \leq 3$ are the entries of the signature tensor, and $\delta(U)$ is its symmetric rank.

By Fact 21 the order- ≤ 3 moments are a complete invariant of the gate modulo diagonal Cliffords. The moment map $\mathbb{F}_2^{2^n-1} \rightarrow \mathbb{F}_2^{n+\binom{n}{2}+\binom{n}{3}}$ has kernel $\text{RM}(n-4, n)^*$, and restricted to the parities of weight at most three it is triangular in the subset order, hence already surjective (Proposition 30 solves it explicitly). In particular some residue, the canonical one, uses no parity of weight four or more. A residue of minimum weight may need such parities, as for two disjoint CCZ gates, whose canonical residue has weight 14 while $\delta = 13$.

Definition 22 (pure-cubic gate, cubic moment form, radical, quadratic matrix). A gate is *pure-cubic* if $s_A = 0$ for all $1 \leq |A| \leq 2$, and a *pure quadratic* if $s_A = 0$ for all $|A| \in \{1, 3\}$. A pure-cubic gate's *cubic moment form* is the tensor with $T_{abc} = s_{\{a,b,c\}}$ for distinct a, b, c and $T_{abc} = 0$ otherwise. For $w \in \mathbb{F}_2^n$ the matrix $T(w)_{qr} = \sum_a w_a T_{aqr}$ is the *slice* of T in direction w . The *radical* of T is the set of directions whose slice vanishes,

$$R = \left\{ w \in \mathbb{F}_2^n : \sum_a w_a T_{aqr} = 0 \ \forall q, r \right\}, \quad d = \dim R,$$

and T is *non-degenerate* when $d = 0$. For a general gate the *quadratic moment matrix* is $Q \in \mathbb{F}_2^{n \times n}$ with $Q_{ab} = s_{\{a,b\}}$ for $a \neq b$ and $Q_{aa} = s_{\{a\}}$, of rank $r = \text{rank}_{\mathbb{F}_2} Q$. Placing the moments of orders one and two at coincident indices gives the full moment tensor $s_{\{a,p,q\}}$, $a, p, q \in [n]$, whose radical is

$$R^* = \left\{ w \in \mathbb{F}_2^n : \sum_a w_a s_{\{a,p,q\}} = 0 \ \forall p, q \right\}, \quad d^* = \dim R^*.$$

Given the moments s_A , computing d , d^* and r is Gaussian elimination on at most n^2 rows of length n , so $O(n^3)$ operations on n -bit words ($O(n^4)$ bit operations).

The *unitary stabilizer nullity* of a diagonal level-three gate is $\nu = n - \log_2 |A|$, where $A \leq \mathbb{F}_2^n$ is the *shift group* of directions a for which $X^a U X^a$ equals U times a diagonal Pauli, up to phase [14, 28]. The circuit model is Hadamard-free Clifford+ T , built from CNOT and T with diagonal Clifford gates free. A lower bound on δ also bounds such circuits with arbitrarily many clean ancillas (Corollary 84).

Theorem 38 uses one further classical fact [38].

Lemma 23 (uniform minimum distance). *For every $n \geq 4$, $\text{RM}(n-4, n)^*$ has minimum distance 15.*

Proof. $\text{RM}(n-4, n)$ has minimum distance $2^{n-(n-4)} = 2^4 = 16$, attained by the indicators of 4-flats. Puncturing at the all-zeros coordinate deletes one symbol: a minimum-weight flat through the origin loses one 1, leaving 15. Puncturing removes at most one 1 from any codeword, so no nonzero codeword falls below 15. Hence the minimum distance is 15, independent of n . $\square$

B The cubic moment form and the isotropy floor

This appendix proves the bounds summarized in Figure 1, whose single-gate case is worked in §2.1.

Fix a representation of a gate, with odd parities $y_1, \dots, y_t$. Let $Y \in \mathbb{F}_2^{t \times n}$ be the matrix whose rows are the y_j , and let $c_a \in \mathbb{F}_2^t$ be its a -th column, the indicator of which parities contain qubit a . The key identities are

$$\langle c_a, c_b \rangle = \sum_j (y_j)_a (y_j)_b = M_{\{a,b\}}, \quad \langle c_a, c_a \rangle = \sum_j (y_j)_a = M_{\{a\}}, \quad (10)$$

and, for the triple products with a, b, q distinct,

$$\sum_j (c_a)_j (c_b)_j (c_q)_j = M_{\{a,b,q\}} = T_{abq}. \quad (11)$$

B.1 The pure-cubic case

Lemma 24 (isotropic columns). *For a pure-cubic gate, every representation has $t \geq 2 \text{rank } Y$.*

Proof. By (10), a pure-cubic gate has $\langle c_a, c_b \rangle = M_{\{a,b\}} = 0$ for $a \neq b$ and $\langle c_a, c_a \rangle = M_{\{a\}} = 0$. So $\mathcal{C} = \text{span}\{c_a\}$ is totally isotropic, and Lemma 11 gives $\dim \mathcal{C} \leq t/2$. Since column rank equals row rank, $\text{rank } Y = \dim \mathcal{C} \leq t/2$. $\square$

That a pure-cubic synthesis matrix has columns of even weight and even pairwise overlap is Campbell and Howard's Lemma 3 with vanishing quadratic matrix [13], and it is also implicit in the triorthogonal codes of Bravyi and Haah [34]. Step 1 of Theorem 26 is the flattening bound on the symmetric rank of the signature tensor of Heyfron and Campbell [17]. The new step is to combine it with isotropy, which doubles it for every pure-cubic gate. On controlled unitaries this recovers part of [13, Thm. 5]. The extra one comes from the all-ones vector (Lemma 25), in the spirit of the parity step of their converse [13, App. C].

The cubic part bounds the rank from below, and a parity refinement adds one in the non-degenerate case.

Lemma 25 (radical contraction of the all-ones vector). *Let $\mathcal{C} = \text{span}\{c_a\} \subseteq \mathbb{F}_2^t$ for a pure-cubic gate, and suppose the all-ones vector $\mathbf{1} \in \mathbb{F}_2^t$ lies in $\mathcal{C}$, say $\mathbf{1} = \sum_{a \in S} c_a$ with $S \subseteq [n]$. Then $\mathbf{1}_S \in R$. That is, $\sum_{a \in S} T_{pqa} = 0$ for all $p, q \in [n]$.*

Proof. Fix p, q . Evaluate the dot product of columns c_p, c_q against $\mathbf{1}$ coordinatewise. Because $\mathbf{1}$ is all ones, for each coordinate j we have $(c_p)_j(c_q)_j \cdot \mathbf{1}_j = (c_p)_j(c_q)_j$, so

$$\sum_j (c_p)_j(c_q)_j = \sum_j (c_p)_j(c_q)_j \left(\sum_{a \in S} c_a \right)_j = \sum_{a \in S} \sum_j (c_p)_j(c_q)_j(c_a)_j = \sum_{a \in S} T_{pqa},$$

using $\mathbf{1} = \sum_{a \in S} c_a$ in the first step and the triple-product identity $\sum_j (c_p)_j(c_q)_j(c_a)_j = T_{pqa}$ of (11) in the last. When a coincides with p or q the triple product degenerates to an order- ≤ 2 moment, which vanishes for a pure-cubic gate, matching $T_{pqa} = 0$ at coincident indices. The left-hand side is $\langle c_p, c_q \rangle = M_{\{p,q\}} = 0$ for $p \neq q$ (pure-cubic), and for $p = q$ it is $\sum_j (c_p)_j \mathbf{1}_j = \langle c_p, \mathbf{1} \rangle = \sum_{a \in S} \langle c_p, c_a \rangle = 0$ as well, since every $\langle c_p, c_a \rangle = M_{\{p,a\}}$ vanishes. Hence $\sum_{a \in S} T_{pqa} = 0$ for all p, q , i.e. $\mathbf{1}_S$ contracts T to zero, which is exactly $\mathbf{1}_S \in R$. $\square$

Theorem 26 (cubic lower bound). *Let U be a pure-cubic gate on n qubits whose cubic tensor has radical dimension d . Then $\delta(U) \geq 2(n - d)$, and $\delta(U) \geq 2n + 1$ if $d = 0$.*

Proof. Let $V = \text{span}\{y_j\}$ be the row space and $\rho = \dim V = \text{rank } Y$.

Step 1 (rank versus radical). If $w \in V^\perp$ then w is orthogonal to every row, so $\sum_a w_a c_a = 0$ in $\mathbb{F}_2^t$. Hence for all p, q ,

$$\sum_a w_a T_{pqa} = \sum_j (c_p)_j(c_q)_j \left(\sum_a w_a c_a \right)_j = 0,$$

so $w \in R$. Thus $V^\perp \subseteq R$, giving $\dim V^\perp \leq d$ and $\rho \geq n - d$. With Lemma 24, $t \geq 2\rho \geq 2(n - d)$.

Step 2 (the extra $+1$ when $d = 0$). Suppose $d = 0$ and, for contradiction, that some representation has $t = 2n$. Then $\dim \mathcal{C} = \rho = n = t/2$, so by Lemma 11 the inclusion $\mathcal{C} \subseteq \mathcal{C}^\perp$ is an equality: $\mathcal{C} = \mathcal{C}^\perp$ is maximal isotropic. The all-ones vector satisfies $\langle \mathbf{1}, c_a \rangle = M_{\{a\}} = 0$ for every a , so $\mathbf{1} \in \mathcal{C}^\perp = \mathcal{C}$. Write $\mathbf{1} = \sum_{a \in S} c_a$ with S nonempty (nonempty because $\mathbf{1} \neq 0$ for $t \geq 1$). By Lemma 25, $\mathbf{1}_S \in R = \{0\}$, forcing $S = \emptyset$, a contradiction. Hence no representation has $t = 2n$, and since $t \geq 2n$ already, every representation has $t \geq 2n + 1$. Taking the minimum, $\delta(U) \geq 2n + 1$. $\square$

B.2 The general case

For an arbitrary gate the column Gram matrix is the quadratic moment matrix Q , of rank r . Restricting to the radical of Q would lose $2r$. The proof below loses only r , because on the even-weight part of $\mathcal{C}$ the form is alternating.

Theorem 27 (general lower bound). *For any diagonal level-three gate on n qubits, with quadratic moment matrix of rank r and $d^* = \dim R^*$, where $R^* = \{w : \sum_a w_a s_{\{a,p,q\}} = 0 \ \forall p, q\}$ is the radical of the full moment tensor (coincident indices give the order- ≤ 2 moments),*

$$\delta(U) \geq 2(n - d^*) - r.$$

Proof. By (10) the Gram matrix of the columns is Q . Put $\mathcal{C} = \text{span}\{c_a\}$ and $\rho = \dim \mathcal{C} = \text{rank } Y$. Over $\mathbb{F}_2$ one has $\langle v, v \rangle = \sum_j v_j$, which is additive, so $\lambda := \langle \cdot, \cdot \rangle|_{\text{diag}}$ is linear on $\mathcal{C}$ and $\lambda(c_a) = M_{\{a\}}$. Put $\eta = 0$ if $\lambda \equiv 0$ on $\mathcal{C}$ (equivalently Q has zero diagonal) and $\eta = 1$ otherwise, and set $\mathcal{C}_0 = \mathcal{C} \cap \ker \lambda$, of dimension $\rho - \eta$. On $\mathcal{C}_0$ the form is alternating, of some rank r_0 . Every symmetric bilinear form over $\mathbb{F}_2$ is either alternating or congruent to $I_r \oplus 0$ [38]. In the first case $\eta = 0$, r is even and $r_0 = r$, and in the second, restricting I_r to the even-weight hyperplane gives $r_0 = r - 2$ for r even and $r_0 = r - 1$ for r odd. An alternating form of rank r_0 on $\mathcal{C}_0$ has a totally isotropic subspace S of dimension $\dim \mathcal{C}_0 - r_0/2$ (its radical together with a Lagrangian of the nondegenerate quotient).

Since $S \subseteq \ker \lambda$, every vector of S has even weight, so S is a self-orthogonal binary code and Lemma 11 gives $\dim S \leq t/2$. In the first two cases $\dim S = \rho - r/2$, so $t \geq 2\rho - r$. In the third case, r odd, $\dim S = \rho - (r+1)/2$ and the same count gives only $t \geq 2\rho - r - 1$. We show that this value is not attained, so that $t \geq 2\rho - r$ in every case. Suppose $t = 2 \dim S$. Then $\dim S^\perp = t - \dim S = \dim S$, and since $S \subseteq S^\perp$ this forces $S = S^\perp$. As $S \subseteq \mathcal{C}_0 \subseteq \mathcal{C}$, taking orthogonal complements gives $\mathcal{C}^\perp \subseteq S^\perp = S \subseteq \mathcal{C}$, so $\mathcal{C}^\perp = \mathcal{C} \cap \mathcal{C}^\perp$ is the radical of the dot product restricted to $\mathcal{C}$. The ambient form on $\mathbb{F}_2^t$ is non-degenerate, so $\dim \mathcal{C}^\perp = t - \rho$, while the radical of the form on $\mathcal{C}$ has dimension $\rho - r$ by Lemma 12, because the Gram matrix of the spanning set $\{c_a\}$ is Q . Hence $t = 2\rho - r$, which contradicts $t = 2 \dim S = 2\rho - r - 1$. Therefore $t \geq 2 \dim S + 1 = 2\rho - r$ when r is odd. Since an alternating form has even rank, odd r forces $\eta = 1$ in every representation. For the rank step, if $w \in V^\perp$ then $\sum_a w_a c_a = 0$, so $\sum_a w_a s_{\{a,p,q\}} = \sum_j c_p c_q (\sum_a w_a c_a) = 0$ for all p, q . The sum runs over all a , so coincident indices contribute order- ≤ 2 moments via $c_p^2 = c_p$, and thus $w \in R^*$, giving $\rho \geq n - d^*$ and hence $t \geq 2\rho - r \geq 2(n - d^*) - r$.¹ $\square$

For a pure-cubic gate the order- ≤ 2 moments vanish, so $r = 0$ and $d^* = d$, and Theorem 27 reduces to the first part of Theorem 26. In general the isotropy floor is never below ν (Corollary 35).

Remark 28 (2-adic grading). The cubic and quadratic parts carry the coefficients 4 and 2 in $\phi = 4f_3 + 2f_2 + f_1$. A CNOT relabelling replaces each variable by an $\mathbb{F}_2$ sum, which differs from the integer sum by a carry, $u + v = (u \oplus v) + 2uv$. Since $4z \bmod 8$ depends only on $z \bmod 2$, carries do not affect the cubic part, so the pure-cubic locus is closed under $GL(n, \mathbb{F}_2)$ and δ is constant on each orbit. Since $2u + 2v = 2(u \oplus v) + 4uv$, a relabelling converts quadratic terms into cubic ones, so the pure-quadratic locus is not GL -closed, and we do not expect a bound depending on r alone to be tight on the quadratic side.

Remark 29 (comparison with stabilizer nullity). Stabilizer nullity ν is the standard efficiently computable lower bound on T -count. For the non-degenerate pure-cubic gates it gives $\nu = n$ (for example $\nu(\text{CCZ}) = 3$), so Theorem 26 is a factor-two improvement at the same $O(n^3)$ cost. No magic monotone under stabilizer operations with feedforward can match it, since such a monotone also bounds adaptive circuits, in which CCZ costs four T gates [12] (Proposition 101). The bounds are exact on the named families of Appendices C–D for which a construction matches them.

B.3 An upper bound, also in $O(n^3)$

The moment conditions can be solved from the top down, which gives an upper bound that is also computable in $O(n^3)$. It uses no parity of weight four or more (§A.5).

Proposition 30 (canonical residue ceiling). *Let U be any diagonal level-three gate with moments s_A , $1 \leq |A| \leq 3$. Define z on the parities of weight at most three by solving downwards,*

$$z_A = s_A \quad (|A| = 3), \quad z_A = s_A \oplus \bigoplus_{y \supsetneq A, |y|=3} z_y \quad (|A| = 2), \quad z_A = s_A \oplus \bigoplus_{y \supsetneq A, |y| \leq 3} z_y \quad (|A| = 1),$$

and $z_y = 0$ for $|y| \geq 4$. Then z is a residue for U , so

$$\delta(U) \leq W(s) := \text{wt}(z),$$

¹Off the pure-cubic locus the full-tensor radical R^* is needed. Substituting the cubic radical d for d^* gives a quantity that is not in general a valid lower bound once the quadratic moments are nonzero. For example, the three-qubit gate with $s_{\{0,2\}} = s_{\{1,2\}} = s_{\{0,1,2\}} = 1$ and all other moments 0 has $d = 0$, $d^* = 1$ and $r = 2$, so $2(n - d) - r = 4$ while $\delta = 3$.

and $\delta(U) = W(s)$ whenever $W(s) \leq 7$. For a pure-cubic gate with monomial set F this is $W(s) = w(F) := \#\{A : 1 \leq |A| \leq 3, \#\{S \in F : A \subseteq S\} \text{ odd}\}$, and for a pure quadratic with moment graph E it is $e + \text{odd}$, the edge count plus the number of odd-degree vertices.

Proof. By §A.5 the moments of order ≤ 3 determine the coset, and parities of weight at most three suffice. Solving downwards, at each $|A|$ the supersets of A have already been fixed, so z_A is determined and by construction $M(A) = \bigoplus_{y \supseteq A} z_y = s_A$ for every A with $1 \leq |A| \leq 3$. Hence z lies in the coset of U and $\delta \leq \text{wt}(z)$, δ being the minimum weight over that coset. Equality below 8 is Theorem 38. The construction takes $O(n^3)$ time and involves no decoding.

For the pure-cubic specialization, $z_A = [A \in F]$ at $|A| = 3$. At $|A| = 2$, $s_A = 0$ gives $z_A = \#\{S \in F : A \subseteq S\}$, and at $A = \{a\}$ the weight-three parities through a contribute $\deg_F(a)$ while the weight-two parities contribute $\sum_{b \neq a} \#\{S \in F : \{a, b\} \subseteq S\} = 2 \deg_F(a) \equiv 0$, so $z_{\{a\}} = \deg_F(a)$ and the total weight is $w(F)$. For a pure quadratic the weight-three level is empty, $z_{\{a, b\}} = E_{ab}$ and $z_{\{a\}} = \deg_E(a)$, giving $e + \text{odd}$. $\square$

The cubic and quadratic formulas are two cases of one construction, and unlike $w(F)$ the ceiling $W(s)$ applies to mixed-degree phase polynomials. Together with Theorem 27, every diagonal level-three gate satisfies

$$2(n - d^*) - r + [\text{pure-cubic, } d < n] \leq \delta(U) \leq W(s).$$

Both ends are computable in $O(n^3)$ from the moments s_A , which a phase polynomial with t terms yields in $O(tn^3)$ time, and the $+1$ on the pure-cubic locus is Proposition 32. When the two ends coincide they determine δ .

Example 31 (the ceiling on two overlapping blocks). Take the overlapping layer $f = x_1x_2x_3 \oplus x_3x_4x_5$, two CCZ blocks sharing qubit 3.

The phase is $\phi = 4f$, so $s_{123} = s_{345} = 1$ and every other moment is 0 (the order-one and order-two moments of any $(-1)^f$ vanish, since a cubic monomial's seven parities meet each singleton four times and each pair twice, and its linear and quadratic terms are Clifford). Proposition 30 takes the parity A exactly when $z_A = 1$, working down the levels:

A	s_A	taken above A	z_A	gate
123, 345	1	0	1	T
the other eight triples	0	0	0	—
12, 13, 23, 34, 35, 45	0	1	1	$T^\dagger$
the other four pairs	0	0	0	—
1, 2, 4, 5	0	3	1	T
3	0	6	0	S

Twelve parities are taken, against $7 + 7 = 14$ for the two blocks built separately. The saving is at qubit 3, which receives a T from each block and so an S . The ceiling is not optimal here, since $\delta = 11$, the fan-out of Theorem 43 at $m = 2$, so the bracket is $[11, 12]$.

B.4 Exactness of the floor for $n \leq 6$

The exactness claims of Appendices C–D are stated on named families. For small n one can instead check every cubic gate, because the classes are finite. By Remark 28 the pure-cubic locus is closed under $GL(n, \mathbb{F}_2)$, and δ is constant on orbits. It therefore suffices to consider $GL(n, \mathbb{F}_2)$ -orbits of cubic forms modulo quadratics, the classical classification of $\text{RM}(3, n)/\text{RM}(2, n)$, known completely

n	orbit representative	orbit size	d	$2\nu + 1$	δ	$w(F)$	$\text{low}(f)$	
4	$x_0x_1x_2$	15	1	7	7	7	15	floor
5	$x_0x_1x_4 \oplus x_0x_2x_3$	868	0	11	11	12	15	floor
5	$x_0x_1x_2$	155	2	7	7	7	31	floor
6	$x_0x_1x_5 \oplus x_0x_2x_4 \oplus x_1x_2x_3$	468 720	0	13	15	15	7	ceiling
6	$x_0x_4x_5 \oplus x_1x_2x_3$	357 120	0	13	13	14	14	floor
6	$x_0x_2x_4 \oplus x_0x_2x_5 \oplus x_0x_3x_5 \oplus x_1x_2x_5 \oplus x_1x_3x_4$	166 656	0	13	17	17	0	ceiling
6	$x_0x_1x_4 \oplus x_0x_2x_3$	54 684	1	11	11	12	31	floor
6	$x_0x_1x_2$	1 395	3	7	7	7	63	floor

Table 3: Every $GL(n, \mathbb{F}_2)$ -orbit of nonzero cubic forms for $4 \leq n \leq 6$ (at $n = 3$ the only orbit is CCZ, with $\delta = 7$): the floor $2\nu + 1$ of Proposition 32, the exact T -count, and the ceiling $w(F)$ of Proposition 30. Orbit size, d and δ are invariants, while $w(F)$ is not and is evaluated on the sparse representative shown. The last column records which bound δ attains for that presentation. The column $\text{low}(f)$ counts the nonzero directions a with $\text{rank } T(a) \leq 2$ (§2.4.1). On the $n = 6$ rows it decreases as δ increases.

to $n = 9$ [40] and to $n = 10$ [37]. Table 3 is the resulting census for $n \leq 6$, computed by orbit enumeration under transvections together with exact decoding.

The floor is exact on every orbit for $n \leq 5$, hence on every cubic form. At $n = 6$ it is exact on 3 of the 5 orbits, but the two it misses are the largest and the third largest. Weighted by orbit size, the floor is exact on 413,199 of the 1,048,575 nonzero cubic forms at $n = 6$, or 39.4%.

The two-disjoint-CCZ orbit has $\delta = 13 < 14$, the non-additivity of Theorem 39. The three-term orbit has $\delta = 15 = 2n + 3$, the first failure of the non-degenerate $2n + 1$. The five-term orbit at $n = 6$ is non-degenerate, so its floor is 13, but $\delta = 17$. Since the table lists every orbit, a gap of 4 is the largest among pure-cubic gates at $n = 6$. No invariant of the radical alone can account for it, since $d = 0$ on all three non-degenerate $n = 6$ orbits while δ takes the three values 13, 15, 17.

On the sparse representatives shown, δ equals the floor or the ceiling in every row. Appendix D.4 gives the brackets on the application families.

The formula $\nu = n - \log_2 |A|$ for a phase-polynomial gate is the diagonal specialization of a result of Jiang and Wang [14]. The radical of a cubic form is a standard invariant in the classification of cubic forms [40]. The next proposition identifies A with it, which gives $\nu = n - d$ and extends the +1 of Theorem 26 to every d .

Proposition 32 (stabilizer nullity equals $n - d$, and a uniform +1). *For a pure-cubic gate the unitary stabilizer nullity is $\nu = n - d$, and if its cubic tensor is nonzero ($d < n$) then $\delta \geq 2\nu + 1 = 2(n - d) + 1$.*

Proof. The unitary stabilizer nullity of a diagonal gate $U = \text{diag}(\omega^\phi)$ is $\nu = n - \log_2 |A|$, where $A \leq \mathbb{F}_2^n$ is the group of shift directions a for which conjugation $X^a U X^{-a}$ equals U times a diagonal Pauli, up to phase. Concretely, $a \in A$ iff the discrete derivative $\Delta_a \phi(y) := \phi(y \oplus a) - \phi(y)$ is of the form $c + 4\ell(y)$ with $c \in \mathbb{Z}_8$ constant and ℓ affine over $\mathbb{F}_2$ (so that $X^a U X^{-a}$ differs from U only by a Pauli). That A is a group and $\nu = n - \log_2 |A|$ is the diagonal specialization of [14], whose $\mathcal{P}_U = U \mathcal{P}_n U^\dagger \cap \mathcal{P}_n$ restricts on a diagonal gate to exactly the X -shifts recorded by A . We compute A .

Step 1 ($A = R$). Take the homogeneous representative $\phi = 4f$ with f the cubic form of tensor T (the linear and quadratic parts are 0 for a pure-cubic gate, up to a diagonal Clifford factor D , and $\nu(UD) = \nu(U)$ because conjugation by D maps Paulis to Paulis). Then $\Delta_a \phi = 4 \Delta_a f$, and

since $4z \bmod 8$ depends only on $z \bmod 2$ the derivative can be taken over $\mathbb{F}_2$, where

$$\Delta_a f(y) = f(y \oplus a) - f(y) = \underbrace{\sum_{p < q} T(a)_{pq} y_p y_q}_{\text{quadratic in } y} + (\text{affine in } y),$$

because differencing a degree-3 form in direction a lowers the degree to 2, and its quadratic part is the bilinear form $(p, q) \mapsto T(a)_{pq} = \sum_b a_b T_{bpq}$, the slice of T in direction a . Thus $\Delta_a \phi = 4\Delta_a f$ has the required form $c + 4\ell(y)$ iff that quadratic part vanishes identically, i.e. iff $T(a)_{pq} = 0$ for all p, q , which is the condition $a \in R$. Both inclusions follow: $a \in A \Rightarrow \text{quadratic part} = 0 \Rightarrow a \in R$, and $a \in R \Rightarrow \Delta_a f \text{ affine} \Rightarrow a \in A$. Hence $A = R$, so $|A| = 2^d$ and $\nu = n - \log_2 |A| = n - d$.

Step 2 (the $+1$). Align R to the last d coordinates by a CNOT relabelling (an invertible $\mathbb{F}_2$ -linear change of the x -variables, which preserves δ and d and transforms the moments covariantly). Since $T_{apq} = 0$ whenever any index lies in R , the cubic f has no monomial meeting a radical coordinate, so up to a diagonal Clifford factor, which changes neither δ nor d , $U = U' \otimes I_d$ with U' a non-degenerate pure-cubic gate on the first $n - d$ qubits. Then $\delta(U) = \delta(U')$: a representation of U' is one of U (giving $\leq$), and restricting any representation of U by setting the d radical inputs to 0 (Lemma 16) turns each parity into a parity of the first $n - d$ variables or a constant, without increasing the odd count (giving $\geq$). Hence $\delta(U) = \delta(U') \geq 2(n - d) + 1$ by the non-degenerate case of Theorem 26, i.e. $\delta(U) \geq 2\nu + 1$. $\square$

The purity hypothesis in the first claim of Proposition 32 can be removed, with d replaced by d^* . The $+1$ cannot, since CS has $\nu = 2$ and $\delta = 3$. Via the carry identity, membership in A reduces to the moment contraction that defines the full-tensor radical R^* .

Lemma 33 (the quadratic rank is bounded by the tensor corank). $R^* \subseteq \ker Q$, and hence $r \leq n - d^*$.

Proof. Taking $p = q = a$ in the defining condition of R^* collapses the coincident indices to order- ≤ 2 moments and gives $\sum_b w_b s_{\{a,b\}} = 0$ for every a , with $s_{\{a,a\}} = s_{\{a\}}$. That is $(Qw)_a = 0$. Hence $R^* \subseteq \ker Q$ and $d^* \leq n - r$. $\square$

Proposition 34 (the shift group is the full-tensor radical). *For every diagonal level-three gate, $A = R^*$. Hence $\nu = n - d^*$.*

Proof. Write $\phi = \sum_y \kappa_y \chi_y$ with $\chi_y(x) = \langle y, x \rangle \in \{0, 1\} \subset \mathbb{Z}$ and $\kappa_y \in \mathbb{Z}_8$. Since $\chi_y(x \oplus w) = \chi_y(x)$ when $\langle y, w \rangle = 0$ and $\chi_y(x \oplus w) = 1 - \chi_y(x)$ otherwise, we have the exact identity over $\mathbb{Z}$

$$\Delta_w \phi(x) = C_w - 2L_w(x), \quad C_w := \sum_{\langle y, w \rangle=1} \kappa_y, \quad L_w := \sum_{\langle y, w \rangle=1} \kappa_y \chi_y.$$

So $w \in A$, i.e. $\Delta_w \phi \equiv \text{const} + 4(\text{affine}) \pmod{8}$, holds iff

$$L_w \equiv \text{const} + 2(\text{affine}) \pmod{4}. \tag{12}$$

Split $\kappa_y = \varepsilon_y + 2h_y$ with $\varepsilon_y = \kappa_y \bmod 2$ the residue bit, and put $\mathcal{T}_w := \{y : \langle y, w \rangle = 1, \varepsilon_y = 1\}$. Then $L_w = S_w + 2D_w$ with $S_w = \sum_{y \in \mathcal{T}_w} \chi_y$ and $D_w = \sum_{\langle y, w \rangle=1} h_y \chi_y$. Modulo 4 the second term is $2(D_w \bmod 2)$, and $D_w \bmod 2$ is an $\mathbb{F}_2$ -linear function of x because each χ_y is, so the even part of κ contributes only an allowed $2(\text{affine})$ and never obstructs (12). For the odd part use the carry identity for $\{0, 1\}$ -valued u_i ,

$$\sum_i u_i \equiv \left(\bigoplus_i u_i \right) + 2 \sum_{i < j} u_i u_j \pmod{4},$$

which gives $S_w \equiv \chi_{\sigma_w} + 2q_w \pmod{4}$ with $\sigma_w := \bigoplus_{y \in \mathcal{T}_w} y$ and $q_w := \sum_{y < y' \in \mathcal{T}_w} \chi_y \chi_{y'} \pmod{2}$.

The mod-2 level is Q . The a -th bit of σ_w is $\sum_{y \in \mathcal{T}_w} y_a = \sum_{y: y_a=1} \varepsilon_y \langle y, w \rangle = \sum_b w_b \sum_{y \supseteq \{a,b\}} \varepsilon_y = \sum_b w_b s_{\{a,b\}} = (Qw)_a$. If $\sigma_w \neq 0$ then $L_w \pmod{2} = \chi_{\sigma_w}$ is a nonconstant function of x , so (12) fails. Hence $Qw = 0$ is necessary.

The mod-4 level is the tensor. Assume $\sigma_w = 0$, so $L_w \equiv 2(q_w + \text{linear})$ and (12) holds iff q_w is affine. Expanding $\chi_y \chi_{y'} = \sum_{a,b} y_a y'_b x_a x_b$ over $\mathbb{F}_2$ and using $x_a^2 = x_a$, the coefficient of $x_a x_b$ with $a \neq b$ in q_w is

$$\sum_{\{y,y'\} \subseteq \mathcal{T}_w} (y_a y'_b + y_b y'_a) = \sum_{y,y' \in \mathcal{T}_w, y \neq y'} y_a y'_b = \sigma_a \sigma_b - \sum_{y \in \mathcal{T}_w} y_a y_b = \sum_{y \in \mathcal{T}_w} y_a y_b,$$

using $\sigma_w = 0$ in the last step. As before $\sum_{y \in \mathcal{T}_w} y_a y_b = \sum_e w_e \sum_{y \supseteq \{a,b,e\}} \varepsilon_y = \sum_e w_e s_{\{a,b,e\}}$. Thus q_w is affine iff $\sum_e w_e s_{\{a,b,e\}} = 0$ for all $a \neq b$.

Together with the $a = b$ conditions from the mod-2 level, $w \in A$ iff $\sum_e w_e s_{\{a,b,e\}} = 0$ for all a, b , which is exactly $w \in R^*$. Hence $A = R^*$ and $\nu = n - \log_2 |A| = n - d^*$. $\square$

In particular ν is computable from the moments s_A in $O(n^3)$ time by Gaussian elimination. On the pure-cubic locus the order- ≤ 2 moments vanish, $R^* = R$, and Proposition 34 recovers Proposition 32. Off it, d^* cannot be replaced by the cubic radical dimension d (see the footnote to Theorem 27).

Corollary 35 (the floor dominates stabilizer nullity). *For every diagonal level-three gate, $2(n - d^*) - r \geq \nu$, with equality iff $r = n - d^*$.*

Proof. By Proposition 34, $\nu = n - d^*$, so $2(n - d^*) - r - \nu = (n - d^*) - r$, which is ≥ 0 by Lemma 33. $\square$

Remark 36 (the worst-case ratio is $\Omega(n)$). By Proposition 34, $\nu = n - d^* \leq n$, so for every diagonal level-three gate

$$2(n - d^*) - r \leq 2n, \quad 2\nu + 1 \leq 2n + 1.$$

Every such floor is therefore $O(n)$. The worst-case ancilla-free $\{\text{CNOT}, T\}$ T -count is by contrast quadratic, $\Theta(n^2)$. For the upper bound, write the cubic part as $\sum_a 4x_a q_a$ with q_a a quadratic form in the variables after a . A CNOT relabelling of those variables puts q_a in the normal form $\sum_i z_{2i-1} z_{2i}$ plus a linear part, and a coefficient-4 phase ignores carries, so $4x_a q_a$ costs at most $7\lfloor (n-1)/2 \rfloor$ (one CCZ per pair, the rest Clifford). The quadratic part costs at most $e + \text{odd} \leq \binom{n}{2} + n$ by the pure-quadratic case of Proposition 30, the linear part at most n , and Lemma 81 adds these. The lower bound is a count: the $2^{\binom{n}{3}}$ cubic forms give distinct gates, while fewer than 2^{nt+1} residues have weight at most t , so some gate needs $t \geq (\binom{n}{3} - 1)/n$. Hence

$$\max_{U \text{ on } n \text{ qubits}} \frac{\delta(U)}{\text{isotropy floor}(U)} = \Omega(n),$$

and the argument uses nothing about the floor except that its value is $O(n)$. Any such quantity (stabilizer nullity, the isotropy floor, or any constant multiple of either) is off by a factor $\Omega(n)$ on worst-case cubic gates in the ancilla-free model. Exactness results of the kind in Appendices C–D can therefore hold only where $\delta = \Theta(n)$. The counting argument guarantees a ratio above 2 only from $n = 28$.

The worst case of Remark 36 requires gates with $\Omega(n^2)$ monomials, by Corollary 37 below. Bounding the density gives an approximation guarantee, certified for each instance because both ends are computed.

Corollary 37 (sparse gates are approximated within a constant factor). *Let U be a non-Clifford diagonal level-three gate. Write $F = \{A : |A| = 3, s_A = 1\}$ for its cubic support and $E = \{A : |A| = 2, s_A = 1\}$ for its quadratic support. Then the ceiling of Proposition 30 obeys*

$$W(s) \leq 4|F| + |E| + n,$$

and in particular $w(F) \leq 4|F| + n$ on the pure-cubic locus. Consequently

$$1 \leq \frac{\delta(U)}{2(n - d^*) - r} \leq \frac{4|F| + |E| + n}{2(n - d^*) - r}.$$

Proof. The canonical residue z is supported on parities of weight at most three. At weight three $z_A = s_A$, contributing $|F|$. At weight two $z_A = s_A \oplus \bigoplus_{y \supseteq A, |y|=3} z_y$, which can be nonzero only if $A \in E$ or A lies inside some element of F , and each element of F contains three pairs, contributing at most $|E| + 3|F|$. At weight one there are n parities. Summing gives the bound, and the pure-cubic case is $E = \emptyset$. The lower end is Theorem 27 and the upper is Proposition 30. The denominator is positive. If $R^* = \mathbb{F}_2^n$ then every moment of order at most three vanishes and U is Clifford by Fact 21, so $d^* < n$ and Corollary 35 gives $2(n - d^*) - r \geq \nu = n - d^* \geq 1$. $\square$

Sparsity typically comes from bounded locality. Say the gate is D -local if every qubit lies in at most D elements of $F \cup E$. This is a condition on the moment supports, which is what the count above uses, and it can be checked in $O(n^3)$. Counting incidences gives $3|F| + 2|E| \leq Dn$, hence $4|F| + |E| \leq \frac{4}{3}Dn$, so a gate with $d^* = 0$, whose floor is at least $\nu = n$, has

$$\frac{\delta(U)}{\text{floor}} \leq \frac{(\frac{4}{3}D + 1)n}{n} = \frac{4}{3}D + 1,$$

a constant independent of n . For a local Hamiltonian with cubic and quadratic term sets H and J (Appendix D), $|E| \leq |J| + 3|H|$ gives the same conclusion with a worse constant. On the pure-cubic locus the ratio is at most $(\frac{4}{3}D + 1)/2$, and a layer of disjoint CCZs has certified ratio tending to $7/6$. For a cost layer in the Ising form the ceiling is the term count (Proposition 52), and the ratio is at most $D/2 + 1$ when D bounds the number of terms of $H \cup J$ containing a qubit, the $+1$ accounting for the linear terms.

B.5 Rigidity and non-additivity

The next statement is immediate from the Amy–Mosca correspondence [3] and the minimum distance 15 of Lemma 23. Several exact values rest on it.

Theorem 38 (rigidity radius). *If a diagonal level-three gate on $n \geq 4$ qubits has a residue of Hamming weight $w \leq 7$, then $\delta = w$.*

Proof. For any nonzero codeword $c \in \text{RM}(n - 4, n)^*$, the triangle inequality and Lemma 23 give

$$\text{wt}(\text{Res}_2 \oplus c) \geq \text{wt}(c) - w \geq 15 - w \geq 8 > w.$$

So the residue itself is the strict minimum-weight coset leader, and by (9), $\delta = w$. The threshold $w \leq 7 = \lfloor (15 - 1)/2 \rfloor$ is the unique-decoding radius. $\square$

For $n \leq 3$ the code $\text{RM}(n - 4, n)^*$ is zero, every gate has a single residue, and $\delta = w$ for every w . The statement therefore holds for all n , which is how it is used at $n = 3$.

For example $\delta(T) = 1$, $\delta(\text{CS}) = 3$ and $\delta(\text{CCZ}) = 7$ for all n . Above weight seven a weight-15 codeword can lower the residue weight. Campbell and Howard observed that T -count is often strictly subadditive and gave both upper bounds below [13, Thm. 6, Exs. III.3–III.4]. The lower bounds follow from restriction and from Theorem 26.

Theorem 39 (non-additivity). *T-count is not additive over variable-disjoint diagonal level-three gates:*

$$\delta(\text{CCZ} \oplus T) = 7 = \delta(\text{CCZ}) \quad (n \geq 4), \quad \delta(\text{CCZ} \oplus \text{CCZ}) = 13 < 14 = 2\delta(\text{CCZ}) \quad (n = 6).$$

Proof. $\text{CCZ} \oplus T$. Restricting any representation to the three CCZ qubits (setting the other inputs to 0) gives a representation of CCZ with no more odd parities, so $\delta \geq \delta(\text{CCZ}) = 7$. For the matching upper bound, the residue has weight 8: the seven CCZ parities (nonempty subsets of $\{1, 2, 3\}$) and the singleton $\{4\}$. Take the weight-15 codeword $c_y = \prod_{i \geq 5} (1 \oplus y_i)$, the indicator of the 4-flat $\text{span}\{e_1, e_2, e_3, e_4\}$ punctured at the origin. It is 1 at all 15 nonzero parities supported on $\{1, 2, 3, 4\}$, hence at all eight support points, leaving $\text{wt}(\text{Res}_2 \oplus c) = 15 - 8 = 7$.

$\text{CCZ} \oplus \text{CCZ}$. On its six qubits this is pure-cubic with $d = 0$, so Theorem 26 gives the lower bound $2 \cdot 6 + 1 = 13$, and the $6m + 1$ construction of Theorem 40 at $m = 2$ gives a representation of weight 13. This is strictly below $2\delta(\text{CCZ}) = 14$. $\square$

C Toffoli layers: disjoint, overlapping, fan-out

This appendix determines the T -count of Toffoli layers whose blocks are disjoint or share qubits, and gives one splicing construction that attains the floor whenever the merges can be routed along a forest of blocks in which each block has at most three neighbors.

C.1 Optimality of the Toffoli-layer cost

A Toffoli gate is a CCZ conjugated by Hadamards on the target, so a parallel Toffoli layer of m gates on disjoint qubits is, up to Clifford gates, a layer of m disjoint CCZ gates on $n = 3m$ qubits. Write $\delta_m = \delta(\bigoplus_{i=1}^m \text{CCZ}_i)$. At $m = 1$ the seven- T circuit is standard [5], and Gosset, Kliuchnikov, Mosca and Russo [2] proved it optimal among ancilla-free Clifford+ T circuits by exact search. Campbell and Howard [41] state the count $6m + 1$ in the synthillation setting. Their argument proves the upper bound $\tau[U_{\text{CCZ}}^{\otimes N}] \leq 6N + 1$ [13, Cor. 1, Sec. IV.A] and reports the value 13 at $N = 2$ as known [13, Ex. IV.2] without a proof. Stabilizer nullity is additive under tensor product [14, 28], so it gives $3m$. The lower bound of Theorem 40 for $m \geq 2$ is new.

Theorem 40 (Toffoli-layer optimality). *For every $m \geq 1$, $\delta_m = 6m + 1$, and every $\{\text{CNOT}, T\}$ circuit for the CCZ layer with clean ancillas uses at least $6m + 1$ T gates (Corollary 84). Hence no Clifford+ T circuit for the parallel Toffoli layer whose Hadamards all sit in the two target layers uses fewer than $6m + 1$ T gates.*

Proof. Lower bound. The layer is pure-cubic, with $T_{pqr} = 1$ exactly when $\{p, q, r\}$ is a block. Its radical is trivial: if $w \in R$, taking q, r to be two qubits of a block forces w to vanish on the third, and ranging over the three pairs of each block gives $w = 0$. Hence $d = 0$, and Theorem 26 gives $\delta_m \geq 6m + 1$.

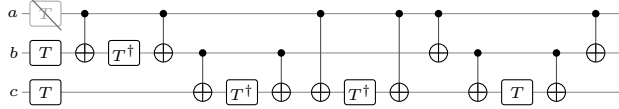
Upper bound. Campbell and Howard's construction gives $\delta_m \leq 6m + 1$ [13, Cor. 1]. The same weight also follows from Proposition 47 with $s = 0$ and F edgeless, which splices the m blocks by case (i) of Lemma 46 into a representation of weight $7 + 6(m - 1) = 6m + 1$. Since the lower and upper bounds agree, $\delta_m = 6m + 1$. $\square$

Implemented separately, the m gates cost $7m$. Figure 2 draws the construction at $m = 2$.

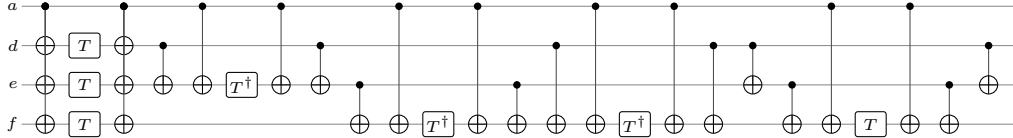
Corollary 41 (controlled-parity layers). *A controlled-parity gate applies $(-1)^{x_a x_b} \bigoplus_{i \in P} x_i$ for a nonempty qubit set P disjoint from $\{a, b\}$. A single controlled-parity gate has $\delta = 7$ for every $|P|$, and a layer of m such gates on disjoint qubit sets has $\delta = 6m + 1$.*

the first block, CCZ_{abc} : seven T s

this one is deleted; the block below supplies its phase



the second block, CCZ_{def} , with every T bracketed by CNOTs from wire a



every parity in this block now contains a , and the seven coefficients of a CCZ sum to 1 (2), so this block leaves one T 's worth of phase on wire a , which is the gate deleted above

seven T s, less the one deleted above: the second block costs six. In general $7 + 6(m - 1) = 6m + 1$

Figure 2: The $6m + 1$ construction, drawn at $m = 2$: bracketing the second block onto a qubit of the first removes one gate, so the pair uses thirteen T s where two independent blocks would use fourteen. The circuit restores its wires and implements $\text{CCZ}_{abc}\text{CCZ}_{def}$ (checked by simulation, Appendix L).

Proof. Let C be the Clifford that, in each block, XORs the parity qubits P_i onto one representative wire. Conjugating the layer by C replaces each controlled-parity by a CCZ on three wires, on disjoint triples. Hence the layer equals $C^{-1}(\bigoplus_i \text{CCZ}_i)C$, and since δ is Clifford bi-invariant its T -count is $\delta(\bigoplus_i \text{CCZ}_i) = 6m + 1$ by Theorem 40. For $m = 1$ the block is a single CCZ , giving $\delta = 7$. $\square$

Its nullity is $\nu = 3$ for every $|P|$.

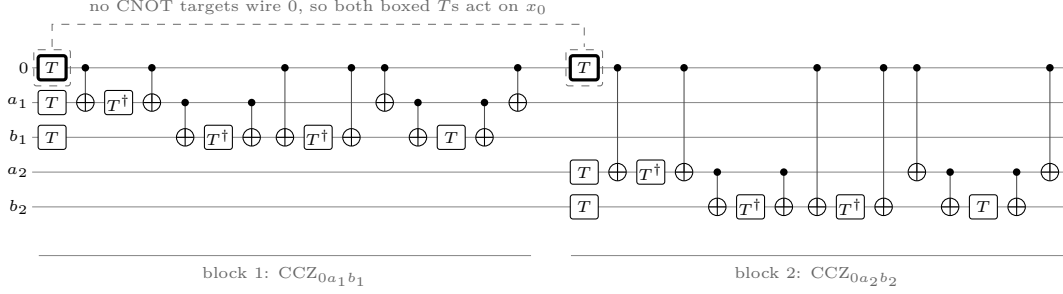
C.2 Overlapping layers: the cost of shared qubits

The parallel Toffoli layer has $\delta_m = 6m + 1$, but the Toffoli blocks of practical oracles rarely act on disjoint qubits. Controlled layers and swap tests share a common control among many gates, and adders chain their Toffolis through shared carries. This subsection determines the T -count of such layers. Blocks that pairwise share at most one qubit keep the tensor non-degenerate, and each shared incidence lowers the floor by 2 (Corollary 45). If all blocks share a common pair of qubits, the layer is Clifford-equivalent to a single CCZ . The resulting counts, summarized in Figure 4, are

$$6m + 1 \xrightarrow{\text{share one qubit}} 4m + 3 \xrightarrow{\text{share a pair}} 7.$$

Lemma 42 (linear overlap keeps the tensor non-degenerate). *Let $B_1, \dots, B_m$ be distinct CCZ blocks (triples) covering n qubits, such that every pair of qubits lies in at most one block (the blocks form a linear 3-uniform hypergraph: pairwise intersections have size at most one). Then the union's cubic tensor is non-degenerate, so*

$$\delta\left(\prod_i \text{CCZ}_{B_i}\right) \geq 2n + 1 = 2(3m - s) + 1,$$



$TT = S$ is Clifford, so of the fourteen T s emitted twelve remain

a different arrangement of the same two blocks reaches 11, which is both $6m + 1 - 2s$ and $4m + 3$ here (Theorem 43)

Figure 3: How sharing lowers the T -count. Two CCZ blocks both act on wire 0, and their circuits run one after the other. Each block places one of its T s on wire 0 itself. Wire 0 is only ever a CNOT control, so it carries x_0 throughout and the two boxed T s act on the same parity. They combine into S , a Clifford gate, and the pair is removed.

where $s = 3m - n$ is the number of merges, the qubit-incidences in excess of one per qubit.

Proof. Let $w \in R$ and let p be any qubit, lying in some block $B = \{p, q, r\}$. The pair $\{q, r\}$ is contained in B and, by linearity, in no other block, so the radical relation at (q, r) is $\sum_a w_a T_{aqr} = w_p = 0$. This holds for every covered qubit, so $R = 0$ and Theorem 26 applies with $d = 0$. $\square$

For the fan-out pattern, in which one control is shared by all blocks, a construction attains the floor of Lemma 42 for every m .

Theorem 43 (fan-out layer: exact cost $4m + 3$). *Let $U_m = \prod_{i=1}^m \text{CCZ}_{0,a_i,b_i}$ be m CCZ gates sharing the single control qubit 0, on $n = 2m + 1$ qubits. Then*

$$\delta(U_m) = 4m + 3 \quad \text{for every } m \geq 1.$$

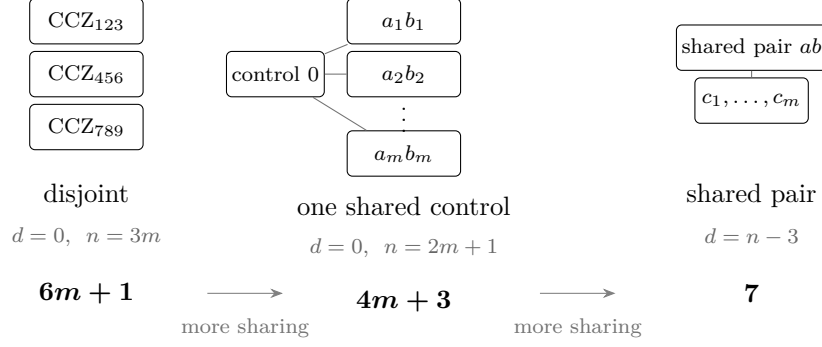
Proof. Lower bound. The blocks pairwise share only qubit 0, a linear hypergraph, so Lemma 42 gives $\delta \geq 2n + 1 = 4m + 3$.

Upper bound. Take the tree F_0 for the shared qubit 0 to be the path $B_1, \dots, B_m$, so F has maximum degree two, and apply Proposition 47 with $s = m - 1$. Each of the $m - 1$ splices is case (ii) of Lemma 46, so the weight is $7 + 4(m - 1) = 4m + 3$, which equals the floor. $\square$

Sharing one control among m Toffolis thus saves $2(m - 1)$ T -gates against the parallel layer, and by Lemma 42 no implementation saves more. Campbell and Howard obtained this count and its optimality from their controlled-unitary theorem [13, Thm. 5]. Here it is the floor of Lemma 42, attained by splicing. More generally, the floor comes within one of their exact count on every controlled diagonal Clifford.

Proposition 44 (the floor on controlled unitaries). *Let $U = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U_t^2$, controlled on a qubit h , with U_t a diagonal level-three gate on the other $n - 1$ qubits, and let μ be the least T -count of U_t up to CCZ gates. Suppose U is not Clifford. If U is pure-cubic, its isotropy floor is $2\mu + 1$. Otherwise it is 2μ .*

Proof. Over $\mathbb{Z}$, $\chi_y = \sum_{\emptyset \neq A \subseteq y} (-2)^{|A|-1} \prod_{a \in A} x_a$, by induction from $u \oplus v = u + v - 2uv$. If ϕ_A is the coefficient of $\prod_{a \in A} x_a$ in a phase polynomial, then $\phi_A \equiv (-2)^{|A|-1} \sum_{y \supseteq A} c_y \pmod{8}$ for $|A| \leq 3$, and the syndrome is $s_A = (\phi_A / 2^{|A|-1}) \pmod{2}$.



the radical dimension d is the dial: $d = 0$ costs scale with m ; $d = n - 3$ collapses the layer

Figure 4: Sharing qubits interpolates the cost of a CCZ layer, and the radical dimension measures the position of a layer in that interpolation. Disjoint blocks give $d = 0$ on $n = 3m$ qubits and cost $6m + 1$ (Theorem 40). Fanning every block out of one shared control keeps $d = 0$ but drops n to $2m + 1$, so the cost falls to exactly $4m + 3$ (Theorem 43); each single-qubit merge saves two T -gates (Corollary 45). Sharing a full pair sends d to $n - 3$, and the layer is Clifford-equivalent to a single gate, $\delta = 7$, for every m (Remark 49).

The phase of U_t is $g = \sum_i g_i x_i + \sum_{i < j} g_{ij} x_i x_j + \sum_{i < j < k} g_{ijk} x_i x_j x_k$ with every g_{ij} even and every $g_{ijk} \equiv 0 \pmod{4}$, so U_t^2 has phase $2 \sum_i \alpha_i x_i + 4 \sum_{i < j} \beta_{ij} x_i x_j$ with $\alpha_i = g_i \pmod{4}$ and $\beta_{ij} = g_{ij}/2 \pmod{2}$. The phase of U is therefore

$$\phi = x_h \left(2 \sum_i \alpha_i x_i + 4 \sum_{i < j} \beta_{ij} x_i x_j \right),$$

and its only nonzero syndromes are $s_{\{h,i\}} = a_i := \alpha_i \pmod{2}$ and $s_{\{h,i,j\}} = \beta_{ij}$. Let M be the symmetric $(n-1) \times (n-1)$ matrix with $M_{ii} = a_i$ and $M_{ij} = \beta_{ij}$. It is the matrix of order- ≤ 2 syndromes of U_t , and $M \neq 0$ because U is not Clifford.

A product of CCZ gates changes only order-3 syndromes, and every order-3 syndrome is that of such a product (Fact 21). Hence μ is the least number of parities of $\mathbb{F}_2^{n-1}$ with order- ≤ 2 moments M , that is, the least number of rows of a matrix Y with $Y^\top Y = M$ in the convention of Appendix B, the quantity $\mu[U_t]$ of [13]. By Lempel's theorem [31] this is $\text{rank } M$ when M has a nonzero diagonal entry and $\text{rank } M + 1$ otherwise. When U is pure-cubic every a_i vanishes, so M is alternating, its rank is even and μ is odd. The floor $2\mu + 1$ is then consistent with the exact count of [13, Thm. 5].

A vector $w = (w', w_h)$ lies in R^* when $\sum_b w_b s_{\{b,p,q\}} = 0$ for all p, q . Taking $p, q \neq h$ gives $w_h M_{pq} = 0$, so $w_h = 0$. Taking $p = h$ and $q = j \neq h$ then gives $(Mw')_j = 0$. Taking $p = q = h$ gives $a \cdot w' = 0$, which follows from $Mw' = 0$, since $a \cdot x = x^\top Mx = 0$ for every $x \in \ker M$. Hence $R^* = \ker M \times \{0\}$ and $n - d^* = \text{rank } M + 1$.

If $a \neq 0$, the gate is not pure-cubic, its quadratic moment matrix is nonzero only in row and column h , where it equals a , so $r = 2$, and the floor is $2(\text{rank } M + 1) - 2 = 2 \text{rank } M = 2\mu$. If $a = 0$, the gate is pure-cubic with $R = R^*$, so by Proposition 32 its floor is $2(n - d) + 1 = 2 \text{rank } M + 3$. Then M has zero diagonal and $\mu = \text{rank } M + 1$, so the floor is $2\mu + 1$. $\square$

Campbell and Howard's controlled-unitary theorem [13, Thm. 5] gives the exact count 2μ when μ is even and $2\mu + 1$ when μ is odd. The floor is therefore exact on every pure-cubic controlled unitary, the fan-out included, and on every controlled unitary with μ even. It is one below the exact count otherwise. The missing unit is the parity step of their converse [13, App. C].

Corollary 45 (the sharing floor). *A layer of m distinct CCZ blocks with linear overlap covering n qubits, with $s = 3m - n$ merges, has $\delta \geq 6m + 1 - 2s$.*

This is Lemma 42 with $n = 3m - s$, so each merge lowers the floor by 2. It is not attained on every linear pattern (Remark 48), but the following construction attains it on a large class. Its case (i) is the gate-synthesis construction of Campbell and Howard's subadditivity theorem [13, Thm. 6] in the language of parity sets. Case (ii) and the routing of Proposition 47 are our formulation, and the single shared control of case (ii) is already covered by their fan-out count.

Lemma 46 (splicing). *Call a set S of parities pure-cubic odd if $|S|$ is odd and $M_A(S) = 0$ for $|A| \in \{1, 2\}$, and call its block set $B(S) = \{A : |A| = 3, M_A(S) = 1\}$. The set $\text{CCZ}(B)$, the seven nonempty subsets of a block, is pure-cubic odd with $|\text{CCZ}(B)| = 7$ and block set $\{B\}$. Let P, Q be pure-cubic odd on supports $X = \bigcup_{s \in P} s$ and $Y = \bigcup_{s \in Q} s$ with $|X \cap Y| \leq 1$, let $y \in P$ with $y \cap Y = \emptyset$, and write $Q \oplus y = \{s \cup y : s \in Q\}$ for the lift of Q by y . Then*

$$P \star_y Q := (P \setminus \{y\}) \triangle (Q \oplus y)$$

is pure-cubic odd with block set $B(P) \sqcup B(Q)$, and

1. *if $X \cap Y = \emptyset$ then $|P \star_y Q| = |P| + |Q| - 1$.*
2. *if $X \cap Y = \{c\}$ (so $c \notin y$, as $y \cap Y = \emptyset$), $\{c\} \in Q$ and $y \cup \{c\} \in P$, then the single coincidence $\{c\} \cup y = y \cup \{c\}$ cancels and $|P \star_y Q| = |P| + |Q| - 3$.*

Proof. Since $y \cap Y = \emptyset$, each lifted parity $s \cup y$ is a disjoint union. A block lies inside the support that carries it and $|X \cap Y| \leq 1$, so $B(P)$ and $B(Q)$ are disjoint. The lift satisfies $M_A(Q \oplus y) = [A \setminus Y \subseteq y] \cdot M_{A \cap Y}(Q)$, with $M_\emptyset(Q) := |Q| \bmod 2$. If $A \cap Y = \emptyset$ the lifted term is $[A \subseteq y]$, which cancels against the deletion of y . This is where oddness of $|Q|$ is used. If $A \cap Y \neq \emptyset$ then the deleted y contributes nothing, since y misses Y . If moreover $|A \cap Y| \leq 2$, which includes every A with $A \setminus Y \neq \emptyset$, then $M_{A \cap Y}(Q) = 0$ because Q is pure-cubic. Otherwise $A \subseteq Y$ and the lifted term is $M_A(Q)$. Hence $M_A(P \star_y Q) = M_A(P) + [A \subseteq Y] M_A(Q)$ for $1 \leq |A| \leq 3$, which gives pure-cubicity and the block set $B(P) \sqcup B(Q)$. A symmetric difference changes the size by an even number, so $|P \star_y Q| \equiv |P| - 1 + |Q| \equiv 1 \pmod{2}$.

It remains to count. Distinct $s \in Q$ give distinct lifts, since each s lies in Y and y misses Y . A lift $s \cup y$ can equal a parity of P only if $s \subseteq X$, that is $s \subseteq X \cap Y$. In case (i) this is impossible, so the symmetric difference is a disjoint union of size $(|P| - 1) + |Q|$. In case (ii) it forces $s = \{c\}$, and the lift $\{c\} \cup y$ equals the parity $y \cup \{c\}$ of P , which differs from the deleted y because $c \notin y$. That single pair cancels, and the size is $(|P| - 1) + |Q| - 2$. $\square$

Proposition 47 (tree-shaped overlap attains the sharing floor). *Let a layer of m blocks with linear overlap have s merged incidences, and suppose the merges can be realized along a spanning forest F of the block set (for each shared qubit q a tree F_q , with edges labelled q , on the blocks containing q , such that the union F is a forest) of maximum degree three (a condition of the port tables in the proof). Splicing along F by Lemma 46, taking case (ii) on each of the s merge edges and case (i) on the remaining $m - 1 - s$ splices, yields a representation of weight*

$$7 + 4s + 6(m - 1 - s) = 6m + 1 - 2s.$$

By Corollary 45 this weight is optimal, so $\delta = 6m + 1 - 2s$.

Proof. Root each tree of F at any block. Since F has maximum degree three, the root has at most three children and every other block at most two. Order the blocks so that the components of F

come one after another and every block follows its parent. Start from $\text{CCZ}(B)$ for the first block and splice in the others one at a time, taking P to be the current set and $Q = \text{CCZ}(B)$ for the new block B , of support $Y = B$. Lemma 46 keeps the current set pure-cubic odd and adds B to its block set, so the final set has the moments of the layer and represents it (Fact 21). Case (i) adds 6 to the weight and case (ii) adds 4. The tree F_q has one edge fewer than the number of blocks containing q , so F has $3m - n = s$ edges, and there are $m - 1 - s$ splices between components. The weight is therefore $7 + 4s + 6(m - 1 - s)$, once each splice is shown to meet the hypotheses of its case.

The first block of a component shares no qubit with the placed blocks, since blocks in different components are disjoint, so case (i) applies with any placed y , which misses B . Now let B be the child of W in F_c . Suppose a placed block A contains a qubit $q \neq c$ of B . Then $A \neq W$, since W already meets B in c and the overlap is linear. The tree F_q joins B to A by a path whose first edge is labelled q . The placed blocks of the component form a subtree of F that contains W and not B , so B also reaches A by a path whose first edge is BW , labelled c . These are two different paths in the forest F , a contradiction. Hence the placed support meets B in $\{c\}$ alone, and any placed y with $c \notin y$ misses B .

It remains to find y with $c \notin y$ and $y, y \cup \{c\}$ both in the current set. Let E_W be the lift used when W was placed, that is the y of its splice, with $E_W = \emptyset$ for the very first block. The first block of a later component is placed by case (i), so its E_W is the y used there. For a block with a parent let c_W be the qubit joining them. Until the component of W is complete, the current set contains $\sigma \cup E_W$ for every $\sigma \in \text{CCZ}(W)$ except $\sigma = \{c_W\}$ and the two parities of each port of W (defined below) already used by a child, since ports are disjoint and a later splice deletes only its own y and the parity $y \cup \{c\}$ it cancels. By the count in Lemma 46, the lifts added at each splice differ from every parity already present except the one cancelled, so parities emitted for different blocks stay distinct and no splice below another block removes a parity of W . Call a pair $(\sigma \cup E_W, \sigma \cup \{c\} \cup E_W)$ with $\emptyset \neq \sigma \subseteq W \setminus \{c\}$ a port of W for c . It is a valid choice of y , since E_W is the y of W 's own splice and so is disjoint from W by the hypothesis of Lemma 46. It suffices that the children of W can be given disjoint ports. Write $W = \{a, b, c\}$ for a root. Three children through c use the three ports of c . Two through c and one through a use $(\{b\}, \{b, c\})$, $(\{a, b\}, \{a, b, c\})$ and $(\{c\}, \{a, c\})$. One through each qubit uses $(\{b\}, \{b, c\})$, $(\{c\}, \{a, c\})$ and $(\{a\}, \{a, b\})$. Every other case is one of these, or a subset of one, after relabelling. For a block $W = \{c_W, u, v\}$ with a parent, the parity $\{c_W\} \cup E_W$ is gone and W has at most two children. Two through c_W use $(\{u\}, \{u, c_W\})$ and $(\{v\}, \{v, c_W\})$. Two through u use $(\{v\}, \{u, v\})$ and $(\{c_W, v\}, \{c_W, u, v\})$. One through c_W and one through u use $(\{u\}, \{u, c_W\})$ and $(\{v\}, \{u, v\})$. One through u and one through v use $(\{v\}, \{u, v\})$ and $(\{c_W, u\}, \{c_W, u, v\})$. The remaining cases follow by exchanging u and v . In every case each port omits the missing parity, and each is lifted by E_W . $\square$

Besides the parallel Toffoli layer ($s = 0$) and the fan-out through a common control ($s = m - 1$), which recover Theorems 40 and 43, the proposition covers the *ripple-carry chain* $B_i = \{c_{i-1}, u_i, c_i\}$ on $n = 2m + 1$ qubits. Its merge structure is a path, so the CCZ layer on these blocks has T -count $4m + 3$ for every m .

Writing m_C , n_C and $s_C = 3m_C - n_C$ for a connected component C of the block-overlap graph, the edges of F inside C number s_C . A connected component always has $s_C \geq m_C - 1$, so the construction needs $s_C = m_C - 1$ for every C , and in addition a choice of the trees F_q whose union has maximum degree at most three. The second condition is not implied by the first (see the paragraph after Conjecture 111), and the port tables of the proof cover at most three neighbors per block. The global inequality $s \leq m - 1$ does not suffice: a triangle of blocks placed beside a disjoint fourth block satisfies it while its triangle component does not.

Remark 48 (the sharing floor fails once the merge structure must cycle). Once some component has $s_C \geq m_C$ no forest exists, and the floor can fail to be attained. The smallest instance is a triangle of blocks, $\{0, 1, 2\}, \{0, 3, 4\}, \{1, 3, 5\}$ on $n = 6$. The overlap is linear and the radical is trivial, so Corollary 45 gives $6m + 1 - 2s = 13$, but exact Reed–Muller coset decoding returns $\delta = 15$. The canonical residue has weight 15 and is its own coset leader, so the ceiling of Proposition 30 certifies the same value from above and the bracket is $[13, 15]$ with δ at its upper end. Up to relabelling the only other linear covering pattern at $n \leq 6$ with $s \geq m$, the four blocks $\{0, 1, 2\}, \{0, 3, 4\}, \{1, 3, 5\}, \{2, 4, 5\}$, also has floor 13 and $\delta = 15$, while every pattern at $n \leq 6$ with $s \leq m - 1$ attains the floor. Whether the componentwise count decides attainment in general is Conjecture 111.

Remark 49 (layers sharing a pair of qubits). If all blocks share a pair of qubits, the tensor is degenerate. For m blocks $\{a, b, c_i\}$ through a common pair $\{a, b\}$, the radical relations give $R = \{w \text{ supported on the } c_i \text{ with } \sum_i w_{c_i} = 0\}$, of dimension $n - 3$. The gate is pure-cubic, so the floor $2(n - d) + 1$ of Proposition 32 equals 7. The layer is Clifford-equivalent to a single CCZ (Corollary 41), so $\delta = 7$ for every m and the floor is attained.

D Phase oracles, factorizable cubics and field multiplication

A cubic phase gate is the elementary cost layer of a 3-local QAOA circuit. The *Boolean-oracle* form $U_f|x\rangle = (-1)^{f(x)}|x\rangle$ takes $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ a homogeneous cubic, for example the cubic form $\sum_{\{a,b,c\} \in E} x_a x_b x_c$ of a 3-uniform hypergraph. The *Ising* form $\exp(-i\gamma \sum_S w_S Z_S)$ has single-qubit fields, pairwise couplings and 3-body terms indexed by a 3-uniform interaction hypergraph H . When every γw_S lies in $\frac{\pi}{8}\mathbb{Z}$ it is a diagonal level-three gate whose parities are the terms S . A Max-3-XORSAT objective is of this form, with only 3-body terms. (In Boolean-oracle form its clauses are linear over $\mathbb{F}_2$, so $(-1)^f$ would be a Pauli.)

Corollary 50 (floors for cubic phase gates). *Let $f \neq 0$ and let d_f be the radical dimension of f . The $\{CNOT, T\}$ T -count of U_f is at least $2(n - d_f) + 1$, which is $2n + 1$ when f is non-degenerate.*

Proof. The phase f is homogeneous of degree 3, so $s_A = 0$ for $|A| \leq 2$ and U_f is pure-cubic with cubic tensor equal to the coefficient tensor of f . The claim is then Proposition 32, and d_f is computed by Gaussian elimination in $O(n^3)$ time. $\square$

D.1 Weighted and inhomogeneous layers

A cost Hamiltonian used in practice carries single-qubit fields and pairwise couplings alongside its 3-body terms, so its phase gate is not pure-cubic. The general bound of Theorem 27 applies without change, because the order- ≤ 3 moments of either cost-layer model can be computed directly from the interaction hypergraph. For a 3-uniform interaction hypergraph H with fields $h \subseteq [n]$ and couplings $J \subseteq \binom{[n]}{2}$, the moments are as follows. In the weighted case H, J and h denote the sets of terms S with γw_S an odd multiple of $\frac{\pi}{8}$, since a term with an even multiple is a Clifford phase on its parity.

$$s_{\{a,b,c\}} = [\{a, b, c\} \in H], \quad s_{\{a,b\}} = J_{ab} \oplus \text{codeg}_H(a, b), \quad s_{\{a\}} = h_a \oplus \deg_J(a) \oplus \deg_H(a),$$

all modulo 2, so the cubic moment form is the adjacency tensor of H and the quadratic moment matrix Q collects the couplings and codegrees.

Corollary 51 (floor for a general cost layer). *A 3-local QAOA cost layer, in either the Boolean-oracle or the Ising model, weighted and with fields and couplings whose angles are multiples of*

$\pi/8$, is a diagonal level-three gate whose $\{CNOT, T\}$ T -count satisfies $\delta \geq 2(n - d^*) - r$. Here $d^* = \dim R^*$ is the radical dimension of the full moment tensor and $r = \text{rank}_{\mathbb{F}_2} Q$, both computable in $O(n^3)$ time from H, J, h .

Proof. Immediate from Theorem 27 applied to the residue of the cost layer, whose order- ≤ 3 moments are the displayed hypergraph quantities. $\square$

In the Boolean-oracle form with f a homogeneous cubic the order- ≤ 2 moments vanish, so $d^* = d_f$ and $r = 0$, and the corollary recovers Corollary 50 up to the $+1$ of Proposition 32.

The same moments give an upper bound, and for a cost layer it equals the number of non-Clifford terms in the Hamiltonian.

Proposition 52 (the ceiling of a cost layer is its term count). *For a 3-local Ising cost layer with interaction hypergraph H , couplings J and fields h , the canonical residue of Proposition 30 is supported exactly on the terms, with $z_{\{a,b,c\}} = [\{a,b,c\} \in H]$, $z_{\{a,b\}} = J_{ab}$ and $z_{\{a\}} = h_a$. Hence*

$$\delta \leq W(s) = |H| + |J| + |h|,$$

the number of non-Clifford terms in the cost Hamiltonian.

Proof. Solve downwards as in Proposition 30. At order three $z_{\{a,b,c\}} = s_{\{a,b,c\}}$ is the indicator of H . At order two, $\bigoplus_{c \notin \{a,b\}} z_{\{a,b,c\}}$ counts the triples of H through the pair $\{a,b\}$, which is $\text{codeg}_H(a,b)$ modulo 2, so

$$z_{\{a,b\}} = s_{\{a,b\}} \oplus \text{codeg}_H(a,b) = J_{ab} \oplus \text{codeg}_H(a,b) \oplus \text{codeg}_H(a,b) = J_{ab}.$$

At order one the two remaining sums are $\bigoplus_{b \neq a} z_{\{a,b\}} = \deg_J(a)$ and $\bigoplus_{b < c} z_{\{a,b,c\}} = \deg_H(a)$ modulo 2, so $z_{\{a\}} = h_a \oplus \deg_J(a) \oplus \deg_H(a) \oplus \deg_J(a) \oplus \deg_H(a) = h_a$. Each degree and codegree correction carried by the moments is cancelled by the corresponding sum in the solve, and z vanishes above weight three, so $\text{wt}(z) = |H| + |J| + |h|$. $\square$

D.2 Cubics that factor

When the cubic factors as a linear form times a quadratic, the floor is attained. With $\ell = u_0$ a factorizable cubic is $u_0 q$, which after a CNOT relabelling is a controlled unitary with control u_0 and a diagonal Clifford target, so the exact value is that of the controlled-unitary theorem of Campbell and Howard [13, Thm. 5].

Corollary 53 (factorizable cubics are exact). *Let $f = \ell \cdot q$ on n qubits, with ℓ a nonzero linear form, q quadratic and f of degree three. Then $\delta(U_f) = 2(n - d_f) + 1$.*

Proof. Choose coordinates with $\ell = u_0$. Up to a diagonal Clifford factor, which changes neither δ nor d_f , $U_f = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes D$ with $D = (-1)^{q(1,u')}$ a product of CZ and Z gates on the other $n - 1$ qubits. Replacing each CZ by CS and each Z by S gives a diagonal level-three gate U_t with $U_t^2 = D$. The gate U_f is pure-cubic (Corollary 50) and not Clifford, so by Proposition 44 and its proof its floor $2(n - d_f) + 1$ equals $2\mu + 1$ with μ odd. For odd μ the controlled-unitary theorem [13, Thm. 5] gives $\delta(U_f) = 2\mu + 1$. $\square$

When the reduced adjacency is invertible, the proposition below gives an explicit representation of weight $2n + 1$ from the even-weight subspace. This class contains the star and the complete cubic (Corollary 55). A set of parities represents f when its moments match the syndrome of f (Fact 21).

Proposition 54 (the even-subspace representation). *Let $f = \ell \cdot q$ on $n \geq 3$ qubits, with ℓ a nonzero linear form and q quadratic. Choose coordinates with $\ell = u_0$, put $N = n - 1$, and let $\text{Adj} \in \mathbb{F}_2^{N \times N}$ be the adjacency matrix of $\bar{q} = q|_{u_0=0}$ on $u_1, \dots, u_N$. If Adj is invertible over $\mathbb{F}_2$, then f is non-degenerate and*

$$\delta(U_f) = 2n + 1.$$

Proof. The cubic monomials of f are the $u_0 u_j u_k$ with $\text{Adj}_{jk} = 1$, so the cubic tensor has $T_{0jk} = \text{Adj}_{jk}$ and no entry supported in $u_1, \dots, u_N$ alone. Since Adj is alternating and invertible, N is even.

The monomials of f of degree at most two give a diagonal Clifford factor, which changes neither δ nor d , so only the cubic part matters. The floor. Let $w \in R$. Contracting against the pair (u_0, u_j) gives $(\text{Adj } w')_j = 0$ for every j , where $w' = (w_1, \dots, w_N)$. As Adj is invertible, $w' = 0$. Contracting against a pair (u_j, u_k) leaves $w_0 \text{Adj}_{jk} = 0$, and $\text{Adj} \neq 0$, so $w_0 = 0$. Hence $d = 0$ and Corollary 50 gives $\delta(U_f) \geq 2n + 1$.

The construction. Let $\mathcal{E} \subseteq \mathbb{F}_2^{N+1}$ be the even-weight subspace, of dimension N . On $\mathcal{E}$ the dot product is alternating ($v \cdot v = \text{wt}(v) \bmod 2 = 0$) and non-degenerate: its radical is $\mathcal{E} \cap \mathcal{E}^\perp$, and $\mathcal{E}^\perp = \langle \mathbf{1} \rangle$ with $\text{wt}(\mathbf{1}) = N + 1$ odd, so $\mathbf{1} \notin \mathcal{E}$. Thus $\mathcal{E}$ is a symplectic $\mathbb{F}_2$ -space of dimension N . Since Adj is an invertible alternating form on $\mathbb{F}_2^N$ and any two non-degenerate alternating forms of equal dimension over $\mathbb{F}_2$ are isometric, there is an isomorphism $\mathbb{F}_2^N \rightarrow \mathcal{E}$ carrying Adj to the dot product. Let $c_1, \dots, c_N \in \mathcal{E}$ be the images of the standard basis, so that $\text{wt}(c_j)$ is even and $c_j \cdot c_k = \text{Adj}_{jk}$ for $j \neq k$.

Let V be the $(N + 1) \times N$ matrix with columns c_j and let $\rho_r = \{j : V_{rj} = 1\}$ be its rows. Emit

$$\mathcal{S} = \{\{u_0\}\} \cup \{\rho_r \cup \{u_0\}\}_{r=1}^{N+1} \cup \{\rho_r\}_{r=1}^{N+1}.$$

No ρ_r is empty, because the c_j span $\mathcal{E}$ and $\mathcal{E}$ lies in no coordinate hyperplane $\{v_r = 0\}$, as $\mathbf{e}_r + \mathbf{e}_s \in \mathcal{E}$. The ρ_r are pairwise distinct, because $\rho_r = \rho_s$ would put $\mathbf{e}_r + \mathbf{e}_s$ in $\mathcal{E}^\perp = \langle \mathbf{1} \rangle$, impossible for $N + 1 \geq 3$. Hence the three groups are disjoint and $|\mathcal{S}| = 1 + (N + 1) + (N + 1) = 2n + 1$.

The moments. Write $A = A_0 \cup A'$ with $A_0 = A \cap \{u_0\}$ and $A' \subseteq \{u_1, \dots, u_N\}$. If $u_0 \notin A$ then each ρ_r with $A' \subseteq \rho_r$ is counted twice, as ρ_r and as $\rho_r \cup \{u_0\}$. Hence $M_A \equiv 0$, as required because every block of f contains u_0 . If $u_0 \in A$ then only $\{u_0\}$ (when $A' = \emptyset$) and the sets $\rho_r \cup \{u_0\}$ with $A' \subseteq \rho_r$ contribute, so $M_A = [A' = \emptyset] + \#\{r : A' \subseteq \rho_r\}$. With $|A| \leq 3$ there are three cases. The case $A' = \emptyset$ gives $1 + (N + 1) = N + 2 \equiv 0$ because N is even. The case $A' = \{j\}$ gives $\text{wt}(c_j) \equiv 0$. The case $A' = \{j, k\}$ gives $c_j \cdot c_k = \text{Adj}_{jk}$, which is 1 exactly on the blocks $\{u_0, u_j, u_k\}$ of f . Hence $\mathcal{S}$ is a representation of f of weight $2n + 1$, equal to the floor. The value $\delta(U_f) = 2n + 1$ is also Corollary 53. $\square$

Invertibility of Adj requires n odd, the parity condition already carried by the star and complete-cubic rows of Proposition 66. Corollary 55 shows that both families satisfy the hypothesis.

Corollary 55 (the star and the complete cubic). *For every odd n , the star $S_n = \sum_{0 < j < k < n} x_0 x_j x_k$ and the complete cubic $e_3 = \sum_{i < j < k} x_i x_j x_k$ satisfy $\delta = 2n + 1$.*

Proof. For S_n take $\ell = x_0$. Then $\text{Adj} = J + I$ is the adjacency of K_N , and $(J + I)^2 = NJ + I = I$ over $\mathbb{F}_2$ for N even, so Adj is invertible. Unwinding the construction gives the explicit representation

$$\mathcal{S}(S_n) = \{\{0\}\} \cup \{\{0, j\}\}_{j=1}^N \cup \{\{j\}\}_{j=1}^N \cup \{\{0, \dots, N\}\} \cup \{\{1, \dots, N\}\},$$

of weight $2n + 1$, from the choice $c_j = \mathbf{e}_j + \mathbf{e}_{N+1}$. For e_3 , note $e_3 = e_1 \cdot e_2$ over $\mathbb{F}_2$: in $(\sum_i x_i)(\sum_{j < k} x_j x_k)$ each triple arises three times and each pair twice. Hence e_3 factors with $\ell = e_1$,

and the substitution $u_0 = \sum_i x_i$, $u_j = x_{j+1}$ is a CNOT change of basis carrying e_3 to the star. Pulling the star representation back through it gives

$$\mathcal{S}(e_3) = \{\{i\}\}_{i \in [n]} \cup \{[n] \setminus \{i\}\}_{i \in [n]} \cup \{[n]\},$$

of weight $2n + 1$: directly, $M_{\{i\}} = 1 + (n - 1) + 1 = n + 1$, $M_{\{i,j\}} = (n - 2) + 1 = n - 1$ and $M_{\{i,j,k\}} = (n - 3) + 1 = n - 2$, which for odd n are 0, 0, 1 as required. $\square$

D.3 Field-multiplication oracles

Proposition 56 (floor for field-multiplication oracles). *Let $U_k|x, y, z\rangle = (-1)^{z \cdot (x * y)}|x, y, z\rangle$ be the multiplication phase oracle of $\text{GF}(2^k)$ on $n = 3k$ qubits ($*$ the field product, $\cdot$ the coordinate dot). Then U_k is pure-cubic with a non-degenerate tensor, and $\delta(U_k) \geq 6k + 1$.*

Proof. The phase is the trilinear form $z \cdot (x * y)$, homogeneous cubic, so the gate is pure-cubic with $T_{x_q y_r z_l} = [\text{bit } l \text{ of } e_q * e_r]$ and no other entries across registers. Let $w = (w_x, w_y, w_z) \in R$. Contracting against the pairs (y_r, z_l) gives $0 = \sum_q (w_x)_q T_{x_q y_r z_l} = \text{bit } l \text{ of } (w_x * e_r)$ for all r, l , so $w_x * e_r = 0$ for every basis element. A field has no zero divisors, so $w_x = 0$. Symmetrically $w_y = 0$, and contracting against (x_q, y_r) gives $w_z \cdot (e_q * e_r) = 0$ with the products spanning $\text{GF}(2^k)$ (for fixed r the products $e_q * e_r$ form a basis, as multiplication by $e_r \neq 0$ is invertible), so $w_z = 0$. Hence $d = 0$ and Theorem 26 gives $\delta \geq 2n + 1 = 6k + 1$. $\square$

Remark 57 (the $\text{GF}(4)$ multiplier). At $k = 1$ the oracle is CCZ and the floor 7 is tight. At $k = 2$ the lower bounds are the nullity 6, the isotropy floor $6k + 1 = 13$ and the slice-rank floor $G(2) = 14$ (Proposition 60). Exact Reed–Muller coset decoding shows that the merged phase-polynomial residue of weight 17 is its own coset leader, so $\delta(U_2) = 17$ and the term-by-term expansion is optimal once parities are merged. The unmerged schoolbook expansion has weight 35.

Since $\delta(U)$ is the symmetric rank of the signature tensor (Appendix A), the value 17 was already established by AlphaTensor-Quantum [16], whose Z3 search finds no symmetric decomposition of that tensor with fewer terms. The coset decoding reproduces it independently. It certifies published circuits optimal within a natural class. The reversible multiplier $|x, y, z\rangle \mapsto |x, y, z \oplus x * y\rangle$ is U_k conjugated by Hadamards on the output register, so no multiplier circuit whose Hadamards all sit in those two layers uses fewer than $\delta(U_k)$ T gates (Appendix E). The best published T -count for the $\text{GF}(4)$ multiplier is 17, both from a subquadratic multiplier with the FASTODD optimizer [25] and from AlphaTensor-Quantum [16].

The floor $6k + 1$ uses only that the radical is trivial. The slices of the field tensor give more, because a slice is a Gram matrix on the support of a codeword.

Lemma 58 (a slice is a Gram matrix on a codeword’s support). *Let $Y \in \mathbb{F}_2^{t \times n}$ be a representation of a pure-cubic gate, with columns c_a . For $w \in \mathbb{F}_2^n$ write $c(w) = \sum_a w_a c_a$ and $S(w) = \{j : c(w)_j = 1\}$. Then*

$$T(w) = Y^\top \text{diag}(c(w)) Y = Y_{S(w)}^\top Y_{S(w)},$$

and whenever $c(w) \neq 0$,

$$\text{wt } c(w) \geq \text{rank } T(w) + 2.$$

Proof. Entrywise, $T(w)_{bc} = \sum_a w_a T_{abc} = \sum_j (\sum_a w_a (c_a)_j) (c_b)_j (c_c)_j = \sum_{j \in S(w)} (c_b)_j (c_c)_j$, which is the identity. Hence $\text{rank } T(w) \leq \text{rank } Y_{S(w)}$. Isotropy gives $\langle c(w), c_a \rangle = 0$ for every a , that is $\sum_{j \in S(w)} y_j = 0$. The rows of $Y_{S(w)}$ sum to zero, so $\text{rank } Y_{S(w)} \leq |S(w)| - 1$ when $S(w) \neq \emptyset$. Finally $|S(w)|$ is even because $\langle c(w), c(w) \rangle = 0$, and $\text{rank } T(w)$ is even because $T(w)$ is alternating, its diagonal entries $\langle c(w), c_b \rangle$ vanishing by isotropy. An even number at most $|S(w)| - 1$ is at most $|S(w)| - 2$. $\square$

Corollary 59 (the rank spectrum caps the column code). *Let U be pure-cubic with $d = 0$. For any representation by t parities the column code $\mathcal{C} = \{c(w)\}$ is an even self-orthogonal $[t, n]$ code with $\mathbf{1} \in \mathcal{C}^\perp \setminus \mathcal{C}$, and for every r it has at most $\#\{w \neq 0 : \text{rank } T(w) \leq r - 2\}$ nonzero codewords of weight at most r . Hence $\delta(U)$ is at least the least t at which the Delsarte linear program in the weight distribution of $\mathcal{C}$ is feasible under these constraints [38].*

Proof. Since $d = 0$, $w \mapsto c(w)$ is injective by (7), so the caps are Lemma 58. Evenness, self-orthogonality and $\mathbf{1} \in \mathcal{C}^\perp$ are the isotropy (4), and $\mathbf{1} \notin \mathcal{C}$ is Lemma 25 with $d = 0$. The program imposes that the MacWilliams transform of the weight distribution is nonnegative and dominates it, since $\mathcal{C} \subseteq \mathcal{C}^\perp$. $\square$

At $n = 6$ the program gives 13, 14 and 14 on the three non-degenerate orbits, whose T -counts are 13, 15 and 17 (Appendix L).

Proposition 60 (the slice-rank floor for field oracles). *Every nonzero slice of the multiplication tensor of $\text{GF}(2^k)$ has rank $2k$. Hence*

$$\delta(U_k) \geq G(k) := \sum_{i=0}^{3k-1} \left\lceil \frac{2k+2}{2^i} \right\rceil \geq 7k + 2 - \lceil \log_2(2k+2) \rceil,$$

with $G(k) = 7k$ whenever k is a power of two, and $G(2), \dots, G(8) = 14, 20, 28, 34, 41, 47, 56$.

Proof. Choose the trace-dual basis on the output register, so that the moment tensor is $M(x, y, z) = \text{Tr}(xyz)$. For $u = (u_x, u_y, u_z)$ the slice is the symmetric bilinear form

$$T(u)(v, v') = \text{Tr}(u_x(v_y v'_z + v'_y v_z) + u_y(v_x v'_z + v'_x v_z) + u_z(v_x v'_y + v'_x v_y)).$$

Since the trace form is non-degenerate, v lies in its kernel iff $u_x v_y = u_y v_x$, $u_x v_z = u_z v_x$ and $u_y v_z = u_z v_y$ as field elements. If $u_z \neq 0$ these say $v_x = u_x v_z / u_z$ and $v_y = u_y v_z / u_z$, and the first equation then holds automatically. Thus the kernel is the line $\text{GF}(2^k) \cdot u$, of dimension k over $\mathbb{F}_2$, and the same holds with the roles of the registers exchanged when $u_z = 0$. Hence $\text{rank } T(u) = 3k - k = 2k$ for every $u \neq 0$.

Now take any representation with t odd parities. Its columns span a code $\mathcal{C}$ of dimension $3k$, since $d = 0$ (Proposition 56 and (7)). The code is even and self-orthogonal by isotropy, and by Lemma 58 every nonzero codeword has weight at least $2k + 2$. The Griesmer bound for a binary $[t, 3k, 2k + 2]$ code [38] gives $t \geq G(k)$. For the inequality, let $m = \lceil \log_2(2k + 2) \rceil$. The terms with $i \geq m$ equal 1 and number $3k - m$, while the terms with $i < m$ sum to at least $(2k + 2)(2 - 2^{1-m}) \geq 4k + 2$. For $k = 2^j$ the terms are $2k + 2$ at $i = 0$ and $2^{j+1-i} + 1$ for $1 \leq i \leq j$, followed by a single 2 and then ones. They sum to $7k$. The listed values of $G(k)$ are direct evaluations of the sum. $\square$

Remark 61 (linear growth of field-oracle T -counts). The floors $6k + 1$ and $G(k)$ are linear while the explicit circuits are not: schoolbook multiplication uses $\Theta(k^2)$ CCZs and Karatsuba-style circuits $\Theta(k^{1.58})$ [42, 43]. The T -count itself is nevertheless linear in k , as the tensor rank shows. Write $\mu_2(k)$ for the tensor rank of $\text{GF}(2^k)$ multiplication as an $\mathbb{F}_2$ -bilinear map. The T -count is within a factor 7 of it,

$$\mu_2(k) \leq \delta(U_k) \leq 7\mu_2(k).$$

For the lower bound, restrict a representation's cubic tensor to index triples with one leg in each register. The restriction is the multiplication tensor, so the number of parities is at least its rank. For the upper bound, a rank- μ decomposition $T = \sum_{i=1}^\mu a_i \otimes b_i \otimes c_i$ rewrites the phase as

Family	n	floor	ceiling	δ	
GF(2) oracle U_1	3	7	7	7	determined
line f_5	5	11	11	11	determined
hub chain $\sum_i x_0 x_i x_{i+1}$	5	11	11	11	determined
Toffoli layer $m = 1$	3	7	7	7	determined
fan-out layer $m = 1$	3	7	7	7	determined
Toffoli layer $m = 2$	6	13	14	13	floor
fan-out layer $m = 2$	5	11	12	11	floor
line f_6	6	13	14	13	floor
hub chain, $n = 6$	6	11	12	11	floor
line f_6 plus CS on qubits 0, 1	6	10	13	11	interior
wrapped cycle C_6	6	13	18	17	interior
complete cubic e_3	6	11	20	11	floor
GF(4) oracle U_2	6	13	17	17	ceiling

Table 4: Two-sided brackets, both ends $O(n^3)$ from the phase polynomial. The floor is Theorem 27 plus the +1 of Proposition 32 on the pure-cubic locus. The ceiling is Proposition 30. Exact δ is by decoding, for comparison only. Rows above the rule have a degenerate bracket, so their cost is fixed without decoding. For the GF(4) row the slice-rank bound of Proposition 60 gives 14.

$\sum_i (a_i \cdot x)(b_i \cdot y)(c_i \cdot z)$, a product of μ gates, each a CCZ on three linear forms in different registers (a CCZ conjugated by CNOT gates) contributing seven parities. Since $\mu_2(k) = \Theta(k)$ over $\mathbb{F}_2$ for every extension degree, by a Chudnovsky-type algorithm on a descent of a Garcia–Stichtenoth tower [44, 45], we get $\delta(U_k) = \Theta(k)$. The floors therefore have the right order of growth, while the ceiling W_k of Remark 69 grows quadratically. The known constants in the linear bound are too large to compete with the tabulated ceilings.

The circuits reported for these multipliers keep their Hadamards in the two output layers, so after the conjugation of Remark 57 the floor $G(k)$ of Proposition 60 applies to them:

k	2	4	6	8	10	16	32	64	256
floor $G(k)$	14	28	41	56	69	112	224	448	1792
best published	17	39	77	123	196	425	1255	3817	34731
ratio	1.2	1.4	1.9	2.2	2.8	3.8	5.6	8.5	19.4

The published values are the best of AlphaTensor-Quantum without gadgets and FASTTODD, alone or after the subquadratic multiplier [16, 25, 46]. Since the true cost is $\Theta(k)$ (Remark 61), the subquadratic multipliers are eventually a growing factor from optimal. At the tabulated sizes it is open whether the gap lies in the circuits or in the floor (Open Problem 10).

D.4 Two-sided brackets on the application families

Every floor in this section has a matching ceiling from Proposition 30 at the same $O(n^3)$ cost, so each family can be quoted as an interval. Table 4 does this across the families above. In five of the thirteen rows the interval is a single point, and there the T -count is determined by two cubic-time evaluations. These five rows are only two gates up to CNOT relabelling, since the first row and the last two are each a single CCZ, and f_5 and the hub chain at $n = 5$ lie in the one non-degenerate orbit at $n = 5$.

The wrapped cycle C_6 , an ordinary translation-invariant cost layer, is pure-cubic with $R = 0$ and $\delta = 17 = 2n + 5$, so it lies in the five-term orbit of Table 3, and its gap of four above the floor

is the largest among pure-cubic gates at $n \leq 6$.

E Level-three permutation gates

The overlap results of §C.2 extend to a larger class through the structural theory of He, Robitaille and Tan [15], who characterize all permutation gates in the third level C_3 of the Clifford hierarchy. They prove that every such gate is, up to left and right Clifford permutations, a product of Toffolis in *staircase form* (each $\text{TOF}_{i,j,k}$ has $i < j < k$ and the targets are applied in nondecreasing order), and that staircase C_3 permutations are in bijection with descending multiplications: commutative associative products on $\mathbb{F}_2^n$ with $e_i e_i = 0$ and $e_i e_j \in \text{span}\{e_k : k > j\}$ for $i < j$. Writing $e_i e_j = \sum_k c_{ij}^k e_k$, the gate $\text{TOF}_{i,j,k}$ appears in the staircase exactly when $c_{ij}^k = 1$. The constants c_{ij}^k are symmetric in i, j but not in all three indices. What transfers is their support, since $\mathcal{B} = \{\{i, j, k\} : c_{ij}^k = 1\}$ is a set of unordered triples and the cubic moment form is its indicator, $T_{ijk} = [\{i, j, k\} \in \mathcal{B}]$. For a C_3 permutation π with staircase triples $\mathcal{B}$, the CCZ layer $D_\pi = \prod_{B \in \mathcal{B}} \text{CCZ}_B$ on the same triples is the *diagonal shadow* of π , defined relative to the chosen staircase presentation. Write $\delta_T(V)$ for the least number of T gates in any Clifford+ T circuit for a gate V , internal Hadamards included. A Clifford gate contains no T gate, so $\delta_T(CVC') = \delta_T(V)$ for all Cliffords C, C' , and $\delta_T(D) \leq \delta(D)$ for a diagonal gate D . Whether equality holds is the open question of Appendix H. The bounds of Appendix B and §C.2 apply to D_π . They give efficiently computable floors on the diagonal shadow. For a semi-Clifford π these bound every synthesis whose Hadamards lie in the target layers, and they bound $\delta_T(\pi)$ in the regimes of Appendix H. On a characterized subclass they determine the optimum within that layering exactly.

E.1 Semi-Clifford permutations: exact shadow counts and layered optimality

A staircase Toffoli product is *mismatch-free* if no qubit used as a target is used as a control by any of its Toffolis, which makes its Toffolis pairwise commute. He–Robitaille–Tan show that a C_3 permutation is *semi-Clifford* exactly when it is a mismatch-free Toffoli product up to left and right Clifford permutations [15, Lem. B.1, Cor. B.6]. For this class the floors apply to the permutation itself.

Proposition 62 (transfer to semi-Clifford permutations). *Let π be a semi-Clifford C_3 permutation, presented (up to Clifford) as a mismatch-free staircase product of Toffolis on the index triples $\mathcal{B}$, with diagonal shadow D_π . Then π is Clifford-equivalent to D_π , so*

$$\delta_T(\pi) = \delta_T(D_\pi) \leq \delta(D_\pi).$$

Proof. Mismatch-free means each target qubit k is never a control, so all Toffolis targeting k can be conjugated together: $\prod_{i,j} \text{TOF}_{i,j,k} = H_k(\prod_{i,j} \text{CCZ}_{i,j,k})H_k$, and since no later gate controls on k the two H_k factors commute past the rest of the circuit to the boundary. Doing this at every target qubit writes $\pi = C D_\pi C'$ with C, C' products of Hadamards on the target qubits, composed with the Clifford permutations of the presentation, i.e. Clifford. Hence $\delta_T(\pi) = \delta_T(D_\pi)$, and $\delta_T(D_\pi) \leq \delta(D_\pi)$ because a Hadamard-free circuit is a Clifford+ T circuit. $\square$

Every floor of Appendix B and §C.2 therefore bounds every circuit CVC' for a semi-Clifford π in which V is Hadamard-free and C, C' are the target Hadamard layers composed with the Clifford permutations of the presentation, since then $V = D_\pi$, and it bounds $\delta_T(\pi)$ wherever it bounds $\delta_T(D_\pi)$, as in the regimes of Appendix H.

Corollary 63 (optimal Toffoli-network syntheses). *Let the triples $\mathcal{B}$ of a semi-Clifford C_3 permutation form a linear hypergraph (pairwise intersections of size ≤ 1) covering n qubits, with $m = |\mathcal{B}|$ and $s = 3m - n$. If the shared incidences can be routed along a forest of blocks in which each block has at most three neighbors, then*

$$\delta(D_\pi) = 6m + 1 - 2s,$$

and the representation of Proposition 47, conjugated by the target Hadamards of Proposition 62, is a circuit for π with $6m + 1 - 2s$ T gates that is optimal among circuits whose Hadamards all lie in those two layers.

Proof. The shadow D_π is a linear-overlap CCZ layer, so Lemma 42 gives the floor and Proposition 47 supplies a matching representation under the routing hypothesis. Optimality among the stated circuits is the paragraph after Proposition 62. $\square$

The parallel Toffoli layer and the fan-out layer are the cases $s = 0$ and $s = m - 1$ (Theorems 40 and 43). The ripple-carry chain is mismatched, since each carry is a target and then a control, so the corollary does not apply to it, although its shadow still has $\delta(D_\pi) = 4m + 3$ by Proposition 47. The routing hypothesis is needed, because the bound $6m + 1 - 2s$ is not always attained once the merges must cycle (Remark 48). The general case is Conjecture 111.

E.2 General C_3 permutations and the Gottesman–Mochon gate

For a C_3 permutation that is not semi-Clifford, the staircase has a mismatch and the Toffolis do not commute. The permutation is then not Clifford-equivalent to a diagonal gate, and $\delta_T(\pi)$ need not equal $\delta(D_\pi)$. The diagonal shadow D_π is still a diagonal level-three gate, and the floor bounds it in $O(n^3)$.

Corollary 64 (floor for the shadow of a C_3 permutation). *Let $\pi \in C_3$ be a permutation with staircase triples $\mathcal{B}$, with $\mathcal{B}$ nonempty (equivalently π not Clifford), and diagonal shadow D_π of cubic radical dimension d . Then $\delta(D_\pi) \geq 2(n - d) + 1$, computable from $\mathcal{B}$ in $O(n^3)$ word operations (Appendix A).*

Proof. D_π is pure-cubic with tensor supported on $\mathcal{B}$, so Proposition 32 applies. $\square$

By Proposition 32 the stabilizer nullity of the shadow is $\nu = n - d$, so the floor is $2\nu + 1$. For a non-semi-Clifford π it bounds the shadow only. The permutation’s own $\delta_T(\pi)$ is open, since Proposition 82 applies only to circuits that implement a diagonal gate. The first known non-semi-Clifford member of C_3 is the Gottesman–Mochon gate G , reported in [21], which He–Robitaille–Tan identify (their Proposition 5.8) as Clifford-conjugate to the smallest gate U_3 of their non-semi-Clifford family.

Proposition 65 (a bracket on the Gottesman–Mochon shadow). *Let U_3 be the gate above, Clifford-conjugate to the Gottesman–Mochon gate, and let D_{U_3} be its diagonal shadow, the product of CCZ on the six triples $\{1, 2, 3\}$, $\{1, 4, 5\}$, $\{2, 4, 6\}$, $\{3, 4, 7\}$, $\{2, 5, 7\}$, $\{1, 6, 7\}$. Then $d = 0$, $\nu = 7$ and*

$$15 \leq \delta(D_{U_3}) \leq 28.$$

Proof. The eighteen pairs inside the six triples are distinct, so the triples form a linear hypergraph, and they cover all seven qubits. Lemma 42 gives $d = 0$ and $\delta(D_{U_3}) \geq 2 \cdot 7 + 1 = 15$, and Proposition 32 gives $\nu = n - d = 7$. For the upper bound, expand each CCZ into its seven parities and merge. Every pair and every triple occurs once, contributing $18 + 6$ odd parities. The singleton $\{a\}$ occurs once for each triple containing a , which is three times for $a \in \{1, 2, 4, 7\}$ and twice for $a \in \{3, 5, 6\}$, so four singletons remain odd. The merged residue has weight $4 + 18 + 6 = 28$. $\square$

F The cubic catalogue, further application floors, and benchmark certification

This appendix collects the cubic families behind Table 4, two further application floors, and the benchmark certification. Each family's floor comes from evaluating the radical relation $\sum_a w_a T_{aqr} = 0$ at a well-chosen pair (q, r) .

Proposition 66 (the standard cubic families). *The radical and floor columns hold for the stated range of n , with R the radical and $d = \dim R$. The tightness column records the values established after the proof (Corollary 55, Remark 67) and by computation (Appendix L).*

Family	radical	floor	tightness
line $f_n = \sum_i x_i x_{i+1} x_{i+2}$	$R = 0, n \geq 5$	$\geq 2n + 1$	$= 2n + 1$ for $5 \leq n \leq 12$
wrapped cycle C_n ($n \geq 5$)	$R = 0, n \geq 5$	$\geq 2n + 1$	equality at $n = 5$, not $n = 6$
star $S_n = \sum_{0 < j < k < n} x_0 x_j x_k$	$R = 0, \text{ odd } n$	$\geq 2n + 1$	$= 2n + 1$, all odd n
complete cubic e_3	$R = 0$ odd n ; $\langle \mathbf{1} \rangle$ even	$\geq 2n + 1$ / $\geq 2n - 1$	$= 2n + 1$ odd n ; $= 2n - 1$ even n
complete e_k , level k	$R = 0 \iff n \equiv k \pmod{2}$	$\geq 2n$ / $\geq 2(n - 1)$	—

Proof. The cubic floors are Proposition 32, $2(n - d) + 1$, applied to the value of d computed below, and the e_k floor is the level- k bound $\delta_k \geq 2(n - d_k)$ of Proposition 70. Line. For an interior j take $(q, r) = (j - 1, j + 1)$: the only a making $\{a, j - 1, j + 1\}$ a consecutive triple is $a = j$, so $w_j = 0$. At the endpoint, $(1, 2)$ gives $w_0 + w_3 = 0$, and $w_3 = 0$ because 3 is interior when $n \geq 5$, so $w_0 = 0$. The other endpoint is symmetric. (At $n = 4$ the direction $w_0 = w_3 = 1$ survives and $d = 1$.) Cycle. Every index is interior, so the same pair $(j - 1, j + 1)$ forces $w_j = 0$ for every j . Star. A pair (j, k) with $j, k \geq 1$ is covered only by $\{0, j, k\}$, giving $w_0 = 0$. The pair $(0, j)$ then gives $\sum_{a \geq 1, a \neq j} w_a = 0$, so all w_a ($a \geq 1$) equal a constant c with $(n - 2)c = 0$, forcing $c = 0$ for odd n . Complete cubic. Here the relation at (q, r) is $\sum_{a \notin \{q, r\}} w_a = (\sum_a w_a) - w_q - w_r = 0$. Comparing (q, r) with (q, r') gives $w_r = w_{r'}$, so $w = c\mathbf{1}$ and the relation becomes $(n - 2)c = 0$ in $\mathbb{F}_2$. Odd n forces $c = 0$. Even n leaves c free, so $\mathbf{1} \in R$ and $d = 1$. Complete e_k . The same computation at a $(k - 1)$ -set Q forces all w_a equal to a constant c with $(n - k + 1)c = 0$, so the order- k radical vanishes exactly when $n \equiv k \pmod{2}$, and $k = 3$ is the previous case. $\square$

The star and e_3 rows hold with equality for every odd n by Corollary 55, since both families factor as a linear form times a quadratic with invertible reduced adjacency. For e_3 at even n the radical is $\langle \mathbf{1} \rangle$, and the construction of Remark 67 attains the floor $2n - 1$. For the line, exact decoding gives equality at $n = 5, 6$, and for $7 \leq n \leq 12$ a randomized descent finds representations of weight $2n + 1$ whose order- ≤ 3 moments match f_n (Appendix L). Equality for all n is open. The wrapped cycle attains its floor at $n = 5$ but not at $n = 6$, where $\delta(C_6) = 17 = 2n + 5$ (Table 4).

Remark 67 (the complete cubic at even n). In the notation of Proposition 54, at even n the complete cubic has $R = \langle \mathbf{1} \rangle$ and $d = 1$, and $\text{Adj} = J + I$ on N odd is singular of corank one, so Proposition 54 does not apply. The same construction with one column folded, $c_j = \mathbf{e}_j + \mathbf{e}_N$ for $j < N$ and $c_N = \mathbf{e}_1 + \cdots + \mathbf{e}_{N-1}$, yields the set

$$\left\{ \{i, n\} \right\}_{i=1}^{n-1} \cup \{[n] \setminus \{i, n\}\}_{i=1}^{n-1} \cup \{[n]\}$$

of weight $2n - 1$. It represents e_3 for every even $n \geq 4$: every singleton has moment n , every pair $n - 2$ or 2 and every triple $n - 3$ or 1, which are 0, 0 and 1 modulo 2. The gate is pure-cubic, so Proposition 32 gives the floor $2(n - 1) + 1 = 2n - 1$, and hence $\delta(e_3) = 2n - 1$ for every even n . Exact decoding agrees at $n = 4$ and $n = 6$, with 7 and 11.

F.1 Further application floors

Remark 68 (IQP sampling cores). An IQP circuit $H^{\otimes n} D H^{\otimes n}$ has the Clifford+ T count $\delta_T(D)$ of its diagonal core (Appendix E). When D is a non-degenerate cubic phase gate, Theorem 26 therefore bounds by $2n+1$ every circuit $H^{\otimes n} D' H^{\otimes n}$ for it with D' Hadamard-free. With internal Hadamards the bound persists in the regimes of Appendix H.

Remark 69 (brackets for $\text{GF}(2^k)$ multiplication, $k \leq 6$). In the polynomial basis of an irreducible polynomial, the merged canonical residue of U_k has a weight W_k , obtained by expanding each cubic monomial into its seven parities. It is the ceiling $w(F)$ of Proposition 30. A change of irreducible is a field isomorphism, hence a CNOT circuit, so $\delta(U_k)$ does not depend on the basis and the best such ceiling is the minimum over the irreducibles of degree k . With Proposition 56,

$$6k + 1 \leq \delta(U_k) \leq W_k, \quad (W_1, \dots, W_6) = (7, 17, 36, 58, 90, 123),$$

minimized over 2, 1, 2, 3, 6, 9 irreducibles, with the radical trivial (as at every k). The ceiling grows like the schoolbook $\Theta(k^2)$, while $\delta(U_k)$ is linear (Remark 61). For $k \geq 3$ better ceilings come from subquadratic multipliers optimized by FASTTODD, with T -count 23, 43, 61, 83 at $k = 3, \dots, 6$ [25, Tab. 3], and from AlphaTensor-Quantum, which reaches 39 at $k = 4$ and 77 at $k = 6$ (the table after Proposition 60). Each bounds $\delta(U_k)$ by the Clifford conjugation of Remark 57, so

k	1	2	3	4	5	6
isotropy floor $6k + 1$	7	13	19	25	31	37
best known	7	17	23	39	61	77

with the best known value exact at $k = 1$ and at $k = 2$ (by coset decoding, Remark 57).

F.2 Certifying the optimizers

This subsection is the record behind §2.3.4. The method uses PyZX 0.10.3's `phase_block_optimize`, PyZX's TODD pipeline, on each circuit of the Feynman suite (commit d2c382a2). The run takes every circuit with input T -count at most 400, one circuit per process under a four-minute cutoff. Of the 21 circuits left out, 18 exceed the T -count limit and three (`gf2^6_mult`, `gf2^7_mult` and `qcla_adder_10`) reach the cutoff. The scripts, logs and per-block records are in the repository [47] and are listed in Appendix L. TODD shuffles matrix columns at random, so each circuit is run at seeds 0 to 4. The tables report seed 4, selected as the seed whose outputs have the lowest total T count. Across the five seeds the floor certifies between 186 and 193 blocks and nullity between 111 and 114.

The optimized circuit is partitioned with PyZX's own greedy block rule. Each block's phase polynomial is bounded with the floor $2(n-d^*)-r$, or $2\nu+1$ on the pure-cubic locus, and exact coset decoding is added whenever a block has at most six active qubits. A certified block has T -count equal to δ of its diagonal, so for this block partition and these Hadamard positions no $\{\text{CNOT}, T\}$ resynthesis of the block uses fewer T gates. Every exact value falls inside the bracket. The twelve multipliers and `mod5_4` are the only circuits of the suite that are mismatch-free as a whole. On `mod5_4` a Hadamard on one wire turns the circuit into a CNOT network times a diagonal gate with $d = 2$, whose floor $2(5-2)+1 = 7$ is the optimum AlphaTensor-Quantum reports from a Z3 search [16].

In the $\text{GF}(2^k)$ multipliers every Toffoli targets the output register, which is never used as a control. The files group the Toffolis into two blocks, each conjugated by Hadamards on the output register, with the CNOT network of the modular reduction on that register between them.

Conjugated by those Hadamards the reduction network is again a CNOT network, so the circuit is a Hadamard layer, a CNOT network times a diagonal gate, and a Hadamard layer. It maps $|x, y, z\rangle \mapsto |x, y, Mz \oplus x*y\rangle$ for an invertible M , which is the multiplier up to a CNOT network on the output, and so it is Clifford-equivalent to the field oracle U_k (Remark 57). The floors of U_k therefore bound every circuit for the multiplier whose Hadamards stay in the two outer layers. The structure and the radical are checked for all twelve benchmark multipliers, $k = 4$ to 256 (Appendix L).

circuit	n	T in	T out	certified	by ν	$\sum$ floor	block total	max gap	decoded
adder_8	24	399	250	8/20	5	214	246	5	1(1/1)
barenco_tof_10	19	224	174	16/33	8	129	146	1	32(15/32)
barenco_tof_3	5	28	20	2/5	1	17	20	1	5(2/5)
barenco_tof_4	7	56	42	4/9	2	33	38	1	9(4/9)
barenco_tof_5	9	84	64	6/13	3	49	56	1	13(6/13)
csla_mux_3	15	70	45	4/5	1	43	45	2	0(0/0)
csum_mux_9	30	196	76	0/3	0	73	76	1	0(0/0)
fprenorm	10	112	94	12/17	5	83	92	2	10(9/10)
gf2^4_mult	12	112	50	0/1	0	25	50	25	0(0/0)
gf2^5_mult	15	175	82	0/1	0	31	82	51	0(0/0)
grover_5	9	336	223	29/49	27	185	205	1	47(29/46)
ham15-low	17	161	123	13/21	3	111	119	1	20(13/19)
hwb6	7	105	80	8/14	2	70	78	2	11(6/9)
mod5_4	5	28	8	2/2	1	8	8	0	2(2/2)
mod_mult_55	9	49	29	2/4	1	22	27	4	3(1/2)
mod_red_21	11	119	83	12/16	3	76	81	2	14(12/14)
qcla_com_7	24	203	116	3/8	2	97	116	11	0(0/0)
qft_4	5	69	67	34/37	31	64	67	1	37(34/37)
rc_adder_6	14	77	52	8/12	4	48	52	1	5(4/5)
tof_10	19	119	87	15/18	8	84	87	1	12(10/12)
tof_3	5	21	17	2/4	1	15	17	1	4(2/4)
tof_4	7	35	27	3/6	2	24	27	1	6(3/6)
tof_5	9	49	37	5/8	3	34	37	1	7(5/7)
vbe_adder_3	10	70	32	5/5	0	28	28	0	1(1/1)
all 24				193/311	113	1563	1800		239

Table 5: Every circuit of the suite that PyZX’s phase-block optimizer finishes within the cutoff, at seed 4. The ten larger multipliers and `cycle_17_3`, `ham15-high`, `ham15-med`, `hwb8`, `hwb10`, `hwb11`, `hwb12`, `mod_adder_1024`, `mod_adder_1048576`, `qcla_mod_7` and `qcla_adder_10` were skipped for size or reached the cutoff. T in and T out are the counts before and after optimization. Certified counts blocks whose floor equals their count, and by ν those that stabilizer nullity alone certifies. The block total is the output’s count after parities are merged inside each block. Max gap is the largest per-block difference between that count and the floor. The last column has the form $a(b/c)$: a blocks small enough to decode exactly, b of them at the floor and c of them at the optimizer’s count, so $a - c$ blocks were left suboptimal.

Since the floor is never below ν (Corollary 35), every block that nullity certifies is certified by the floor too. Exact decoding finds five suboptimal blocks, one each in `grover_5`, `ham15-low` and `mod_mult_55` and two in `hwb6`, and saves 8 T gates on them. The two largest gaps, 51 on `gf2^5_mult` and 25 on `gf2^4_mult`, are on multipliers, where the single block is Clifford-equivalent to the field oracle U_k and its floor is $6k + 1$ (Proposition 56).

Table 6 gives the distance from the floor to the exact T -count on every block small enough to decode, for both optimizers (discussed in §2.3.4). On TODD’s output the total gap between block

counts and floors is 237 T gates. It splits as 81 of floor slack on the 239 decoded blocks, 8 of excess over the optimum on the five suboptimal ones, and 148 on the 72 blocks of 7 to 28 active qubits that decoding cannot reach, of which 35 carry a nonzero gap.

optimizer	decoded blocks	distance 0	distance 1	distance 2
TODD (seed 4)	239	159	79	1
phase teleportation	276	221	47	8

Table 6: The distance from the floor to the exact T -count, over every block with at most six active qubits, where exact coset decoding is affordable. Distance 0 means the floor is exact on the block.

The certificate does not depend on the optimizer. Phase teleportation [19], PyZX’s `teleport_reduce`, changes only the location and value of phases, so its output keeps the input’s Hadamard layering and is bounded with the same block rule (Table 7). It is deterministic and matches the input’s unitary on the three small circuits checked. TODD’s outputs were not checked for unitary equivalence. On the 24 circuits both optimizers finish, its output has 1607 T gates against TODD’s 1878, and the floor certifies 248 of 318 blocks against 193 of 311.

circuit	n	T in	T out	certified	by ν	$\sum$ floor	decoded
adder_8	24	399	173	15/19	5	163	7(6/7)
barenco_tof_10	19	224	100	33/33	23	100	32(32/32)
barenco_tof_3	5	28	16	5/5	2	16	5(5/5)
barenco_tof_4	7	56	28	9/9	5	28	8(8/8)
barenco_tof_5	9	84	40	13/13	8	40	12(12/12)
csla_mux_3	15	70	62	1/5	1	50	1(0/1)
csum_mux_9	30	196	84	0/3	0	73	1(1/0)
fprenorm	10	112	94	7/17	6	80	16(6/16)
gf2^4_mult	12	112	68	0/3	0	36	1(0/1)
gf2^5_mult	15	175	115	1/5	1	66	0(0/0)
gf2^6_mult	18	252	150	0/5	0	68	1(0/1)
gf2^7_mult	21	343	217	1/7	1	100	1(0/1)
grover_5	9	336	166	44/48	34	162	48(44/48)
ham15-low	17	161	97	12/22	5	87	22(12/22)
hwb6	7	105	75	6/14	1	67	14(6/14)
mod5_4	5	28	8	0/1	0	7	1(1/0)
mod_mult_55	9	49	35	2/5	1	31	5(2/4)
mod_red_21	11	119	73	12/17	5	68	15(10/15)
qcla_adder_10	36	238	162	3/7	1	149	0(0/0)
qcla_com_7	24	203	95	7/8	1	94	0(0/0)
qft_4	5	69	67	32/37	30	62	37(32/37)
rc_adder_6	14	77	47	11/12	5	46	10(9/10)
tof_10	19	119	71	17/18	1	70	17(16/17)
tof_3	5	21	15	3/4	1	14	4(3/4)
tof_4	7	35	23	5/6	1	22	6(5/6)
tof_5	9	49	31	7/8	1	30	8(7/8)
vbe_adder_3	10	70	24	6/6	2	24	4(4/4)
all 27			2136	252/337	141	1753	276

Table 7: The same certificate on the output of phase teleportation. Columns as in Table 5. The block total equals T out on every row and is omitted.

G Higher levels and other gate sets

This appendix extends the floor in two directions: to every level of the Clifford hierarchy, and to the gate sets Clifford+CS and Clifford+CCZ.

G.1 Higher levels of the Clifford hierarchy

For pure level- k gates the isotropy floor holds at every level k (phase mod 2^k , with diagonal hierarchy gates classified in [39]), but its tightness results are specific to $k = 3$. Past $k = 3$ the radical floor $2(n - d_k)$ is not tight in general (Proposition 72), for the structural reason of Remark 74. Throughout, for $k \geq 4$ the count δ_k is Amy–Mosca’s $R_Z(2\pi/2^k)$ -count [3], minimum-distance decoding in $\text{RM}(n - k - 1, n)^*$, which at $k = 3$ is the T -count and at $k = 4$ the $\sqrt{T}$ -count over $\{\text{CNOT}, \sqrt{T}\}$ with lower-level phases free. For $k \geq 4$ it is therefore not the T -count, and every statement below about δ_k concerns this count.

Proposition 70 (the floor at level k , unconditional). *Let U be a pure level- k diagonal gate (order- $\leq (k - 1)$ moments vanish) and d_k the radical dimension of its order- k moment tensor. Then $\delta_k(U) \geq 2(n - d_k)$.*

Proof. Every residue realizing U carries U ’s order- $\leq k$ moments (residues differ by codewords of $\text{RM}(n - k - 1, n)^*$, which have vanishing moments), so its columns are isotropic and $V^\perp$ lies in the order- k radical. The argument of Theorem 26 then gives $|\mathcal{S}| \geq 2 \text{rank } Y \geq 2(n - d_k)$. Completeness of the order- $\leq k$ moments enters only the matching upper bound, never the floor. $\square$

Proposition 71 (the nullity identity is cubic). *For the natural pure level- k gate $U = (-1)^f$, f homogeneous of degree k : $A \subseteq R$, hence $\nu \geq n - d_k$, and Proposition 70 gives $\delta_k \geq 2(n - d_k)$. Each single-qubit diagonal non-Clifford rotation (T , $\sqrt{T}$ or $R_Z(2\pi/2^k)$) raises the nullity by at most one, so ν bounds the total number of such gates. At $k = 3$, $A = R$ and $\nu = n - d$ (Proposition 32). For $k \geq 4$ the inclusion can be strict. For $f = e_4$ on $n = 5$, $A = \{0\}$ and $\nu = 5 > 4 = n - d_4$.*

Proof. $a \in A$ iff $\Delta_a f$ is affine, which forces in particular the top slice $T(a, \cdot, \dots, \cdot)$ to vanish, so $A \subseteq R$. At $k = 3$ the degree-2 part of $\Delta_a f$ is that slice, so its vanishing leaves an affine function and $R \subseteq A$. At $k \geq 4$ the intermediate degrees of $\Delta_a f$ must vanish as well, and this can fail. For $f = e_4$ on $n = 5$ and $a = \mathbf{1}$, every 3-set lies in an even number of 4-sets, so $\mathbf{1} \in R$. Yet $\Delta_{\mathbf{1}} e_4 = 1 + e_2$ has degree 2, so $\mathbf{1} \notin A$. $\square$

The two bounds of Proposition 71 count different things. The floor $2(n - d_k)$ bounds δ_k , in which lower-level phases are free, while ν bounds the number of $\sqrt{T}$ and T gates together. At $k = 3$ both count T gates and the isotropy floor always dominates (Corollary 35). For $k \geq 4$ the floor dominates whenever $d_k \leq n/2$, since $\nu \leq n \leq 2(n - d_k)$, while a homogeneous quartic on $n = 10$ qubits with $d_4 = 6$ has $\nu = 9 > 8 = 2(n - d_4)$ (Appendix L).

Proposition 72 (the radical floor is not tight in general past $k = 3$). *On its $n = k$ qubits, $\delta_k(C^{k-1}Z) = 2^k - 1$ while the floor is $2(n - d_k) = 2k$: the slack $2^k - 2k - 1$ ($= 7, 21, 51$ at $k = 4, 5, 6$) is unbounded in k . At $k = 4$, C^3Z costs 15 $\sqrt{T}$ gates over $\{\text{CNOT}, \sqrt{T}\}$ with lower-level phases free, and at least 15 over $\{\text{CNOT}, \sqrt{T}^{\pm 1}\}$.*

Proof. By the carry identity the residue of $C^{k-1}Z$ is all-ones on the $2^k - 1$ nonzero parities, and at $n = k$ the code of relations is $\text{RM}(-1, k)^* = \{0\}$, so the representation is unique and $\delta_k = 2^k - 1$ with no completeness hypothesis. The order- k tensor is the single top monomial, so $d_k = 0$. The bound over $\{\text{CNOT}, \sqrt{T}^{\pm 1}\}$ holds because that gate set has fewer free gates. $\square$

At $k = 3$ the formula gives a slack of 1, which the $+1$ of Proposition 32 closes.

Corollary 73 (multi-controlled gates). *At $n = 3$ the Toffoli/CCZ T -count is $7 > 2\nu = 6$. For $n \geq 4$, $C^{n-1}Z$ has $\delta_n = 2^n - 1$ and nullity $\nu = n$. At $n = 4$ this is the $\sqrt{T}$ -count 15 of C^3Z .*

Proof. $\delta(\text{CCZ}) = 7$ is Theorem 40 at $m = 1$, and the same value holds for ancilla-free Clifford+ T circuits by exhaustive search [2]. Also $\nu(\text{CCZ}) = 3$. For $n \geq 4$, $\delta_n(C^{n-1}Z) = 2^n - 1$ is Proposition 72 at $k = n$, and $\nu = n$ because no nonzero shift makes the derivative of $x_1 \cdots x_n$ affine. $\square$

For $n \geq 4$ this concerns δ_n . The Clifford+ T T -count of $C^{n-1}Z$, whose circuits need internal Hadamards, is a different quantity and is open (Appendix K).

Remark 74 (why the cubic level is special). The isotropy argument uses that a derivative lowers the degree by one. For a cubic phase the quadratic part of $\Delta_a f$ is the slice $T(a, \cdot, \cdot)$, and quadratic forms over $\mathbb{F}_2$ are classified by rank and Arf invariant [38], so each slice is determined up to equivalence by efficiently computable data. At degree four the corresponding slice is an alternating trilinear form. These are classified only in small dimension, over $\mathbb{C}$ in dimension 9 [48] and over $\mathbb{F}_2$ up to $n = 10$ [40, 37], and their isomorphism problem is believed hard enough to support signature schemes [49]. The floor $2(n - d_k)$ holds at every level, but a tight radical-based bound at $k \geq 4$ would need such a classification.

G.2 Other gate sets: Clifford+CS and Clifford+CCZ

Stabilizer nullity does not rescale with the generator, since $\nu(T) = 1$ and $\nu(\text{CS}) = 2$ while their residues have weights 1 and 3. The moment floor does rescale with the generator. A non-Clifford diagonal generator contributes a residue of some fixed maximal weight, and the floor divides by that weight. We carry this out for two common gate sets, CS (weight 3) and CCZ (weight 7).

Proposition 75 (CS-count floor). *For a diagonal level-three gate U : any Hadamard-free $\{\text{CNOT}, \text{CS}\}$ circuit for U has CS-count $\geq \delta(U)/3$, unconditionally. For measurement-free Clifford+CS circuits with clean ancillas and internal Hadamards, the same floor holds in the regimes proved in Appendix H, and CS-count $\geq (\delta(U) - \bar{\delta}(\kappa))/3$ holds unconditionally for any computable upper bound $\bar{\delta}(\kappa) \geq \delta(\kappa)$ (Proposition 91).*

Proof. A CS acting on the parities $u \cdot x$, $v \cdot x$ applies the phase $2(u \cdot x)(v \cdot x) = (u \cdot x) + (v \cdot x) - ((u \oplus v) \cdot x)$ (the carry identity), i.e. it contributes the odd residue $\{u, v, u \oplus v\}$, the nonzero points of a projective line. We call such a set a *triangle*. A Hadamard-free circuit with c CS gates therefore realizes a residue that is a symmetric difference of c triangles, of Hamming weight at most $3c$. By (9), $\delta(U) \leq 3c$. With internal Hadamards, replace each CS by its standard Hadamard-free 3- T gadget ($T_a T_b \text{CNOT } T_b^\dagger \text{CNOT}$): this preserves the Hadamard structure of the circuit, yielding a measurement-free Clifford+ T circuit with $3c$ T -gates to which Appendix H applies. $\square$

Write $\delta_{\text{CS}}(U)$ and $\delta_{\text{CCZ}}(U)$ for the least CS-count and CCZ-count of a Hadamard-free $\{\text{CNOT}, \text{CS}\}$ or $\{\text{CNOT}, \text{CCZ}\}$ circuit for U , with the diagonal Clifford gates S , CZ and Z free. This is the setting in which Propositions 75 and 77 hold unconditionally. The corollaries below concern these quantities, and they extend to circuits with internal Hadamards in the regimes of Appendix H.

Corollary 76 (CS-counts of CCZ layers). $\delta_{\text{CS}}(\text{CCZ}) \geq \lceil 7/3 \rceil = 3$, matching the known optimal 3-CS Toffoli [50], and a layer of m disjoint CCZ gates has $2m + 1 \leq \delta_{\text{CS}} \leq 3m$.

Proof. The lower bounds are $\lceil 7/3 \rceil$ and $\lceil (6m + 1)/3 \rceil$ by Theorem 40, and the upper bound is m separate conversions. $\square$

Proposition 77 (CCZ-count floor). *For a diagonal level-three gate U : any Hadamard-free $\{\text{CNOT}, \text{CCZ}\}$ circuit for U has $\text{CCZ-count} \geq \delta(U)/7$, unconditionally. For measurement-free Clifford+CCZ circuits with clean ancillas and internal Hadamards, the same floor holds in the regimes proved in Appendix H, and $\text{CCZ-count} \geq (\delta(U) - \bar{\delta}(\kappa))/7$ holds unconditionally for $\bar{\delta}(\kappa)$ as in Proposition 75.*

Proof. A CCZ acting on the parities $u \cdot x, v \cdot x, w \cdot x$ applies the phase $4(u \cdot x)(v \cdot x)(w \cdot x)$, whose carry expansion is supported on the seven combinations $\{u, v, w, u \oplus v, u \oplus w, v \oplus w, u \oplus v \oplus w\}$, a residue set of weight at most 7 (the nonzero points of the span of u, v, w). A circuit with c CCZ gates realizes a symmetric difference of c such sets, of weight at most $7c$, so $\delta(U) \leq 7c$ by (9). With internal Hadamards, replace each CCZ by its Hadamard-free 7- T gadget, giving a Clifford+ T circuit with $7c$ T -gates to which Appendix H applies. $\square$

Corollary 78 (the Toffoli layer cannot compress in CCZ-count below $m = 8$). *For m disjoint CCZ gates, $\delta_{\text{CCZ}} \geq \lceil (6m + 1)/7 \rceil = m - \lfloor (m - 1)/7 \rfloor$. For $m \leq 7$ this equals m , and the trivial construction uses m , so $\delta_{\text{CCZ}}(\bigoplus_{i=1}^m \text{CCZ}_i) = m$. Likewise $\delta_{\text{CCZ}} \geq \lceil (4m + 3)/7 \rceil$ for the fan-out layer and $\delta_{\text{CCZ}}(U_2) \geq \lceil 17/7 \rceil = 3$ for the GF(4) multiplier (Remark 57).*

In CCZ-count the layer therefore admits no saving over separate gates until $m = 8$. These are gate counts for circuits, a different question from the state-conversion rates of [28] and the catalyzed $\text{CCZ} \rightarrow 2T$ factory of Gidney and Fowler [51].

H Internal Hadamards

The preceding appendices bound the T -count of *Hadamard-free* $\{\text{CNOT}, T\}$ circuits, where the gate is the phase polynomial (8). A general Clifford+ T circuit may use internal Hadamards, and then $\delta(U)$ is only an upper bound on the optimal T -count of U . Branches can interfere and produce the same diagonal phase with fewer T -gates, so the $\{\text{CNOT}, T\}$ floor does not transfer unconditionally. With the branch variables treated as indeterminates, the least T -count over circuits with a fixed Hadamard skeleton becomes minimum-weight coset decoding in a punctured Reed–Muller code on $n + h$ variables [3], for which no polynomial-time algorithm is known, and even the number of wires needed for a cubic branch polynomial in Montanaro’s $\{H, Z, \text{CZ}, \text{CCZ}\}$ picture is NP-hard to compute [52]. This section reduces the gap to the single inequality $N_{\text{ker}} \geq \delta(\kappa)$ of Proposition 82 and proves it at maximal output rank (Corollary 83), hence for all circuits with at most two internal Hadamards, and for Clifford corrections.

Each internal Hadamard adds one branching variable, and the amplitude of a fixed output is a sum over the branches. For a diagonal gate on clean ancillas, the branches that return the input form an affine subspace, the fiber (Lemma 79). On a fiber, the phase splits into two kinds of parity. A kernel-free parity is constant along the fiber and is an ordinary $\{\text{CNOT}, T\}$ parity of the output, costing one T . A kernel-meeting parity varies along the fiber. Only these can interfere, and their interference is summarized by a single correction phase κ . The reduction gives $t \geq \delta(U) - \delta(\kappa) + N_{\text{ker}}$.

H.1 The path sum

Let W be a measurement-free Clifford+ T circuit on $n + a$ qubits with h internal Hadamards, implementing a diagonal level-three gate U on the n data qubits with *clean ancillas*: $W|x\rangle|0^a\rangle = \omega^{\phi(x)}|x\rangle|0^a\rangle$ for all $x \in \mathbb{F}_2^n$, where $U|x\rangle = \omega^{\phi(x)}|x\rangle$. Expanding each Hadamard as $H|b\rangle = 2^{-1/2} \sum_y (-1)^{by} |y\rangle$ introduces one *branch variable* $y_i \in \mathbb{F}_2$ per Hadamard and writes the circuit

as a sum over $y \in \mathbb{F}_2^h$:

$$W|x\rangle|0^a\rangle = 2^{-h/2} \sum_{y \in \mathbb{F}_2^h} \omega^{P(x,y)} |G(x,y)\rangle, \quad (13)$$

where P is a phase polynomial in (x,y) whose number of odd coefficients is at most the T -count t of W (each T contributes one), and G is an affine function of (x,y) giving the computational-basis output of each branch. Write $G(x,y) = M_G y + (\text{affine in } x)$ on the clean slice, with $M_G \in \mathbb{F}_2^{(n+a) \times h}$. Its rank $\rho_G = \text{rank } M_G$ is the *output rank* and $V = \ker M_G$ is the *kernel*, of dimension $d_F = h - \rho_G$ (we write d_F for this fiber dimension, keeping d for the radical).

Lemma 79 (branches landing on a fixed output). *On the clean slice the branches that produce the returning output $|x\rangle|0^a\rangle$ form a coset $z_0(x) + V$ for some affine $z_0 : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^h$, the fiber. Diagonality forces $\rho_G \leq h/2$, equivalently $d_F \geq h/2$.*

Proof. Clean return means $M_G y$ equals a fixed value (the affine x -part) for the surviving branches, a linear system whose solution set, when nonempty, is a coset of $V = \ker M_G$. A linear right inverse of M_G supplies one affine solution $z_0(x)$. Summing (13) over that fiber, the returning amplitude is $2^{-h/2} \sum_{v \in V} \omega^{P(x, z_0(x)+v)}$, a sum of 2^{d_F} unit terms, and it must have modulus 1 (diagonality). A sum of N unit terms has modulus $\leq N$, so $2^{-h/2} \cdot 2^{d_F} \geq 1$, i.e. $d_F \geq h/2$. $\square$

Write $R_x(v) := P(x, z_0(x) + v)$ for the phase along the fiber, a function $V \rightarrow \mathbb{Z}_8$.

H.2 The reduction

Split the odd parities of P on the clean slice by their behavior on V . Call a parity *kernel-free* if its branch part is constant on V (so it is the same on every point of the fiber $z_0(x) + V$), and *kernel-meeting* otherwise. Let N_0 and $N_{\ker}$ be the numbers of odd kernel-free and odd kernel-meeting parities, so that

$$t \geq N_0 + N_{\ker}, \quad (14)$$

since every odd parity is of exactly one kind. Evaluating the kernel-free parities, odd and even, along the section $z_0(x)$ gives a Hadamard-free $\{\text{CNOT}, T\}$ phase polynomial in x . Call its phase $\psi(x)$, a phase polynomial in affine forms of x with N_0 odd coefficients. Define the *correction* $\kappa := \phi - \psi : \mathbb{F}_2^n \rightarrow \mathbb{Z}_8$. Equivalently, factoring the constant kernel-free part out of the fiber sum,

$$\sum_{v \in V} \omega^{R_x(v) - \psi(x)} = 2^{h/2} \omega^{\kappa(x)},$$

so κ measures the phase the kernel-meeting content synthesizes by interference. We first check that κ is a legitimate target, so that $\delta(\kappa)$ in Proposition 82 is defined.

Lemma 80 (the correction is a level-three phase). *$\kappa(x) \in \mathbb{Z}_8$ for every x , and κ is the phase polynomial of a diagonal level-three gate. Hence $\delta(\kappa)$ is well-defined as its Amy–Mosca cost.*

Proof. By Lemma 79 the fiber sum has modulus $2^{h/2}$, so $\omega^{\kappa(x)}$ is a unit complex number. The phase ϕ of U and the phase ψ of the Hadamard-free representation read off the section are both $\mathbb{Z}_8$ -valued phase polynomials of degree at most three, and so is their difference κ . The unit phase is therefore the eighth root of unity $\omega^{\kappa(x)}$, and $\delta(\kappa)$ is the cost (9) of the gate with phase κ . $\square$

Lemma 81 (δ is subadditive). *For diagonal level-three gates with phases ϕ_1, ϕ_2 , $\delta(\phi_1 + \phi_2) \leq \delta(\phi_1) + \delta(\phi_2)$.*

Proof. Concatenate optimal $\{\text{CNOT}, T\}$ representations: the union of their parity multisets is a representation of $\phi_1 + \phi_2$, with odd-count at most the sum (merging only lowers it). $\square$

Proposition 82 (Hadamard reduction). *Let W be a measurement-free Clifford+ T circuit with clean ancillas implementing a diagonal level-three gate U , with any number of internal Hadamards. Then*

$$t \geq \delta(U) + (N_{\text{ker}} - \delta(\kappa)).$$

In particular $t \geq \delta(U)$ whenever $N_{\text{ker}} \geq \delta(\kappa)$.

Proof. The kernel-free parities, read on the section, are a $\{\text{CNOT}, T\}$ representation of ψ with N_0 odd coefficients, so $\delta(\psi) \leq N_0$. Since $\phi = \psi + \kappa$, Lemma 81 gives $\delta(\psi) \geq \delta(\phi) - \delta(\kappa) = \delta(U) - \delta(\kappa)$, hence $N_0 \geq \delta(U) - \delta(\kappa)$. Adding N_{ker} to both sides and using (14) gives $t \geq N_0 + N_{\text{ker}} \geq \delta(U) + (N_{\text{ker}} - \delta(\kappa))$. $\square$

Only N_{ker} and $\delta(\kappa)$ remain in the bound. The Amy-Mosca setting is the case $N_{\text{ker}} = 0$ (no Hadamards, $\kappa = 0$). The next two corollaries give regimes where $N_{\text{ker}} \geq \delta(\kappa)$ holds, so that $t \geq \delta(U)$ there.

Corollary 83 (maximal output rank and Clifford corrections). *Suppose $\rho_G = h/2$ (maximal output rank, which requires h even) or the correction is Clifford ($\delta(\kappa) = 0$, e.g. when no kernel-meeting parity is odd). Then $t \geq \delta(U)$. Moreover $\rho_G \geq 1$ whenever $h \geq 1$. Hence no clean diagonal circuit has $h = 1$, and every circuit with $h \leq 2$ satisfies $t \geq \delta(U)$.*

Proof. At $\rho_G = h/2$ the fiber sum has $2^{d_F} = 2^{h/2}$ terms and modulus $2^{h/2}$, so by the equality case of the triangle inequality R_x is constant on V . Then κ is a single $\{\text{CNOT}, T\}$ parity expression evaluated on the section, so $\delta(\kappa) \leq N_{\text{ker}}$. If no kernel-meeting parity is odd, $R_x(v) - \psi(x)$ is a sum of even parities, a $\mathbb{Z}_4$ -valued quadratic phase in (x, v) , and the fiber sum is a Gauss sum of it. Summing out the variables of v one at a time multiplies by a constant times $i^{\pm \ell}$ for a parity ℓ , or by $1 + (-1)^\ell$, which restricts the remaining variables to the affine subspace $\ell = 0$. The restriction cannot depend on x alone, since the fiber sum has modulus $2^{h/2}$ for every x (Lemma 79) and so never vanishes, and an affine reparametrization of the subspace keeps the phase $\mathbb{Z}_4$ -quadratic, so the elimination continues. A parity is quadratic modulo 4 because $u \oplus v = u + v - 2uv$. The value is therefore a constant times $i^{q(x)}$ with q quadratic, so κ is a diagonal Clifford phase and $\delta(\kappa) = 0$. If $\delta(\kappa) = 0$ the inequality is immediate. In both cases Proposition 82 gives $t \geq \delta(U)$. For $\rho_G \geq 1$: a branch variable can leave the affine output G only when a later Hadamard overwrites a wire whose content contains it, since the CNOTs act invertibly on the wire vector. The last Hadamard's branch variable therefore reaches G . Combined with $\rho_G \leq h/2$ (Lemma 79) this excludes $h = 1$, and for $h = 2$ it forces $\rho_G = h/2$. $\square$

The reduction allows any number of clean ancillas. When there are no internal Hadamards, the floor needs no hypothesis.

Corollary 84 (clean ancillas alone do not help). *Let W be a $\{\text{CNOT}, T\}$ circuit on $n+a$ qubits (no internal Hadamards) with $W|x\rangle|0^a\rangle = \omega^{\phi(x)}|x\rangle|0^a\rangle$ for all x , implementing the diagonal level-three gate U . Then its T -count satisfies $t \geq \delta(U)$.*

Proof. A $\{\text{CNOT}, T\}$ circuit applies a linear reversible permutation $L \in GL(n+a, \mathbb{F}_2)$ to the wires together with T -phases on intermediate parities. Clean return $W|x\rangle|0^a\rangle \propto |x\rangle|0^a\rangle$ forces $L(x, 0) = (x, 0)$ for all x , i.e. $L(x, c) = (x + Bc, Dc)$ for some B, D . Every parity the T -gates act on is an affine form in (x, c) . Restricting to the returning slice $c = 0$ turns each into an

affine form in x , so the phase read on that slice, $P(x, 0)$, is a $\{\text{CNOT}, T\}$ phase polynomial in x with at most t odd coefficients (parities that coincide or become even on $c = 0$ only lower the count). Since $W|x\rangle|0^a\rangle = \omega^{P(x,0)}|x\rangle|0^a\rangle = \omega^{\phi(x)}|x\rangle|0^a\rangle$, this is a representation of ϕ , whence $\delta(U) \leq \#\{\text{odd parities of } P(x, 0)\} \leq t$. $\square$

The standard use of a clean ancilla is to compute an AND into it, use it, and uncompute it. Under the block hypotheses of Corollary 86 below, the floor holds in this case. Call a stretch of an ancilla's wire between two consecutive Hadamards a *window* when the ancilla enters it in $|0\rangle$ (a compute window) or leaves it for $|0\rangle$ (an uncompute window), and call its branch variable the *window variable*.

Lemma 85 (an AND window costs four). *Let y be a window variable and y' the branch variable of the next Hadamard on the same wire, and suppose that summing over y forces $y' = g \oplus \ell$ for a non-affine Boolean function g of the other variables and an affine ℓ , that is, the phase depends on y only through $4y(y' \oplus g \oplus \ell)$. Then at least four odd parities contain y .*

Proof. Write the parities containing y as $y \oplus \lambda_p$ with coefficients c_p . Toggling y changes the phase by $\sum_p c_p(1 - 2\lambda_p)$, and this must equal $4(g \oplus \ell)$ plus the coupling $4y'$ modulo 8. Reducing modulo 2 shows that the number of odd c_p is even. Halving the rest of the congruence and reducing modulo 4, the even c_p contribute twice an affine form, and each odd c_p is ± 1 modulo 4, so $\sum_{c_p \text{ odd}} \pm \lambda_p \equiv 2(g + \text{affine}) + \text{const} \pmod{4}$. With no odd coefficient the right side has no non-affine part. With two, the relation modulo 2 forces $\lambda_2 = \lambda_1 \oplus c$, and $\pm \lambda_1 \pm \lambda_2$ is then twice an affine form plus a constant. Hence a non-affine g needs at least four. $\square$

Corollary 86 (AND-compiled circuits). *Let W be a measurement-free Clifford+ T circuit with clean ancillas implementing a diagonal level-three gate U , in which every internal Hadamard belongs to an AND block: a compute window that acts on two data-linear controls (bits that are affine functions of the data inputs, with linearly independent linear parts) and a clean ancilla as the phase-free AND $|a, b, 0\rangle \mapsto |a, b, ab\rangle$, and later an uncompute window, the reverse of a compute template, that returns the ancilla to $|0\rangle$. The compute may end with a diagonal Clifford on the ancilla after its closing Hadamard, as in the temporary AND of [20]. It is then the window together with that Clifford which acts as the phase-free AND, and the Clifford is a middle parity of even coefficient. An ancilla may be reused by a later block after its uncompute. For either window the phase depends on its window variable y only through $4y(y' \oplus y_{\text{in}} \oplus ab \oplus \ell)$, where y' is the next branch variable on the ancilla and ℓ is affine. Here y_{in} is the incoming value of the ancilla. It is 0 for a compute window on a fresh ancilla and the closing branch variable of the earlier uncompute window on a reused one. Inside a window the block's gates are CNOT, X and diagonal gates acting only on the ancilla and its two controls, no other gate touches these three wires, and the window returns the controls to their input values. The block's parities that contain no window variable carry even coefficients once compute and uncompute are combined. Suppose finally that every parity involving an ancilla's value between its two windows carries an even coefficient and involves at most one such value. Then, with A blocks,*

$$t \geq \delta(U) + A.$$

Proof. A window variable is carried by its ancilla, and by controls that the window restores, only for the length of its window, so the parities that contain it come from that window alone and merge with no other parity. By hypothesis the toggle of the phase in y is $4(y' \oplus y_{\text{in}} \oplus ab \oplus \ell)$, and $y_{\text{in}} \oplus \ell$ is affine, which is the hypothesis of Lemma 85 with the non-affine $g = ab$. Each of the two windows of a block, compute and uncompute, therefore carries at least four odd parities, and $t \geq 8A + t_0$,

h	ρ_G	status	reason
1	–	impossible	$\rho_G \geq 1$ (Corollary 83)
2	1	proved	maximal output rank
even $h \geq 4$	$h/2$	proved	maximal output rank
$h \geq 3$	$< h/2$	open	proved for Clifford corrections

Table 8: Status of the inequality $N_{\text{ker}} \geq \delta(\kappa)$ by stratum. For AND-compiled circuits in the open strata, Corollary 86 proves the conclusion $t \geq \delta(U)$ directly and leaves the inequality open.

with t_0 the number of odd parities containing no window variable and no middle value. Parities involving an ancilla’s middle value are even by hypothesis and do not count.

Summing over the window variables collapses each block onto the single branch in which the ancilla holds $g_i(x)$, the AND of its controls. The sum over y forces $y' = y_{\text{in}} \oplus ab \oplus \ell$. The compute writes ab into a $|0\rangle$ ancilla and the uncompute returns it to $|0\rangle$, so $\ell = 0$ on the branch. In particular the closing variable of an uncompute is 0 there, also when the ancilla is reused. Hence U ’s phase is the circuit’s phase polynomial evaluated with the window variables and these closing variables at 0 and each middle value at $g_i(x)$. On this branch the window parities at 0 need not vanish. Inside a window the controls carry a and b on entry and the opening Hadamard leaves the ancilla carrying y . The block’s gates in the window act on these three wires alone and are CNOT, X and diagonal gates, so every wire content in the window lies in $\text{span}\{1, a, b, y\}$. A window parity is therefore $y \oplus \lambda$ with $\lambda \in \text{span}\{1, a, b\}$, and at $y = 0$ it becomes λ . The block’s window-free parities inside its windows lie in the same span, so together with them the window parities at 0 form a phase in the block’s two controls alone. If the ancilla carries $y \oplus \lambda_c$ at the closing Hadamard, that Hadamard also leaves $4y'\lambda_c$ with $\lambda_c \in \text{span}\{1, a, b\}$. This is an even phase on the middle value and is treated with the middle parities below. What survives is the data part M_D of the rest of the circuit, whose odd count is at most t_0 , together with its even middle parities and these block phases. A parity $2c(g_i \oplus \ell)$ with ℓ data-affine expands as $2c g_i + 2c \ell - 4c g_i \ell$, so the middle content is a diagonal Clifford on the data plus, for each block, the phase $g_i(x)(2\alpha_i + 4L_i(x))$ with L_i data-affine. Together with the block phase above it depends on x through the two controls and L_i alone, three affine forms. It is a $\mathbb{Z}_8$ combination of parities of these forms, since the block phase is one, $2\alpha_i g_i = \alpha_i(a + b - a \oplus b)$ for controls a, b , and $4g_i L_i$ is a CCZ phase. A parity and its complement merge, so at most seven parities remain and the phase costs at most 7. By Lemma 81, $\delta(U) \leq t_0 + 7A \leq t - A$. $\square$

The block hypothesis holds for the Toffoli template $HCCZH$ and for the four- T temporary AND [20], and the bound is attained at $m = 1$ by the circuit of Remark 90. The bound holds on 500 random AND-compiled circuits. The hypotheses exclude AND chains, in which one AND controls the next, and odd phases on an AND value, and on 968 such random instances every circuit satisfies $t \geq \delta(U)$ (Appendix L).

Every lower bound of the preceding appendices on a diagonal level-three gate therefore holds against measurement-free Clifford+ T circuits with clean ancillas at maximal output rank, for Clifford corrections, for $h \leq 2$ and for AND-compiled circuits (Corollaries 83 and 86), and by Proposition 108 the $\{\text{CNOT}, T\}$ case extends to returned catalysts whose support has trivial translation stabilizer. The remaining strata of the inequality are indexed by h and the output rank ρ_G , with $1 \leq \rho_G \leq h/2$ (Table 8).

The open strata turn on how much of the kernel is active at once. Order the branch variables by the time their Hadamard fires, and fix the reduced echelon basis of V for that order. In it each basis vector has its pivot at its earliest nonzero coordinate, and every other basis vector vanishes

there, which makes the basis unique. A term of the path sum depends on a basis direction b when its linear form λ in the branch variables has $\lambda(b) = 1$. For a phase term λ is the branch part of its parity, and for a Hadamard it is the branch part of the linear content of its input wire. Call b live from the first to the last time step at which a term depends on it. The *frontier width* is the largest number of kernel directions live at one time. It is a cut-width of the branch variables, and it stays small in compiled arithmetic, where Hadamards are local and short-lived, even when h is large.

Conjecture 87 (width-one conservation). *Every measurement-free circuit of frontier width one with clean ancillas that implements a diagonal level-three gate satisfies $\delta(\kappa) \leq N_{\text{ker}}$, and hence $t \geq \delta(U)$ for every h .*

The conjecture holds on 2153 realizable width-one families (Appendix L). It must use realizability, as the following remark shows.

Remark 88 (realizability and two notions of width). Consider an abstract width-one fiber form $R_x(v) = c_0(x) + \sum_i \alpha_i(x)v_i + \sum_i 2\gamma_i(x)v_iv_{i+1}$, which need not come from a circuit. Let N_{ker} be the number of directions i whose field α_i is odd at some x , and read κ off the fiber sum. The width-one form with $\kappa = 2(x_0x_1 \oplus x_0x_2) = \text{CS}_{01}\text{CS}_{02}\text{CCZ}_{012}$ has $\delta(\kappa) = 3$ and arises with $N_{\text{ker}} = 0$ once the fields or couplings may depend non-affinely on x . Even fields suffice. Take two uncoupled directions with $\alpha_1 = 2$ and $\alpha_2 = 2 + 4(1 \oplus g(x))$, where $g = x_0(x_1 \oplus x_2)$. Neither field is ever odd, and the fiber sum is $(1+i)(1+i(-1)^{1 \oplus g}) = 2\omega^{2g}$, so $\kappa = 2g$. In a circuit neither an x -dependent field of this kind nor an x -dependent coupling is free. Every wire of a $\{\text{CNOT}, H, T\}$ path sum carries an affine function of the data and the branch variables, so a Hadamard contributes only a term $4z\ell$ with ℓ affine. A field $4f(x)v$ with f non-affine, or an x -dependent coupling of two branch variables, then comes from the cubic part of an odd parity containing the branch variables involved. That parity is kernel-meeting and counted by the N_{ker} of Proposition 82. A proof must therefore use how a circuit creates its branch variables. Ji, Liu and Zhou [52] study a related width in Montanaro’s picture [53], where a $\{H, Z, \text{CZ}, \text{CCZ}\}$ circuit is a set of *symbols*, one per Hadamard-delimited wire segment, and the circuit width of a cubic f over $\mathbb{F}_2$ is the fewest wires on which the symbols of some circuit realizing f can be packed. Both widths are interval-overlap quantities, and a tail symbol in their certificate is a branch variable that reaches G , the observation behind $\rho_G \geq 1$ in Corollary 83. Their f is a cubic polynomial over $\mathbb{F}_2$ with $P = 4f$, since every gate of $\{H, Z, \text{CZ}, \text{CCZ}\}$ contributes phases ± 1 only. The correction κ above contains the CS content $2x_0x_1 + 2x_0x_2$, which no $\{H, Z, \text{CZ}, \text{CCZ}\}$ circuit produces, as all four gates are real. The open stratum therefore concerns the mod-4 and mod-8 content of P , which the $\mathbb{F}_2$ polynomial does not record, and symbol packing alone cannot settle it.

Remark 89 (the quantities on one circuit). We trace the quantities of Proposition 82 on $\text{CCZ}(a, b, c)$ built from a Toffoli that writes ab into a clean ancilla u , a $\text{CZ}(u, c)$, and a second Toffoli that erases u (Figure 5).

Here $h = 4$. Each Toffoli is a CCZ between two Hadamards on u , and u must start and end in $|0\rangle$, so none of the four Hadamards can be pushed to the ends, and they introduce the branch bits $y_1, \dots, y_4$. The ancilla ends holding y_4 , so the fiber is $\{y_4 = 0\}$ with $d_F = 3$, and only y_4 reaches the output, so $\rho_G = 1$, below the maximum 2. The first CCZ acts on a, b and y_1 and contributes the seven parities of $\{a, b, y_1\}$, and the second contributes the same seven with y_3 in place of y_1 . The parities a, b and $a \oplus b$ appear twice, so their coefficients become even, with total phase $4ab$. The eight that survive each contain y_1 or y_3 , which the fiber leaves free, so $N_{\text{ker}} = 8$. No odd parity is constant, and the even constant ones give $\psi = 4ab + 2c$, the term $2c$ coming from $\text{CZ}(u, c) = 2u + 2c - 2(u \oplus c)$. So $\kappa = 4abc - 4ab - 2c$ differs from the phase of $\text{CCZ}(a, b, c)$ by a diagonal Clifford and $\delta(\kappa) = 7$. Proposition 82 then gives $t \geq 7 + (8 - 7) = 8$, the circuit’s T -count after merging.

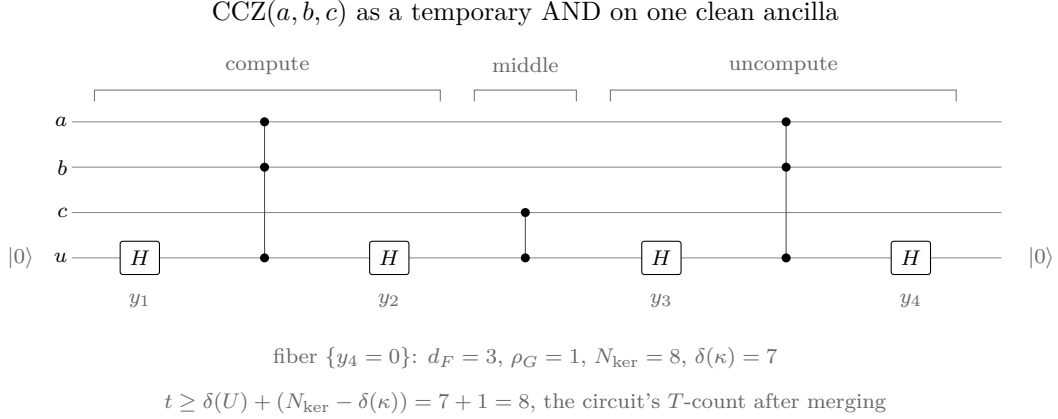


Figure 5: The standard ancilla-based CCZ circuit, built from a temporary AND on one clean ancilla, with the quantities of Remark 89.

Remark 90 (an ancilla-based CCZ circuit in the open stratum). The circuit of Remark 89 lies in the stratum $h = 4$, $\rho_G = 1$, which Table 8 lists as open, and m copies in parallel, implementing the layer of Theorem 40, lie in the stratum $h = 4m$, $\rho_G = m$. Its four Hadamards are internal, since the middle two have T -bearing content on both sides and the outer two act on an ancilla that must start and end in $|0\rangle$, so Lemma 15 removes none of them. The inequality holds on these circuits, with $N_{\text{ker}} = 8$ against $\delta(\kappa) = 7$ at $m = 1$ and 16 against 13 at $m = 2$, and the ripple-carry variant has $\delta(\kappa) = 0$. The circuit is AND-compiled, so Corollary 86 gives $t \geq \delta(U) + m = 7m + 1$ for every m , the 8 it uses at $m = 1$. This settles the family but not the stratum.

Without the hypothesis $N_{\text{ker}} \geq \delta(\kappa)$ a floor remains, reduced by a computed slack.

Proposition 91 (unconditional Hadamard floor with computed slack). *For any measurement-free Clifford+ T circuit with clean ancillas implementing a diagonal level-three gate U , with any number of internal Hadamards,*

$$t \geq \delta(U) - \delta(\kappa) \geq \delta(U) - \bar{\delta}(\kappa),$$

where $\bar{\delta}(\kappa)$ is any upper bound on $\delta(\kappa)$ computed from the path sum.

Proof. Proposition 82 gives $t \geq \delta(U) + (N_{\text{ker}} - \delta(\kappa))$ unconditionally. Drop $N_{\text{ker}} \geq 0$ to get $t \geq \delta(U) - \delta(\kappa)$, then bound $\delta(\kappa) \leq \bar{\delta}(\kappa)$. $\square$

A natural choice of $\bar{\delta}(\kappa)$ is the ceiling $W(s)$ of Proposition 30 applied to κ (Lemma 80). The correction $\kappa = \phi - \psi$ is computable in polynomial time, since ψ is read off the kernel-free parities on the section, and the bound needs no structural hypothesis. It gives $\delta(U)$ exactly when the correction is Clifford.

The two facts behind Corollary 86 do not use the level of the target. An AND chain and the phase versions of a select tree and a QROM are AND-compiled diagonal gates. Each is of level four or higher, so no floor on the gate applies, but the following theorem bounds the compilation class.

Theorem 92 (AND-compiled circuits at every level). *Let W be a measurement-free Clifford+ T circuit with clean ancillas whose internal Hadamards all belong to phase-free AND blocks as in Corollary 86, but with the controls of a block allowed to be earlier AND values and the middle allowed any $\{\text{CNOT}, T\}$ content on data and AND values. Suppose W implements a diagonal gate with phase ϕ . Then*

1. $t \geq 8A + \text{odd}(M)$, where A is the number of blocks and $\text{odd}(M)$ the number of odd parities of the circuit that contain no window variable.
2. if every control of a block is a data input or an earlier AND value, and the Möbius expansion of ϕ over $\mathbb{Z}_8$ has a term $4x_{i_1} \cdots x_{i_k}$ of degree k , then $A \geq k - 3$.

Proof. (i) is the first paragraph of the proof of Corollary 86, which used nothing about the target: window parities of different windows contain different branch variables, no other parity contains one, and Lemma 85 gives four per window. For (ii), the collapse of the window variables leaves $\phi(x) = M(x, g_1(x), \dots, g_A(x))$ with M the middle phase polynomial, where M also collects each block's own parities on that branch, which by the proof of Corollary 86 are parities in its two controls, and with single-wire controls every AND value g_i is a monomial in the data. Call the data inputs and the AND values atoms. A parity of M is an XOR of atoms, and over $\mathbb{Z}$ the XOR expands, by $u \oplus v = u + v - 2uv$, into products of s of its atoms with coefficients $\pm 2^{s-1}$. A term $4x_{i_1} \cdots x_{i_k}$ therefore comes from a product of $s \leq 3$ atoms equal to that monomial, contributed by a parity whose coefficient c satisfies $2^{s-1}c \equiv 4 \pmod{8}$: an odd c for $s = 3$, $c \equiv 2 \pmod{4}$ for $s = 2$, and $c \equiv 4 \pmod{8}$ for $s = 1$. Multiplying those atoms together takes at most two further AND gates, and a monomial of degree k needs $k - 1$ AND gates [54], so $A + 2 \geq k - 1$. $\square$

Corollary 93 (the class-restricted cost of $C^{k-1}Z$). *For $k \geq 4$, the minimum T -count t over measurement-free AND-compiled circuits for the k -qubit gate $C^{k-1}Z$ whose AND controls are data inputs or earlier AND values satisfies*

$$8k - 20 \leq t \leq 8k - 17,$$

the upper bound by an explicit circuit on $k - 3$ clean ancillas.

Proof. The lower bound: by Theorem 92 either $A \geq k - 2$, and then $t \geq 8(k - 2)$, or $A = k - 3$. In the latter case the count in the proof of Theorem 92(ii) is tight. The term $4x_1 \cdots x_k$ comes from three atoms a_1, a_2, a_3 whose product is $x_1 \cdots x_k$, and multiplying them with two further AND gates gives a circuit of $k - 1$ AND gates computing $x_1 \cdots x_k$. In such a circuit join the supports of the two controls of each AND gate by an edge between the variables. The support of every computed monomial is then connected by the edges of the gates below it, so the output needs $k - 1$ edges on its k variables, and with exactly $k - 1$ the edges form a spanning tree. Hence every AND gate joins two controls of disjoint supports and lies below the output, so a_1, a_2, a_3 have disjoint supports covering $[k]$, the $k - 3$ AND gates of the circuit are exactly those computing them, and every AND value lies inside the support of one of them. Hence $a_1 a_2 a_3$ is the only product of at most three atoms equal to $x_1 \cdots x_k$, and for each pair $\{i, j\}$ no single atom and no other pair of atoms equals $a_i a_j$. The block phases that M collects are functions of two controls inside the support of one atom, so their Möbius terms meet a single atom's support and do not enter the coefficients below. Compare Möbius coefficients with the target $4x_1 \cdots x_k$. Modulo 8 the coefficient of $x_1 \cdots x_k$ is $4n_{123}$, where n_{123} counts the odd parities of M containing all three atoms, so n_{123} is odd. Modulo 4 the coefficient of $a_i a_j$ is $2n_{ij}$, where n_{ij} counts the odd parities of M containing a_i and a_j , since products of three or more atoms carry a multiple of 4. The target coefficient is 0, so each n_{ij} is even and exceeds n_{123} , and for each of the three pairs some odd parity contains that pair but not the third atom. These three parities and the n_{123} parities are distinct, so $\text{odd}(M) \geq 4$ and $t \geq 8(k - 3) + 4$. The upper bound: chain $k - 3$ temporary ANDs to $g = x_1 \cdots x_{k-2}$, apply $\text{CCZ}(g, x_{k-1}, x_k)$, uncompute. This circuit uses $8(k - 3) + 7$ T -gates. It is correct for every k , since each temporary AND writes the AND of its two controls into a clean ancilla and the uncomputation restores it [20], and the correctness is also checked by simulation for $k \leq 6$ (Appendix L). $\square$

H.3 The Hadamard hypothesis and block-local estimates

The conclusion $t \geq \delta(U)$ that the hypothesis $N_{\text{ker}} \geq \delta(\kappa)$ delivers is, for layered circuits, equivalent to the correctness of the estimate that block-local T -count optimizers compute.

Block-local optimizers cut a Clifford+ T circuit at its Hadamard layers and write it as $W = B_\ell H^{\otimes n} \dots H^{\otimes n} B_1$ with each block B_i a $\{\text{CNOT}, T\}$ circuit. They optimize each block separately and report the sum. Write

$$E(W) := \sum_{i=1}^{\ell} \delta(B_i)$$

for that *layer-sum estimate*, the number such an optimizer returns. Here $\delta(B)$ of a $\{\text{CNOT}, T\}$ block is the fewest T -gates in a $\{\text{CNOT}, T\}$ circuit implementing the same unitary, and $\delta(U)$ stands for the true Hadamard-free optimum of the gate U the circuit implements.

Proposition 94 (the layer-sum estimate is safe exactly when the Hadamard hypothesis holds). *Fix a diagonal gate U and let W range over the H -layered circuits implementing it. Then*

$$E(W) \geq \delta(U) \text{ for every such } W \iff t(W) \geq \delta(U) \text{ for every such } W.$$

Proof. $t(W) \geq E(W)$ always, since each block's actual T -count is at least that block's own optimum. Hence $E(W) \geq \delta(U)$ implies $t(W) \geq \delta(U)$. Conversely, apply the right-hand side to a circuit realizing the block-local optimum in every block, for which $t(W) = E(W)$. $\square$

The statement that the block-local estimate never falls below $\delta(U)$ is therefore the Hadamard hypothesis for layered circuits, and the estimate is provably safe wherever that hypothesis is proved (Table 8). Outside those regimes Proposition 91, applied to the circuit W' whose blocks are replaced by their optima, still gives $E(W) = t(W') \geq \delta(U) - \delta(\kappa_{W'})$. Lemma 81 does not give $\delta(U) \leq E(W)$ directly, since a block between two Hadamard layers is not a diagonal gate.

I Approximation, measurement and catalysis

This appendix extends the floor to approximate synthesis (§I.1), adaptive measurement with feed-forward (§I.2) and borrowed catalyst states (§I.3). With h internal Hadamards, Appendix H proves $t \geq \delta(U)$ only under hypotheses on the branch phase κ of Proposition 82, called the correction there. We call the cases where these are proved the *proved regimes*: maximal output rank or $\delta(\kappa) = 0$ (Corollary 83, which covers every circuit with $h \leq 2$), and AND-compiled circuits (Corollary 86).

I.1 Resource estimation under approximation: an ε -rigidity

Fault-tolerant compilers stop at $\|U - V\| \leq \varepsilon$ [55, 28]. A Clifford+ T circuit with few T -gates either implements a diagonal target exactly, so that the exact floors apply, or misses it by an amount bounded below in terms of its T -count. The proof uses a Liouville-type gap in the ring $\mathbb{Z}[\omega, 1/\sqrt{2}]$.

Lemma 95 (denominator growth). *Every measurement-free Clifford+ T circuit V on N qubits with t T -gates has entries in $\frac{1}{\sqrt{2}^K} \mathbb{Z}[\omega]$ with $K = \text{sde}(V) \leq 2N(t+1)$, where sde is the smallest denominator exponent.*

The gap below is the standard denominator-exponent argument of exact synthesis [56, 57, 55].

Proof. Group the circuit as $C_0 T_1 C_1 \cdots T_t C_t$ with each C_i Clifford. A Clifford unitary on N qubits has a canonical circuit with at most two Hadamard stages [58], each of at most N Hadamards, and H is the only generator with a denominator $(\sqrt{2})$, so $\text{sde}(C_i) \leq 2N$. T has denominator 1, and $\text{sde}(AB) \leq \text{sde}(A) + \text{sde}(B)$, so $K \leq 2N(t+1)$. $\square$

Lemma 96 (Galois rigidity of ring entries). *Let V be a unitary with entries in $\mathbb{Z}[\omega, 1/\sqrt{2}]$ and $K = \text{sde}(V)$, and let v be an entry of V . Then for every eighth root of unity ω^j : either $v = \omega^j$, or $|v - \omega^j| \geq 2^{-(K+1)}$. Also either $v = 0$, or $|v| \geq 2^{-K}$.*

Proof. The Galois automorphism $\sigma : \omega \mapsto \omega^3$ of $\mathbb{Q}(\omega)$ maps $\sqrt{2} = \omega + \omega^{-1}$ to $-\sqrt{2}$ and commutes with complex conjugation ($x \mapsto \bar{x}$ is the element $\omega \mapsto \omega^7$ of the abelian Galois group), so $\sigma(V)\sigma(V)^\dagger = \sigma(VV^\dagger) = I$. The Galois image $\sigma(V)$ is again unitary, and in particular $|\sigma(v)| \leq 1$ for every entry. Set $\xi := \sqrt{2}^K v - \sqrt{2}^K \omega^j \in \mathbb{Z}[\omega]$, an algebraic integer. If $\xi \neq 0$ then $1 \leq |\text{Nm}(\xi)| = |\xi|^2 |\sigma(\xi)|^2$, while $|\sigma(\xi)| \leq \sqrt{2}^K (|\sigma(v)| + 1) \leq 2\sqrt{2}^K$. Hence $|\xi| \geq (2\sqrt{2}^K)^{-1}$ and $|v - \omega^j| = |\xi|/\sqrt{2}^K \geq 2^{-(K+1)}$. The $v \neq 0$ case is the same with $\xi = \sqrt{2}^K v$ and $|\sigma(\xi)| \leq \sqrt{2}^K$. $\square$

Theorem 97 (ε -rigidity dichotomy). *Let U be a diagonal level-three gate on n qubits and let V be a measurement-free Clifford+ T circuit on $N = n + a$ qubits with clean ancillas and t T -gates. Let $V_{\text{eff}} = (I \otimes \langle 0^a |) V (I \otimes |0^a \rangle)$ be its clean-slice block, and suppose $|(V_{\text{eff}})_{xx'} - U_{xx'}| \leq \varepsilon$ for all x, x' (implied by $\|U - V_{\text{eff}}\| \leq \varepsilon$). The hypothesis fixes the global phase, and a block close to $\omega^j U$ is covered by replacing ϕ with $\phi + j$. If*

$$\varepsilon < 2^{-(2N(t+1)+2)},$$

then $V_{\text{eff}} = U$. Consequently, for every ε ,

$$t \geq \min\left(\delta(U) - \bar{\delta}(\kappa), \frac{\log_2(1/\varepsilon) - 2}{2N} - 1\right),$$

where κ is the branch phase of V (Proposition 82), defined when V is exact, the only case in which the first term is used, and $\bar{\delta}(\kappa)$ is any computable upper bound on $\delta(\kappa)$ (Proposition 91).

Proof. By Lemma 95, $K = \text{sde}(V) \leq 2N(t+1)$, and each clean-slice entry v of V is an entry of a unitary over the ring. For a diagonal entry, the target is $\omega^{\phi(x)}$: by Lemma 96, $|v - \omega^{\phi(x)}| < 2^{-(K+2)} \leq 2^{-(K+1)}$ forces $v = \omega^{\phi(x)}$. For an off-diagonal entry the target is 0 and $\varepsilon < 2^{-K}$ forces $v = 0$. Hence below the stated radius $V_{\text{eff}} = U$ entrywise. Since V is unitary and U is unitary, this gives $V|x\rangle|0^a\rangle = \omega^{\phi(x)}|x\rangle|0^a\rangle$, the exact hypothesis of Appendix H. Proposition 82 and Proposition 91 then give $t \geq \delta(U) - \bar{\delta}(\kappa)$. Otherwise $\varepsilon \geq 2^{-(2N(t+1)+2)}$, which rearranges to the second branch. $\square$

The theorem measures phase-sensitive closeness up to an eighth root of unity. It covers a block close to $\omega^j U$, but not a block close only to $e^{i\theta} U$ for a general θ , which is the global-phase-insensitive distance that compilers usually report. We call $2^{-(2N(t+1)+2)}$ the *exactness radius*. Below it V implements U exactly, so every floor of this paper applies to V in the regimes where Proposition 82 is proved, and there the first branch improves to $\delta(U)$. Outside those regimes only $t \geq \delta(U) - \bar{\delta}(\kappa)$ is unconditional.

Corollary 98 (robust nullity). *In the proved regimes, any measurement-free Clifford+ T circuit ε -approximating a non-Clifford pure-cubic U satisfies*

$$t \geq \min\left(2\nu(U) + 1, \frac{\log_2(1/\varepsilon) - 2}{2N} - 1\right).$$

In the proved regimes this gives $t \geq 4m + 3$ for a fan-out layer of m Toffolis and $t \geq 6m + 1$ for a parallel Toffoli layer, for every circuit within the exactness radius. The $\log(1/\varepsilon)$ shape of the second branch matches the known single-qubit bounds [28, 55]. The per-qubit constant $\frac{1}{2N}$ is not tight and degrades with ancillas, whereas the constant $\frac{1}{7}$ of [28] does not depend on N .

Remark 99 (the crossover accuracy, and randomized constructions). Logarithmic scaling is achievable for structured targets by randomized constructions, which lie outside the theorem. Gosset, Kothari and Zhang [27] implement the n -controlled Toffoli to diamond-norm error ε with $O(\log(1/\varepsilon))$ T -gates, independent of n , and show that a purely unitary implementation needs $\Omega(n)$ T -gates even at constant error. For the parallel Toffoli layer on its $N = 3m$ wires without ancillas (with a ancillas replace N by $3m + a$), any measurement-free circuit with $t \leq 6m$ T -gates has exactness radius $2^{-(2N(t+1)+2)} \geq 2^{-(36m^2+6m+2)}$, so for

$$\varepsilon < \varepsilon_m^* := 2^{-(36m^2+6m+2)}$$

every approximating circuit has $t \geq 6m + 1$ (in the proved regimes), while above ε_m^* only the logarithmic branch applies. The constructions of [27] are classical mixtures of Clifford+ T circuits whose average is close to the target while every member is far from it. A mixture is a channel, so they do not contradict Theorem 97, which concerns a single unitary. Theorem 106 covers channels but requires Choi entries in $\mathbb{Z}[\omega, 1/\sqrt{2}]$, which excludes mixtures with non-dyadic weights.

I.2 The adaptive model: measurement and feedforward

The floors so far hold for measurement-free circuits: exactly for $\{\text{CNOT}, T\}$ (Appendices B–D), and through internal Hadamards in the proved regimes. Fault-tolerant architectures also allow *mid-circuit measurement with classical feedforward*, the mechanism of gate teleportation and magic-state injection. For a non-Clifford pure-cubic gate the three models and their floors are as follows. The first two floors are proved for circuits without internal Hadamards and in the regimes of Appendix H. With internal Hadamards the second also assumes that the ancillas end in a computational-basis state.

$$\underbrace{\text{measurement-free}}_{t \geq 2\nu+1} \subseteq \underbrace{\text{non-adaptive measurement}}_{t \geq 2\nu+1} \subseteq \underbrace{\text{adaptive feedforward}}_{t \geq \nu}$$

Measurement without feedforward leaves the floor unchanged (Proposition 102), Pauli feedforward with a stabilizer record keeps it (Proposition 104), and non-Pauli feedforward can break it (Proposition 101). Every adaptive protocol is bounded by the stabilizer nullity of the *Choi state* $|J_U\rangle = 2^{-n/2} \sum_x \omega^{\phi(x)} |x\rangle|x\rangle$, a magic monotone non-increasing under stabilizer operations, post-selection and classical feedforward [28, 14]. Its channel-level version is the Pauli spectrum of the induced channel [59].

Proposition 100 (the Choi-state nullity is $n - d^*$). *For any unitary U the state stabilizer nullity of $|J_U\rangle$ equals the unitary stabilizer nullity $\nu(U)$ [14, Thm. 22]. Consequently every adaptive implementation of a diagonal level-three gate satisfies*

$$t_{\text{ad}}(U) \geq \nu(U) = n - d^*.$$

Proof. The identity is [14]. We include the short argument. Write $|J_U\rangle = (U \otimes I)|\Omega\rangle$ with $|\Omega\rangle = 2^{-n/2} \sum_x |x\rangle|x\rangle$. For Paulis P, Q on n qubits, $(P \otimes Q)|\Omega\rangle = (PQ^\top \otimes I)|\Omega\rangle$, so

$$(P \otimes Q)|J_U\rangle = (PU \otimes Q)|\Omega\rangle = (PUQ^\top \otimes I)|\Omega\rangle,$$

which equals $\pm|J_U\rangle = \pm(U \otimes I)|\Omega\rangle$ iff $U^\dagger P U = \pm Q^\top$. Thus a Pauli $P \otimes Q$ stabilizes $|J_U\rangle$ iff $U^\dagger P U$ is (proportional to) a Pauli, with Q then determined by P . The stabilizer group of $|J_U\rangle$ is in bijection with $\{P : U^\dagger P U \text{ is Pauli}\}$, whose size defines $\nu(U)$. For diagonal U every Z -string lies in this set and X^a does iff a lies in the shift group A , so up to phases the set is $\{Z^z X^a : z \in \mathbb{F}_2^n, a \in A\}$, of size $2^n|A|$, and $\nu(U) = 2n - \log_2(2^n|A|) = n - \log_2|A|$.

For the value, Proposition 34 identifies the shift group with the full-tensor radical $A = R^*$ for every diagonal level-three gate, so $\nu = n - d^*$, and on the pure-cubic locus $R^* = R$ (Proposition 32). $\square$

In the exact $\{\text{CNOT}, T\}$ model the isotropy floor of a pure-cubic gate is $2\nu + 1$, about twice the adaptive floor ν . This gap is the factor of two of §1.

Proposition 101 (feedforward breaks the factor of two). *No bound $t_{\text{ad}} \geq c\nu - K$ with constants $c > 4/3$ and K holds for adaptive Clifford+ T implementations of all pure-cubic gates. In particular $t_{\text{ad}} \geq 2\nu$ fails.*

Proof. In the exact model $\delta(\text{CCZ}) = 7 = 2\nu + 1$ (Theorem 40 at $m = 1$). The Jones gate-teleportation gadget [12] builds CCZ from four T -gates using one measured ancilla and a conditional Clifford correction, so $t_{\text{ad}}(\text{CCZ}) \leq 4$. Since $\nu(\text{CCZ}) = 3$, this gives $t_{\text{ad}}(\text{CCZ}) < 3c = c\nu(\text{CCZ})$ for every $c > 4/3$, and $c = 2$ is the factor of two. Applying the gadget to each gate of a layer of m disjoint CCZ gates gives $t_{\text{ad}} \leq 4m$, while $\nu = 3m$ by additivity [14], so $4m \geq 3cm - K$ fails for large m whenever $c > 4/3$. $\square$

Whether some multiple $c\nu$ with $1 < c \leq 4/3$ holds under feedforward is open. The Jones correction is a CZ. It is Clifford, so the correction needs no T -gate, and it is not a Pauli, so Proposition 104 does not apply to it.

Proposition 102 (measurement dichotomy). *Let W be a Clifford+ T circuit with clean ancillas implementing a diagonal level-three gate U , possibly with measurements. Call W non-adaptive if no measurement outcome is used before the final readout. Suppose W is non-adaptive with all measurements in the computational basis, and that W either is a $\{\text{CNOT}, T\}$ circuit apart from its measurements or lies in the proved regimes and leaves its ancillas in a computational-basis state. Then $t \geq \delta(U)$, which is at least $2\nu + 1$ for a non-Clifford pure-cubic U . If some outcome is used before the final readout, the bound that holds in general is $t \geq \nu$.*

Proof. If no outcome is used before the final readout, postpone every measurement to the end. A CNOT first copies the measured qubit onto a fresh clean ancilla, and the copy is measured at the end, which leaves the channel unchanged because the result conditions nothing. This uses computational-basis measurements. An X -basis measurement would add a Hadamard before the copy, which can change the stratum of Appendix H. Then W becomes a measurement-free Clifford+ T unitary V_0 on data + clean ancillas followed by terminal measurements whose outcomes are discarded. Since the induced data channel is the unitary U , purity forces $V_0|\psi\rangle|0^a\rangle = (U|\psi\rangle) \otimes |e\rangle$ with a fixed ancilla state $|e\rangle$. For a $\{\text{CNOT}, T\}$ circuit the wire map is affine, so $|e\rangle$ is a computational-basis state, a terminal X -string resets it, and Corollary 84 applies. With internal Hadamards, when $|e\rangle$ is a computational-basis state (in particular whenever the circuit returns its scratch space) a terminal X -string, a Clifford, restores the clean slice and Proposition 82 applies in its proved regimes. In both cases $t \geq \delta(U)$. In the adaptive case $t \geq \nu$ is Proposition 100. $\square$

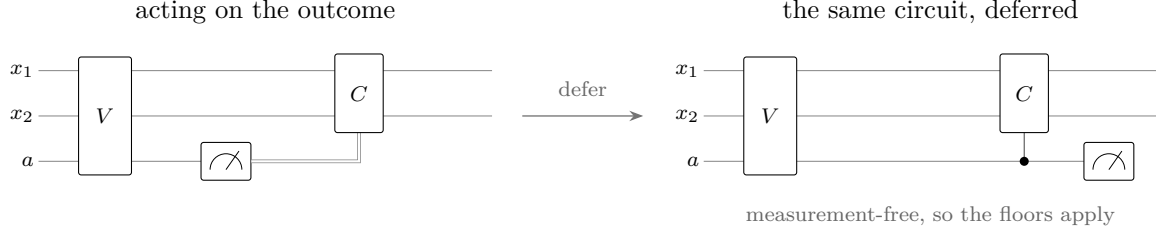


Figure 6: Dilation. Left, the outcome of measuring a selects a correction C . Right, the measurement is deferred and C becomes controlled- C , one Clifford level higher.

I.2.1 Purified feedforward

Purifying the measurement shows which kinds of feedforward break the factor of two, and at what T -cost.

Lemma 103 (Kraus proportionality). *If an adaptive circuit implements a unitary U exactly as its induced channel, then every Kraus operator satisfies $K_m = \lambda_m U$ with $\sum_m |\lambda_m|^2 = 1$. Consequently the deferred-measurement dilation W obeys $W(|x\rangle \otimes |0^{a+M}\rangle) = (U|x\rangle) \otimes |\chi\rangle$ with the record state $|\chi\rangle = \sum_m \lambda_m |m\rangle$ independent of the input.*

Proof. $\sum_m K_m \rho K_m^\dagger = U \rho U^\dagger$ for all ρ means the channel $\rho \mapsto \sum_m (U^\dagger K_m) \rho (U^\dagger K_m)^\dagger$ is the identity, whose Kraus operators are exactly the scalars: $U^\dagger K_m = \lambda_m I$. Completeness gives $\sum |\lambda_m|^2 = 1$, and the dilation statement follows from the definition of W . $\square$

The dilation replaces each classically conditioned correction C by a coherently controlled C (Figure 6). The T -cost of the control follows from the floors of this paper:

- controlled-Pauli (CNOT or CZ) is Clifford and needs no T -gates.
- controlled- S is CS, at 3 T -gates.
- controlled-CZ is CCZ, at 7 (Theorem 40).
- controlled- T is a fourth-level gate, outside the class the floors cover.

Dilating the CZ correction of the Jones construction gives a measurement-free circuit with $4+7 = 11$ T -gates.

Proposition 104 (Pauli feedforward keeps the floor). *Let an adaptive Clifford+ T circuit with t T -gates implement the diagonal level-three gate U exactly, with every feedforward correction a Pauli whose exponent is an affine function of the earlier measurement outcomes (Pauli-frame tracking), and suppose the record state $|\chi\rangle$ of Lemma 103 is a stabilizer state. Then the dilation is a measurement-free Clifford+ T circuit with the same t implementing U with clean ancillas, and $t \geq \delta(U)$ whenever that dilation, including the Clifford that resets the record, lies in the proved regimes.*

Proof. A Pauli correction whose exponent is affine in the outcomes dilates to a product of controlled-Paulis, each controlled by a single record qubit, which is Clifford, so the dilation uses no extra T -gates. A correction conditioned on a product of outcomes would dilate to a multiply-controlled Pauli and is excluded. A Clifford on the record rotates $|\chi\rangle$ to $|0^{a+M}\rangle$, after which W is a clean-ancilla measurement-free circuit for U , and Proposition 82 applies in its proved regimes. That Clifford may add Hadamards and so change the stratum of the dilation. When $|\chi\rangle$ is a computational basis state the reset is a string of X gates and adds none. $\square$

T -teleportation fails both hypotheses: its correction S is not a Pauli, and with the correction normalized so that branch 1 carries the phase ω , its record state $|\chi\rangle = (|0\rangle + \omega|1\rangle)/\sqrt{2}$ is the magic state $|T\rangle$. The dilation inherits the circuit's Hadamards and any added by the reset of the record, so the conclusion carries the regime hypotheses of Proposition 82. In particular Pauli-frame tracking, the correction model of fault-tolerant software, leaves the floor unchanged under those hypotheses.

1.2.2 The adaptive floor on an ε -ball

Lemmas 95 and 96 need a single unitary over $\mathbb{Z}[\omega, 1/\sqrt{2}]$. For a circuit without feedforward the deferred-measurement normal form, with all measurements at the end, still has one. Theorem 106 below handles feedforward.

Corollary 105 (ε -robust floor with terminal measurements). *Let V be a Clifford+ T circuit with t T -gates in deferred-measurement form, with no feedforward (a unitary V_0 on $N = n + a$ qubits over $\mathbb{Z}[\omega, 1/\sqrt{2}]$ followed by terminal measurements), and suppose the clean-slice block of V_0 satisfies the entrywise hypothesis of Theorem 97 for a diagonal level-three U with $\varepsilon < 2^{-(2N(t+1)+2)}$. Then V_0 implements U on that slice exactly and $t \geq \nu(U) = n - d^*$. Under the hypotheses of Proposition 102, $t \geq 2\nu + 1$ for a non-Clifford pure-cubic U .*

Proof. Theorem 97 applies to the unitary V_0 . The two bounds are then Propositions 100 and 102. $\square$

Closeness of the induced channel alone does not imply the block hypothesis. For that weaker hypothesis, and for circuits with intermediate measurement and feedforward, the Galois gap still holds entrywise for the Choi matrix, which is ring-valued for any adaptive Clifford+ T circuit.

Theorem 106 (Choi-matrix ε -rigidity). *Let Λ be the channel on n data qubits induced by any adaptive Clifford+ T circuit with clean ancillas (arbitrary internal Hadamards, mid-circuit measurements, and classical feedforward) using t T -gates and M mid-circuit measurements on $N = n + a$ wires, and let U be a diagonal level-three gate with Choi state $|J_U\rangle$. Set $K' := 4N(t + M + 1) + 4M + 2n$. If the Choi matrix $J(\Lambda)$ satisfies $|J(\Lambda)_{ab} - |J_U\rangle\langle J_U|_{ab}| < 2^{-(K'+1)}$ for every entry, then $\Lambda = U$ exactly, and hence $t \geq \nu(U) = n - d^*$. Consequently, with no hypothesis on ε ,*

$$t \geq \min\left(\nu(U), \frac{\log_2(1/\varepsilon)}{4N} - 2M - 2\right),$$

where ε bounds the Choi-matrix distance (in particular any diamond-norm $\|\Lambda - U\|_\diamond \leq \varepsilon$).

Proof. Write $\Lambda(\rho) = \sum_m K_m (\rho \otimes |0^a\rangle\langle 0^a|) K_m^\dagger$ traced over the ancillas, where the Kraus operators K_m collect the Clifford, T , Hadamard, and measurement-projector factors along outcome branch m . Each K_m has entries in $\mathbb{Z}[\omega, 1/\sqrt{2}]$. The M projectors cut the circuit into at most $t + M + 1$ Clifford segments, each of denominator exponent at most $2N$ as in Lemma 95, and each projector $(I \pm P)/2$ adds at most 2, so $\text{sde}(K_m) \leq 2N(t + M + 1) + 2M$. Each entry of the Choi matrix $J(\Lambda) = (\Lambda \otimes I)(|\Omega\rangle\langle\Omega|)$, $|\Omega\rangle = 2^{-n/2} \sum_x |xx\rangle$, is 2^{-n} times a sum over branches of products of two K_m entries, so it lies in $\frac{1}{\sqrt{2}^{K'}} \mathbb{Z}[\omega]$ with $K' = 2(2N(t + M + 1) + 2M) + 2n = 4N(t + M + 1) + 4M + 2n$. Since $K' \geq 2n$, the target entries (which lie in $\frac{1}{2^n} \mathbb{Z}[\omega]$) lie in the same ring. Summing the 2^M branches cannot raise the exponent, since a sum of ring elements with denominator exponent $\leq K'$ again has denominator exponent $\leq K'$. The Choi matrix $J(\Lambda)$ is a density operator, since $\text{tr } J(\Lambda) = 1$ and $J(\Lambda) \succeq 0$, so $|J(\Lambda)_{ab}| \leq 1$ for every entry. The Galois automorphism $\sigma : \omega \mapsto \omega^3$ commutes with $\dagger$ (complex conjugation is $\omega \mapsto \omega^7$, also in the abelian group), and $\sum_m \sigma(K_m)^\dagger \sigma(K_m) = \sigma(\sum_m K_m^\dagger K_m) = \sigma(I) = I$, so $\{\sigma(K_m)\}$ is again a valid CPTP Kraus set. Hence $\sigma(J(\Lambda)) = J(\Lambda^\sigma)$

is again a Choi density operator and $|\sigma(J(\Lambda)_{ab})| \leq 1$. For any entry $v = J(\Lambda)_{ab}$ and its target $u = |J_U\rangle\langle J_U|_{ab} \in \frac{1}{2^n}\mathbb{Z}[\omega]$, the number $\xi := \sqrt{2}^{K'}(v - u) \in \mathbb{Z}[\omega]$ is an algebraic integer with $|\sigma(\xi)| \leq 2\sqrt{2}^{K'}$, so if $\xi \neq 0$ then $1 \leq |\text{Nm}(\xi)| = |\xi|^2|\sigma(\xi)|^2$ gives $|v - u| \geq 2^{-(K'+1)}$ (exactly as in Lemma 96). Below that radius every entry matches, so $J(\Lambda) = |J_U\rangle\langle J_U|$. Since the Choi map is a bijection, $\Lambda = U$ exactly, and Proposition 100 gives $t \geq \nu(U)$. Otherwise $\varepsilon \geq 2^{-(K'+1)}$, i.e. $\log_2(1/\varepsilon) \leq 4N(t + M + 1) + 4M + 2n + 1 \leq 4N(t + 2M + 2)$, using $n \leq N$, $4M \leq 4NM$, and $2N + 1 \leq 4N$. This rearranges to the second branch. $\square$

Remark 107 (what the Choi extension loses). The exponent K' doubles the unitary exponent because Choi entries are quadratic in the branch operators, and the second branch bounds $t + 2M$ because each projector starts a new Clifford segment. Only ν survives because the factor of two fails under feedforward (Proposition 101), so an approximate $2\nu + 1$ needs a non-adaptive circuit (Theorem 97 or Corollary 105) and, with internal Hadamards, the proved regimes.

I.3 Aperiodic catalysts do not lower the $\{\text{CNOT}, T\}$ floor

A *catalyst* is a register in an arbitrary, possibly entangled and non-stabilizer state, which the circuit may use provided it returns the state unchanged. Catalysts do give savings elsewhere, for example in the catalyzed conversion of a $|\text{CCZ}\rangle$ state into two $|T\rangle$ states [51], and whether magic under stabilizer operations admits robust catalysis is open [60]. For the $\{\text{CNOT}, T\}$ floor, a catalyst whose support has no translation symmetry gives no saving. The adaptive floor $t \geq \nu(U)$ of Proposition 100 needs no such hypothesis. Stabilizer nullity is additive under tensor products and monotone under stabilizer operations [28], so a protocol taking $|\Omega\rangle \otimes |\psi\rangle$ to $|J_U\rangle \otimes |\psi\rangle$ gives $\nu(U) + \nu(|\psi\rangle) \leq \nu(|\psi\rangle) + t$.

Proposition 108 (catalysis does not lower the $\{\text{CNOT}, T\}$ floor). *Let W be a $\{\text{CNOT}, T\}$ circuit on $n + a$ qubits with t T -gates and let $|\psi\rangle$ be any a -qubit state with*

$$W(|x\rangle \otimes |\psi\rangle) = \omega^{\phi(x)} |x\rangle \otimes |\psi\rangle \quad \text{for all } x \in \mathbb{F}_2^n,$$

so that $|\psi\rangle$ is returned exactly and W implements the diagonal level-three gate U on the data. Write $S = \text{supp}(|\psi\rangle) \subseteq \mathbb{F}_2^a$ and let $P = \{p : S + p = S\}$ be its translation stabilizer. If $P = \{0\}$ then $t \geq \delta(U)$.

Proof. As in Corollary 84, $W|z\rangle = \omega^{\Phi(z)}|Lz\rangle$ with $L \in GL(n + a, \mathbb{F}_2)$ and Φ a level-three phase polynomial with t odd parities. Write $L(x, c) = (Ax + Bc, Cx + Dc)$. Comparing supports, for each x the map $c \mapsto L(x, c)$ must send $\{x\} \times S$ into $\{x\} \times S$, so $Ax + Bc = x$ for all x and all $c \in S$. At $x = 0$ this gives $Bc = 0$ for every $c \in S$, and hence $A = I$. The ancilla component is then $c \mapsto Cx + Dc$, and $Cx + D(S) = S$ for every x . At $x = 0$ this gives $D(S) = S$, so $Cx + S = S$, i.e. $Cx \in P$ for all x . By hypothesis $P = \{0\}$, so $C = 0$ and $L(x, c) = (x, Dc)$.

Now fix $c_0 \in S$. Matching coefficients of $|x\rangle|Dc_0\rangle$ on the two sides gives $\omega^{\Phi(x, c_0)}\psi_{c_0} = \omega^{\phi(x)}\psi_{Dc_0}$ for every x , so $\Phi(x, c_0) - \phi(x)$ is a constant independent of x . Substituting the constant $c = c_0$ into Φ turns each parity of (x, c) into a parity of x or a constant, without increasing the number of odd coefficients, so $\Phi(\cdot, c_0)$ is a $\{\text{CNOT}, T\}$ phase polynomial for ϕ with at most t odd parities. Hence $\delta(U) \leq t$.

If $|S|$ is odd then $P = \{0\}$ automatically, since S is a disjoint union of cosets of P and so $|P|$ divides $|S|$. In particular every computational-basis catalyst ($|S| = 1$) is covered. $\square$

Remark 109 (scope). When $P \neq \{0\}$ the argument yields only $\delta(U|_H) \leq t$ on the subspace $H = \ker C$ of codimension at most $\dim P \leq a$, and a catalyst can beat $\delta(U)$ only if the data translate

the borrowed register by Cx . The periodic case does occur, since the $|T\rangle$ catalyst of [51] has full support on its qubit. Whether a periodic catalyst can beat the $\{\text{CNOT}, T\}$ floor is open.

J Consequences for resource estimates

This appendix applies the floors to primitives whose expensive part is a diagonal $\pi/4$ -angle gate: a phase oracle, a Toffoli network or a field-arithmetic block.

Primitive	isotropy floor	prior bound	outcome
3-local Ising or Trotter cost layer	$2(n - d^*) - r$	nullity $n - d^*$	per-step floor; $2(n - d_f) + 1$ in Boolean-oracle form, d_f the radical dimension (Cor. 50, 51)
AND chain of m Toffolis (ripple carry, the chains of unary iteration, Toffoli-chain diffusion)	$4m + 3$ (shadow)	nullity $2m + 1$	exact on the CCZ-layer content (Cor. 45, Prop. 47)
GF(2^k) field multiplier	$6k + 1$	nullity $3k$	$2\nu + 1$; = 17 at $k = 2$ (Prop. 56)

Table 9: The isotropy floor on primitives whose expensive part is diagonal level-three. Every floor is computed in $O(n^3)$ time from the phase polynomial and is never below stabilizer nullity [14]. A row marked shadow bounds only the diagonal CCZ-layer part of a circuit that reuses targets as controls (Appendix E). On that part the floor equals the T -count of a construction (Proposition 47) and is therefore exact. The exact 17 on the field row at $k = 2$ is by decoding (Remark 57). The QFT is absent because its retained phases are quadratic.

The floors bound diagonal content with phases in $\frac{\pi}{4}\mathbb{Z}$, the Toffoli and phase-oracle part of an algorithm’s T -count. Arbitrary-angle rotations $R_z(\theta)$ are handled by number-theoretic synthesis [55], which is complementary, and the two meet at the ε -rigidity of Appendix I.1. A magic-state preparation circuit has no free input register and hence no diagonal gate to bound, but the floor still bounds its Hadamard-free segments block by block (§2.3.4).

Remark 110 (known adders between the floors). The chain row of Table 9 places the known constructions between the floors. The CCZ-layer content of the carry chain of a ripple adder with m Toffolis, such as [22], is a chain in which consecutive blocks share one carry wire. This is a linear-overlap pattern on $n = 2m + 1$ qubits, so its diagonal shadow (Appendix E) has the sharing floor $6m + 1 - 2s$ of Corollary 45, which with $s = m - 1$ shared wires is $4m + 3$, in the $\{\text{CNOT}, T\}$ model (and for measurement-free Clifford+ T circuits in the proved regimes). In the adaptive model the floor on the same shadow is $\nu = n - d = 2m + 1$ (Appendix I.2). Both numbers bound the shadow only, since the carry chain reuses each carry target as the next control and so is not mismatch-free (Appendix E). The T -counts of the adder itself are therefore not ordered against them. Gidney’s adder [20] uses 4 T -gates per temporary AND, about $4m$ in total, with a measure-and-fix-up uncomputation, i.e. adaptive feedforward, and compiling gate by gate uses $7m$. The state-conversion bounds of [28] on the modular adder of two w -bit registers ($\geq w + 1$ T -states, [28, §4.2]) concern the adaptive model. A floor on the adder itself is open.

J.1 The chain, the fan-out layer and the field multiplier

Unary iteration [23], the addressing circuit of QROM and of select operations, is built from AND chains of the ripple-carry form, and Remark 110 applies to each such chain. Its full control tree reuses address wires across branches and is not covered. Work on QROM [24, 61] counts Toffolis

or T gates of the whole reversible circuit, and the floors here bound the diagonal shadow of each chain that circuit contains.

A single control that turns a layer of CZ gates into CCZ gates, and the m Fredkin gates of a swap test, are both the fan-out pattern of Theorem 43, with $\{\text{CNOT}, T\}$ T -count exactly $4m+3$ [13, Thm. 5], for the Fredkins on their diagonal shadow.

The $\text{GF}(2^k)$ multiplication oracle on $n = 3k$ qubits is pure-cubic with trivial radical, because a radical direction would be a field element whose product with every element is zero. Hence $\delta \geq 6k + 1$, and $\delta = 17$ at $k = 2$ (Proposition 56, Remark 57). The floor is per call. It adds across calls only when the multipliers are compiled separately, since T gates can merge across calls (Theorem 39). The binary-curve discrete-logarithm circuit of Banegas, Bernstein, van Hoof and Lange [62] over a field of degree k has $2k + 2$ point additions, each using four multiplications plus two Itoh–Tsujii inversions of $\ell + w - 1$ multiplications, where $\ell = \lfloor \log_2(k - 1) \rfloor$ and w is the Hamming weight of $k - 1$ (their Section 6.2). Each multiplication is the oracle U_k up to Cliffords, so its Hadamard-free cost is at least $G(k)$ (Proposition 60), which is 1138 at $k = 163$ and 3992 at $k = 571$. With each multiplier implemented measurement-free and its Hadamards in the output layers, the multiplication content of the algorithm, compiled as separate multipliers, needs at least 8.2×10^6 T gates at $k = 163$ and 1.4×10^8 at $k = 571$. Multipliers uncomputed by measurement fall under Appendix I.2 instead.

K Further discussion and open problems

This appendix compares the floor with the stabilizer Rényi entropy, lists the open questions raised in the appendices, and records two scope limits: the multi-controlled Z and one Hadamard layer.

K.1 Comparison with the stabilizer Rényi entropy

The amortized stabilizer Rényi entropy gives the bound $t \geq \lceil M_2(|J_U\rangle)/(2 - \log_2 3) \rceil$ [11, Cor. 4], which we evaluated on every $GL(n, 2)$ -orbit of cubic forms up to $n = 6$ (Appendix L). On a pure-cubic gate, let $T(a)$ be the slice that is the symplectic form of the derivative $\Delta_a f$. The quadratic Walsh spectrum is flat, and summing it gives

$$M_2(|J_U\rangle) = n - \log_2 \sum_{a \in \mathbb{F}_2^n} 2^{-\text{rank } T(a)},$$

which we checked against the 4^n Pauli sum. The entropy therefore depends on the ranks of the slices, while the floor depends on the dimension of their span (§2.2.1).

The SRE bound is strictly larger than nullity on every orbit at $n = 3, 4, 5, 6$. Already CCZ gets 4 against $\nu = 3$, from $M_2(|J_{\text{CCZ}}\rangle) = 5 - \log_2 11$. Carrying it into the adaptive model would need monotonicity of M_2 under mixed-state operations, which we have not verified. If it holds, CCZ attains the ratio $c = 4/3$ at which Proposition 101 stops. The floor $2\nu + 1$ exceeds the SRE bound on every orbit at $n \leq 6$ and on every named family. On the disjoint Toffoli layer the SRE bound is $\lceil \sigma m \rceil$ with $\sigma \approx 3.7119$ against $6m + 1$, a ratio tending to 1.616, where the ratio for ν is 2. On the fan-out layer of Theorem 43, $M_2 \rightarrow 3$ and the SRE bound reaches 8 at $m = 4$ and never exceeds it, while $\nu = 2m + 1$ grows, so nullity is larger from $m = 4$ on.

Slices are alternating, so their ranks are at most $2\lfloor n/2 \rfloor$ and M_2 is at most about $n - 1$. The SRE bound is then at most about $2.4094(n - 1)$, and this cap exceeds the floor’s own cap $2n + 1$ from $n = 9$ on. A hill-climb on the rank profile (Appendix L, seed 0, 300 iterations) reaches $M_2/n = 0.758$ at $n = 11$ against the 0.868 a crossing needs there, and the SRE bound stays 2 or

3 below $2\nu + 1$ on the best gate it found for each $n \in \{9, 10, 11\}$. Whether this persists is Open Problem 9.

K.2 Further open questions

Besides the five open problems of Section 4, the appendices raise the following.

1. Does the line family have $\delta = 2n + 1$ for every n (Proposition 66)? The wrapped cycle does not, since $\delta(C_6) = 17$.
2. For a connected component C of the block-overlap graph of a linear-overlap layer, write m_C for its blocks, n_C for its qubits and $s_C = 3m_C - n_C$. Proposition 47 attains $6m + 1 - 2s$ whenever the merges route through a forest of maximum degree three, and Remark 48 shows that a component with $s_C \geq m_C$ can fail. All 194,670 linear covering patterns with $n \leq 10$ and $m \leq 4$ satisfying $s_C = m_C - 1$ admit such a routing.

Conjecture 111 (dichotomy for the sharing floor). *A linear-overlap layer attains $6m + 1 - 2s$ if and only if every connected component of its block-overlap graph satisfies $s_C = m_C - 1$.*

The routing of Proposition 47 does not prove the “if” direction in general. If one qubit lies in three blocks and each other qubit of those blocks lies in one further block ($m = 9, n = 19$), then $s_C = m_C - 1$, but every choice of the trees F_q gives some block four neighbors in F .

3. What is δ on residues of weight 8 to 15, just above the rigidity radius, where adding a codeword of $\text{RM}(n - 4, n)^*$ can lower the weight (Theorem 39)?
4. Does the parallel Toffoli layer compress in CS-count inside $[2m + 1, 3m]$ (Corollary 76)?
5. Can the factor N in the exponent $2N(t + 1)$ of Theorem 97 be removed?
6. Which efficiently computable invariant determines the cost of the three non-degenerate classes at $n = 6$ that share $d = 0$ and cost 13, 15 and 17 (§2.4.1)? The ranks of the slices distinguish the classes but do not by themselves give the cost, and the bounds of Proposition 60 and Corollary 59 reach 14 on the two the floor misses. The derivative gates cannot help, since every coefficient of a derivative phase is even, so each derivative gate is Clifford.
7. Can the higher-degree oracles that resource estimates use, such as QROM and select towers, be bounded tightly? Their phase content exceeds degree three, so this needs a tight level- k bound and a way around the obstruction of Remark 74.
8. Can $\delta(U) + \delta(V) - \delta(U \otimes V)$ be unbounded on disjoint supports? It is positive for some pairs by Theorem 39.

K.3 Why this paper does not bound $C^{n-1}Z$

The exact T -count of the multi-controlled Z lies between the nullity n (Corollary 73) and the known $O(n)$ constructions, and we leave it open for three reasons. First, for $k \geq 4$ the quantity δ_k counts level- k rotations and differs from the T -count, so $\delta_n(C^{n-1}Z) = 2^n - 1$ (Proposition 72) is exponential while the T -count is $\Theta(n)$. Second, even as a level- n count the floor has slack $2^n - 2n - 1$ on this family (Proposition 72), and Remark 74 explains why the radical method has no tight analogue there. Third, $C^{n-1}Z$ is a level- n gate, so every Clifford+ T circuit for it uses internal Hadamards, and Appendix H covers only some Hadamard strata. For $n \geq 4$ and measurement-free AND-compiled circuits whose AND controls are data inputs or earlier AND values, Corollary 93 brackets the cost to within three T gates.

K.4 One Hadamard layer: what is known

The nearest non-diagonal target is $U = D_2 H^{\otimes n} D_1$, whose entries are all $2^{-n/2}$ times eighth roots of unity. Call such a unitary *flat*. Its matrix element is $2^{-n/2} \omega^{P(x,z)}$ with $P(x, z) = \phi_1(x) + \phi_2(z) + 4(x \cdot z)$, a level-three phase polynomial on $2n$ variables. The remarks below rest on proof sketches and numerics.

Remark 112 (nullity adds across a Hadamard layer). Let C be a Clifford whose symplectic matrix has upper-right block B . If B is invertible, as for $H^{\otimes n}$, then $\nu(D_2 C D_1) = \nu(D_1) + \nu(D_2)$. Conjugation by D_1 sends $X^a Z^b$ to a Pauli times a diagonal Clifford whose symplectic block M_a is the contraction of the moment tensor in the proof of Proposition 34, and a stabilizer of the product needs $B M_a B^\top = 0$, hence $M_a = 0$. The hypothesis is needed: $C = H \otimes I$ with $D_1 = D_2 = I \otimes T$ gives $\nu = 0 \neq \nu(D_1) + \nu(D_2) = 2$. Brute force over all Paulis at $n \leq 4$ confirms additivity on every sampled pair with B invertible (Appendix L).

Remark 113 (the minimal-Hadamard floor). A measurement-free circuit with clean ancillas for a flat U needs $h \geq n$ Hadamards, since for fixed x the branch map must reach every z . At $h = n$ the branch is affine in (x, z) , and the argument of Proposition 82 gives $t \geq \delta_{2n}(P')$ for the lifted polynomial, which no longer depends on the circuit. For $U = D_2 H^{\otimes n} D_1$ the term $4x \cdot z$ is a CZ layer, so this equals $\delta(D_1 \otimes D_2)$. For $h = n + e$ the reduction re-runs on $2n$ variables. The case $e = 0$ is the sketch above, $e = 2$ at maximal output rank follows as in Table 8, and $e = 1$ is open. On $\text{CCZ } H^{\otimes 3} \text{CCZ}$ the floor is 13, while the minimum coset weight over the parity sets realizable by a prefix of the Hadamard order is 14 for all six orders, which suggests a cost of 14 at $h = n$. The prefix sets are not exhaustive, so this remains a conjecture.

Remark 114 (nullity is at most $2n$). $\nu(U) \leq 2n$ for every n -qubit unitary, and the same cap holds for every efficiently computable magic measure we know of, each factoring through the $2n$ -qubit Choi state. The block sum of an L -layer circuit is about $L(2n + 1)$, so the gap to any such measure grows linearly in L . The case $n = 1$ is exact: $U_L = T(HT)^L$ has T -count $L + 1$ [63, 56, 57] while $\nu(U_L) \leq 2$. A floor within a constant factor of the block sum must therefore use the layer structure of the circuit.

L Computational verification

The theorems of this paper are proved analytically. Computation was used to check them and to obtain the exact δ values quoted for specific gates. Every script named below is in the accompanying repository [47], whose index `CLAIMS.md` maps each row to a command. Its runner, `reproduce.py`, reruns a row and compares the output with a recorded reference where one is recorded (26 of the 64 rows). The exact δ is computed by Reed–Muller coset decoding (`validate_amdist.py`), which serves as ground truth for the others.

Claim	Range checked	Script
Toffoli layer $\delta_m = 6m + 1$: decode $m \leq 2$	lower bound is Thm. 40, all m	<code>validate_amdist.py</code>
Line family $\delta(f_n) = 2n + 1$: decode $n = 5, 6$; weight- $(2n + 1)$ witnesses $7 \leq n \leq 12$	floor is Prop. 66, all n	<code>validate_amdist.py</code> , <code>greedy_reduce.py</code>
Two-sided brackets on the application families (Tab. 4)	layers $m \leq 5$; QAOA/Ising $n \leq 6$	<code>bracket_table.py</code>

Claim	Range checked	Script
Cubic orbit atlas (Tab. 3): every $GL(n, \mathbb{F}_2)$ -orbit, floor exact on $1/1, 2/2, 3/5$ at $n = 4, 5, 6$ (39.4% of forms at $n=6$); largest gap floor 13 against $\delta = 17$ at $n = 6$	complete orbit enumeration $n \leq 6$ (2^{20} forms at $n=6$)	<code>cubic_orbit_atlas.py</code>
Level- k nullity against $2(n - d_k)$: the homogeneous quartic with $d_4 = 6$ has $\nu = 9 > 8$ (App. G.1)	$n = 10, k = 4$	<code>levelk_dominance.py</code> , <code>levelk_counterexample.py</code>
Width-one conservation $N_{\ker} \geq \delta(\kappa)$, evidence for Conj. 87	2153 realizable families	<code>validate_w1.py</code>
Overlap floors: linear-overlap $d = 0$; decode $m = 2$ ($\delta = 11$); pair-share $d = n - 3$ (Lem. 42, Thm. 43, Rem. 49)	stars/chains $m \leq 12$; 300 random linear hypergraphs	<code>overlap_layers.py</code>
Field oracle: $d=0$, floor $6k + 1$; $\delta(\text{GF}(4)) = 17$ exact (Prop. 56, Rem. 57)	$k \leq 4$ radical; decode $k \leq 2$	<code>field_oracle.py</code>
Slice-rank floor for field oracles: every nonzero slice has rank $2k$; wt $c(w) \geq \text{rank } T(w) + 2$, tight on all 1008 optimal residues of U_2 ; $G(2), \dots, G(8) = 14, 20, 28, 34, 41, 47, 56$; Delsarte program 13, 14, 14 on the $n = 6$ orbits, never above δ (Lem. 58, Cor. 59, Prop. 60)	slices enumerated $k \leq 4$	<code>u2_optimal_residues.py</code> , <code>rank_spectrum_lp.py</code>
Field-oracle brackets $[6k + 1, W_k]$, $W_k = (7, 17, 36, 58, 90, 123)$ minimized over all irreducibles of each degree, $d = 0$ (Rem. 69)	$k \leq 6$; all 2, 1, 2, 3, 6, 9 irreducibles	<code>gf2m_floor.py</code>
Floor on controlled unitaries is 2μ , or $2\mu + 1$ when pure-cubic, and Campbell–Howard $2\mu + (\mu \bmod 2)$ equals exact δ (Prop. 44)	2,248 decoded gates $n \leq 6$; 36,000 gates $n \leq 12$ against the formula	<code>floor_vs_controlled.py</code> , <code>floor_vs_controlled_gap.py</code>
Splicing lemma and the constructions attaining Cor. 45: disjoint $6m + 1$, fan-out $4m + 3$, ripple-carry chain $4m + 3$, all with moment match (Lem. 46, Prop. 47)	$m \leq 9$	<code>splice_engine.py</code>
Attainment of the sharing floor: every pattern with $s \leq m - 1$ attains it, and the two patterns with $s \geq m$ fail by 2 (floor 13 vs $\delta = 15$, Rem. 48)	all linear covering hypergraphs $n \leq 6$ up to relabeling, exact RM decoding	<code>ledger_attainment.py</code>
The componentwise count admits a degree-3 forest routing on every tested pattern (Conj. 111)	exhaustive $n \leq 10, m \leq 4$ (194,670 patterns); sampled $n = 11$	<code>merge_forest_exists.py</code>
Benchmark certification: PyZX <code>phase_block_optimize</code> output partitioned into phase blocks; floor, exact decode (≤ 6 active qubits), certified blocks (§2.3.4, App. F.2)	24 Feynman-suite circuits (optimizer seed 4), 311 blocks, 193 certified (113 by nullity alone); 239 decoded exactly, every value inside its bracket; seeds 0 to 4 from the recorded logs in <code>results/benchmark/seeds</code>	<code>benchmark_certify.py</code> , <code>benchmarks/run_suite.py --seed k</code>

Claim	Range checked	Script
Floor tightness on decodable blocks: distance to the exact T -count $0/1/2$ on $159/79/1$ (TODD) and $221/47/8$ (teleportation) blocks; gap split $81 + 8 + 148 = 237$ (Tab. 6)	515 blocks with ≤ 6 active qubits	<code>benchmark_certify.py</code>
Second optimizer: phase teleportation output certified, 252/337 blocks, 141 by nullity; tensor-checked on three circuits (Tab. 7)	27 circuits, deterministic, recorded run in <code>results/benchmark/teleport_suite.log</code>	<code>benchmarks/run_suite.py</code> <code>--optimizer teleport</code>
$\text{GF}(2^k)$ multipliers as whole circuits: mismatch-free structure, field polynomial ($k \leq 10$), $d = 0$, floor $6k + 1$ (Appendix F.2)	all twelve files, $k = 4$ to 256	<code>gf2k_family.py</code>
Whole-circuit mismatch scan of the suite: the multipliers and <code>mod5_4</code> pass, every other circuit reuses a written wire as a control	45 circuits	<code>mismatch_scan.py</code>
AND-compiled circuits, $t \geq \delta(U) + A$ (Cor. 86); windows carry ≥ 4 odd parities (Lem. 85); the standard temporary-AND circuits	500 random in-class circuits, two templates, shared controls, serial reuse; 968 AND-chain and odd-middle instances with $t \geq \delta(U)$ (<code>--samples 500 --seed 1</code>)	<code>and_gadget_probe.py</code>
AND-compiled circuits at every level: $t \geq 8A + \text{odd}(M)$ (Thm. 92); the $8k - 17$ circuits for $C^{k-1}Z$ simulated exactly (Cor. 93)	300 random AND-chain circuits; $k \leq 6$	<code>and_ladder_floor.py</code>
Nullity additive across a Hadamard layer with B invertible; structural stabilizer count (Rem. 112)	90/90 pairs per full-layer Clifford, 30 random Cliffords, $n \leq 4$	<code>nu_layers.py</code>
Two-layer floor $\delta_{2n}(P) = \delta(D_1 \otimes D_2)$ and bracket; flat circuits at $h = n$ obey $t \geq \delta_{2n}(P')$; prefix-realizable minimum 14 for $\text{CCZH}^{\otimes 3}\text{CCZ}$ (Rem. 113)	$n \leq 4$ exact; 200 circuits at $n = 2, 3$; all six Hadamard orders	<code>twolayer_bracket.py</code> , <code>flat_pathsum.py</code>
Single-qubit $T(HT)^L$ has T -count $L + 1$ and $\nu \leq 2$ (Rem. 114)	36,816 classes, $L \leq 8$	<code>ma_tcount_n1.py</code>
Amortized SRE bound $\lceil M_2(J_U\rangle)/(2 - \log_2 3) \rceil$ against ν and $2\nu + 1$; closed form $M_2 = n - \log_2 \sum_a 2^{-\text{rank } T(a)}$ against the 4^n Pauli sum (§K)	every $GL(n, \mathbb{F}_2)$ -orbit of cubic forms $n \leq 6$; Toffoli and fan-out layers	<code>sre_vs_nullity.py</code>
Hill-climb on the derivative-rank profile for a cubic whose SRE bound exceeds $2(n - d^*) + 1$ (§K)	$n = 6, \dots, 11$	<code>sre_crossover.py</code>

The general lower bounds rest on the isotropy argument, and the upper bounds are explicit constructions whose correctness reduces to a finite check of the order- ≤ 3 moments. Exhaustive computation supplies exact values for specific small gates, such as $\delta(\text{GF}(4)) = 17$ (Remark 57) and the cost 15 in Remark 48. It also supplies the stratum searches of Appendix H, which are quoted as evidence and used in no proof. Further scripts that re-verify proved statements, such as the moment identities, are listed in `CLAIMS.md`.

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AI Disclosure

The isotropy floor, its application to Toffoli layers, and the overall proof framework, including the identification and formulation of the principal theorems and lemmas, were developed entirely by the author. Generative artificial intelligence tools (Claude Opus 4.5, Claude Opus 5.5 and Claude Fable) assisted the author in deriving and refining some of the proofs, in writing the verification code, and in drafting and revising portions of the manuscript. The author reviewed and verified all AI-assisted arguments and text and takes full responsibility for the content of this work.

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